

Department of Information Engineering and Computer Science

Master's Degree in
Artificial Intelligence Systems

Final Dissertation

Interleaved Vision Fusion for Time-Series Foundation Models

Measuring What Vision Contributes to Photovoltaic Ramp Forecasting

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Abstract

Solar power forecasting for newly commissioned generation assets presents a difficult zero-shot transfer challenge. Operational histories at newly connected installations are often too brief to expose seasonal cycles, cloud-induced ramp transitions, or localized weather dynamics. While time-series foundation models provide generalized temporal priors learned across extensive multi-domain corpora, historical generation series record atmospheric attenuation only after clouds pass over the array. Geostationary satellite imagery offers an upstream physical observation of approaching cloud fields and atmospheric optical depth. This thesis examines whether fusing numerical time-series foundation models with visual foundation models improves intra-day forecasting on unseen solar installations, evaluating both in-region fleet generalization and cross-continental transfer between distinct satellite sensor domains.

The investigation relies on an aligned multimodal corpus constructed ad hoc for this thesis, combining surface generation records, numerical atmospheric covariates, and geostationary satellite imagery across 100 United Kingdom installations (Meteosat SEVIRI) and eight United States plants (GOES-16 ABI), maintaining sample-level timestamps, solar zenith masks, and data provenance. The primary benchmark enforces a strict plant-disjoint partition: 69 UK plants for training, 15 for validation, and 14 held-out plants for testing. Forecasters generate six-hour predictions from fourteen days of local numerical context without parameter fine-tuning on evaluation installations. The evaluation protocol macro-averages valid daylight targets across installations, measuring skill as the percentage reduction in normalized root mean squared error relative to Smart Persistence and ramp accuracy through normalized mean absolute error across the top decile of observed power transitions.

Within this benchmark, MMTSFM (Multimodal Time-Series Foundation Model) integrates an adapted Chronos-2 temporal forecaster with a frozen V-JEPA 2.1 video encoder. A systematic fusion ladder compares visual compression strategies: pooled-image fusion with a single global token, spatial-token fusion interleaving 98 spatially addressable visual tokens with the numerical sequence, multi-anchor fusion distributing tokens across five daily observation windows, and full payload retaining dense spatial features across all anchors. On the 14 held-out UK plants, spatial-token fusion achieves 55.10% skill and 0.1440 ramp NMAE across three matched seeds, outperforming the adapted numerical baseline (52.30% skill, 0.1506 ramp NMAE) with consistent improvements at 13 of the 14 test plants.

Systematic diagnostic controls confirm that these forecasting gains reflect authentic visual reliance. Evaluating trained models with visual embeddings set to zero degrades ramp NMAE by 0.0063 for spatial-token fusion, whereas pooled-image fusion exhibits zero mean degradation. Substituting stale imagery from six hours prior or contemporaneous clips from distant installations collapses spatial-token skill from 55.10% to 37.12%, confirming that the cross-attention layers actively rely on temporally aligned, local cloud fields. Furthermore, permuting cached temporal latents changes ramp NMAE by at most 0.00030 across all seeds. This insensitivity indicates that cross-attention operates primarily on unordered spatial and optical depth features within the latent field rather than explicit frame-to-frame advection order, querying visual tokens by spatial and semantic coordinates.

Zero-shot transfer to eight held-out US plants across 41,914 daylight targets exposes a functional shift under joint geographic and sensor distribution changes. Multi-anchor fusion achieves the highest cross-region skill at 10.46%, surpassing the numerical control (8.90%) and spatial-token fusion (8.85%). Distributing visual context across multiple daily timestamps stabilizes point forecasts against sensor discrepancies between SEVIRI and GOES-16 by anchoring predictions across recurrent diurnal states. Simultaneously, spatial-token fusion retains the lowest US ramp NMAE at 0.1692 (compared with 0.1702 for multi-anchor and 0.1864 for the numerical baseline), demonstrating that localized cloud boundaries remain informative for abrupt transitions under sensor shift. Probabilistic assessment indicates that point accuracy improvements do not guarantee calibration: spatial-token fusion reduces Continuous Ranked Probability Score from 0.0574 to 0.0544, yet nominal 80% prediction intervals exhibit undercoverage (73% in the UK and 41% in the US). These findings demonstrate that multimodal

foundation model fusion delivers verifiable gains for zero-shot solar forecasting, while establishing that cross-sensor transfer and probabilistic reliability require paired lesion protocols alongside headline benchmark rankings.

Declaration on the Use of Artificial Intelligence

In compliance with institutional academic integrity policies regarding the use of artificial intelligence in postgraduate research, this thesis discloses the assistance of agentic and generative artificial intelligence tools (specifically large language model coding agents) during its preparation. AI assistance was utilized under human direction across three specific operational areas:

- **AI-assisted coding**: Assisting in the implementation of data preprocessing routines, pipeline orchestration, configuration management through Hydra, and evaluation scripts for benchmark models and diagnostic lesion controls.
- **Scientific literature research**: Assisting in literature discovery, querying recent publications on time-series foundation models and multimodal vision-language architectures (2024–2026), and retrieving primary reference documentation.
- **Prose corrections and revision**: Assisting in grammatical checking, fluency editing, stylistic polishing, structural consistency auditing, and LaTeX formatting refinements.

All conceptual formulation, experimental methodology, baseline selections, analytical interpretations, and final editorial decisions were determined and validated by the author. The candidate assumes full intellectual responsibility for the content, scientific claims, and integrity of the dissertation.

1 Introduction

Many machine-learning and deep-learning forecasters learn to predict an entity's future samples from its earlier observations. When the same entity supplies a long training history, the model can learn recurring temporal patterns together with characteristics specific to that entity. Testing that model on later dates measures how well it extrapolates the familiar series. By contrast, a newly observed entity must be forecast before a comparable training record exists.

A short local history reflects recent operating states but provides little evidence about seasonal cycles, extreme transitions, or plant-specific weather responses. The forecaster must therefore import cross-entity representations learned during pretraining or fleet adaptation and condition them on local context. Neither approach guarantees transfer: models often capture patterns specific to training installations that degrade when evaluated on an unfamiliar plant.

Figure 1.1 illustrates this knowledge transfer to unseen plants in the same region and in a new region.

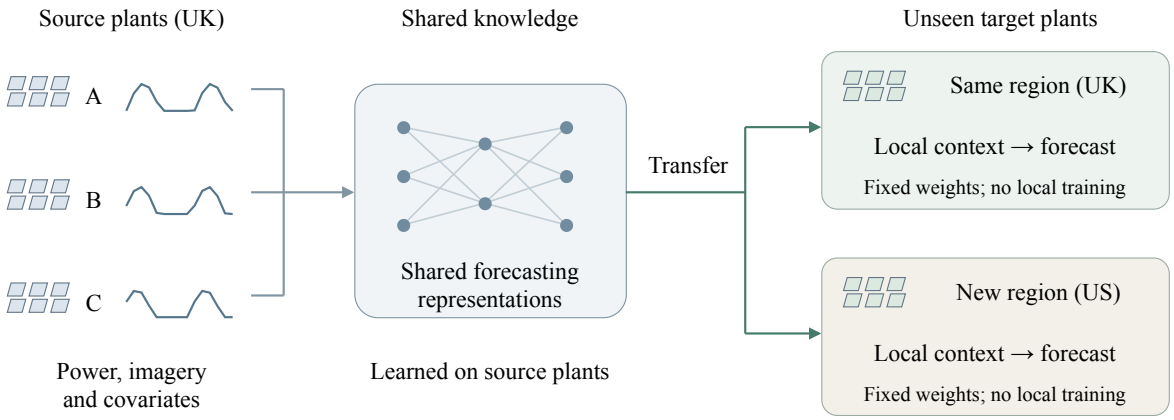


Figure 1.1: Knowledge transfer to unseen plants in the same region and in a new region. Data from UK source plants adapt a shared forecaster. Its learned weights are then applied to held-out UK plants and to US plants without local training. In both settings, fourteen days of local power and environmental context condition a six-hour forecast. Target plants supply inputs but no weight updates. Traces and network symbols are schematic.

Photovoltaic power makes the transfer problem concrete. The sun's position sets a regular daily pattern, but clouds can change output quickly, and the same conditions affect plants differently. Because these cloud-driven changes can occur within the six-hour forecast, recent power shows only what has already happened at the plant. Weather fields describe conditions over a wider area, while satellite images show the nearby cloud field. These sources differ in timing, spatial detail, and format, so the model must combine them for each forecast without a long training history from the target plant.

Figure 1.2 compares clear conditions with a cloudy window containing repeated power ramps at held-out UK plant 11287. The clear-sky reference follows a smooth daily profile, but the cloudy-day measurements rise and fall sharply. Eight half-hour changes exceed ten percentage points of installed capacity, including a change larger than thirty percentage points. The satellite crops show the corresponding spatial observations.

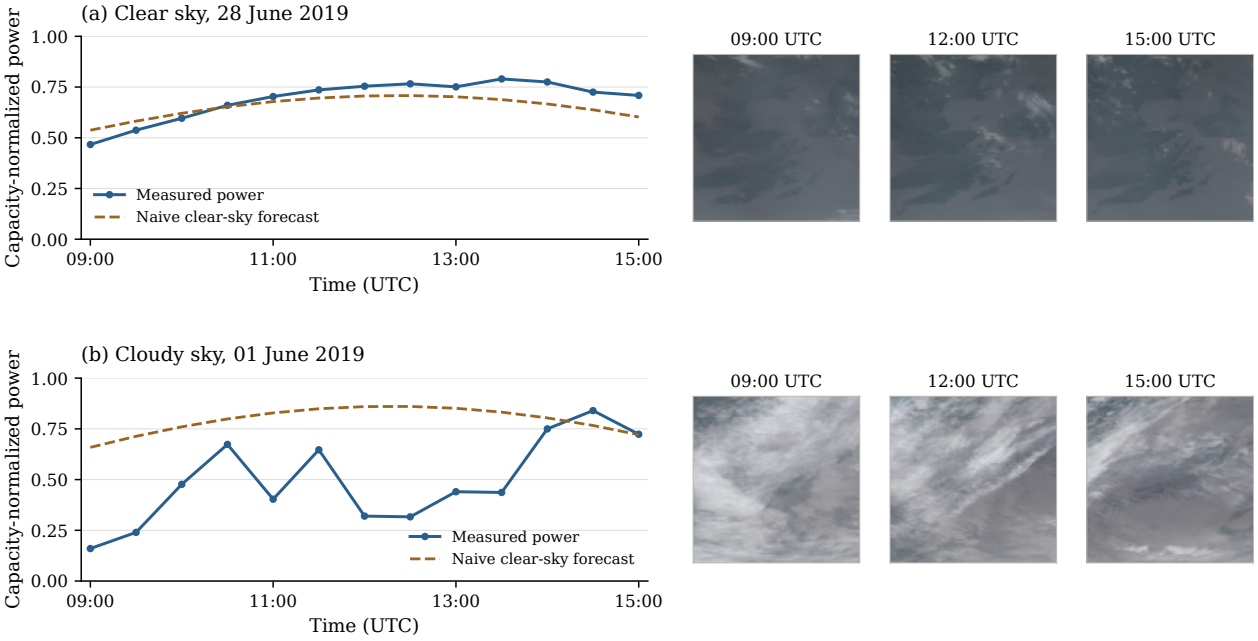


Figure 1.2: Clear-sky and cloudy conditions at UK plant 11287: (a) 28 June 2019; (b) 1 June 2019. Each row pairs a six-hour power trace with satellite observations of the same area. On the left, blue points show half-hourly measurements, joined by solid lines; a value of 0.5 denotes half the installed capacity. The orange dashed line is a naive clear-sky forecast, scaled using clear observations from the preceding fourteen days. The smooth profile in (a) contrasts with the repeated steep rises and falls in (b), which the reference forecast does not capture. On the right, display-enhanced infrared composites show cloud-field structure at 09:00, 12:00, and 15:00 UTC, read from left to right. The corpus is described in Chapter 3.

This thesis studies that challenge by constructing an aligned numerical and satellite corpus, evaluating forecasts on plants excluded from fitting, and testing interfaces between a numerical foundation model and a frozen visual encoder.

1.1 Forecasting from limited history and imagery

Plants can differ in orientation, shading, local weather, and sensor behavior even after power is normalized by capacity. The forecaster must interpret a new plant's short record using patterns learned from other installations. The training, validation, and test sets therefore contain different plants. At each test forecast origin, the model receives fourteen days of recent power and permitted covariates. Models that use vision also receive satellite imagery. The model predicts the next six hours. The test plant supplies observations for that forecast, but no examples for fitting the model or its preprocessing statistics.

Satellite imagery may supply information that has not yet appeared in the plant's power trace. This visual context might help when a cloud system is approaching the array or when the local scene

is represented more precisely than in the numerical covariates. The later controls examine these two explanations separately.

The challenge is to turn the visual field into a useful forecast input. A pretrained video encoder produces features over time and image location; the numerical foundation model processes patches of power and covariates. The thesis compares different interfaces while keeping the underlying numerical lineage and visual source as consistent as the experiments permit.

A higher multimodal benchmark score does not guarantee that the model actively uses the visual inputs. Structural modifications, regularizers, and optimization dynamics can alter predictions even when visual features contribute no useful signal. The evaluation protocol therefore couples benchmark scoring with zeroed-vision controls in fixed weights alongside temporal and spatial image substitutions to confirm whether the forecast genuinely relies on timely, site-matched imagery.

1.2 Formal problem setting

The task is a forecast for one plant from information available at a specified origin. The notation below defines the plant split, input window, target, and scoring rules; Chapter 5 gives the experimental implementation.

1.2.1 Plants, inputs, and forecast windows

Plant identity and forecast time impose different constraints on a prediction. The split determines which installations can shape the fitted model, while the window determines what the model can observe when forecasting at a particular plant.

Let i identify a plant and t a forecast origin on its native power-sampling grid. Denote measured power at that time by $p_{i,t}$ and the plant's audited installed capacity by c_i. Capacity-normalized power is then $y_{i,t} = p_{i,t}/c_i$. At t, the model uses recent normalized power and numerical covariates describing atmospheric conditions, solar geometry, and calendar time. Multimodal methods also use a recent satellite clip when imagery is available.

Let $\mathcal{P}$ denote the complete set of plants, with $\mathcal{P}_{\text{train}}$, $\mathcal{P}_{\text{val}}$, and $\mathcal{P}_{\text{test}}$ denoting the training, validation, and test subsets. These subsets are pairwise disjoint and cover all plants:

$$\mathcal{P} = \mathcal{P}_{\text{train}} \cup \mathcal{P}_{\text{val}} \cup \mathcal{P}_{\text{test}},$$

In particular,

$$\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$$

and the same condition holds for validation against the other two subsets. Model weights, preprocessing statistics, and retrieval stores are fitted without test-plant examples; Chapter 5 specifies the safeguards.

Let H be the number of past power samples, h the number of future power samples, and L the number of satellite frames in a window. The physical history and forecast durations determine H and h at the plant's cadence; L follows the image cadence. Let x denote permitted numerical covariates, v satellite frames, and m validity masks. For plant i at origin t, the window operator $\mathcal{W}(i,t)$ pairs the inputs with the future-power target:

$$\mathcal{W}(i,t) = (y_{i,t-H+1:t}, x_{i,t-H+1:t+h}, v_{i,t-L+1:t}, m_{i,t-H+1:t+h}, y_{i,t+1:t+h}) .$$

The first component is the power history through t; the final component is the target from the following h steps and is withheld from the model. The covariate and mask components cover the indicated history and forecast intervals.

The slice of v is shorthand for the L frames selected by UTC time up to the origin; its indices do not imply that images share the power-sampling grid. Keeping the two clocks separate avoids discarding frames or inventing intermediate ones. Chapter 4 specifies the visual encoder's latent shapes and token selection.

1.2.2 Predictions and scoring

For each window, the forecaster predicts quantiles of future capacity-normalized power, following the direct quantile-regression formulation of Chronos-2 [1]. This study uses nine levels from 0.1 to 0.9. Pinball loss is suited to these outputs because it weights underprediction by the quantile level and overprediction by its complement. Minimizing the loss at a given level therefore estimates that conditional quantile. For level $\tau \in (0, 1)$, target y, and prediction $\hat{y}^\tau$, the loss is

$$\rho_\tau(y - \hat{y}^\tau) = \begin{cases} \tau(y - \hat{y}^\tau) & \text{if } y \geq \hat{y}^\tau, \\ (1 - \tau)(\hat{y}^\tau - y) & \text{otherwise.} \end{cases}$$

Let $\mathcal{Q}$ be the set of nine quantile levels, b index windows in a batch, and k index forecast steps. The target mask $m_{b,k}$ is one for a valid target and zero otherwise. The training objective combines the pinball losses over these indices:

$$\mathcal{L}_{\text{pin}} = \frac{\sum_{b,k,\tau} m_{b,k} \rho_\tau(y_{b,k} - \hat{y}^\tau_{b,k})}{\sum_{b,k} m_{b,k} \, |\mathcal{Q}|}.$$

As in Chronos-2, unavailable targets contribute no loss. Dividing by the number of valid targets and quantile levels keeps the objective's scale from changing merely because a batch has more missing values. Replacing a missing target by zero would instead teach the model that a logging gap is a night-time observation.

The median quantile supplies the point forecast, denoted $\hat{y}$, used by normalized mean absolute error (NMAE) and normalized root mean squared error (NRMSE). The outer quantiles define prediction intervals, while the complete quantile set supports the continuous ranked probability score (CRPS) and calibration analysis. Forecasting the horizon in one pass avoids feeding an earlier prediction back as input to a later step.

For a set of valid forecast points $\mathcal{S}$, normalized mean absolute error and normalized root mean squared error are

$$\text{NMAE} = \frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} |y_{i,t} - \hat{y}_{i,t}|,$$

$$\text{NRMSE} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} (y_{i,t} - \hat{y}_{i,t})^2}.$$

Because y is already capacity normalized, both quantities are dimensionless. Chapter 5 specifies plant weighting, valid-step masking, and the ramp subset used for the co-primary comparison.

Skill score is commonly used in PV forecasting to express improvement over a reference method [36]. Here Smart Persistence is the reference, so the score shows how much NRMSE the model removes beyond a simple forecast evaluated on the same targets:

$$\text{SS} = 100 \left(1 - \frac{\text{NRMSE}_{\text{model}}}{\text{NRMSE}_{\text{SP}}} \right).$$

A score of zero matches Smart Persistence; positive values improve on it, and negative values perform worse. The subscript SP identifies that reference forecast. The factor of 100 makes a value of 55.10 a 55.10 percent reduction in NRMSE.

Ramp NMAE evaluates forecast error during rapid swings in power output. Rapid changes challenge grid operators because steep ramps require fast-acting reserve power. For each plant i, the evaluator computes the absolute change between consecutive valid readings, $\Delta_{i,t} = |y_{i,t} - y_{i,t-1}|$, using the final history step for the first forecast lead. The ramp subset $\mathcal{S}_{\text{ramp}} \subset \mathcal{S}$ contains the valid daylight target steps where $\Delta_{i,t}$ meets or exceeds the plant's 90th percentile threshold γ_i, selecting the top decile of observed changes across the test set. Ramp NMAE is then calculated as the mean absolute error over these ramp instances:

$$\text{Ramp NMAE} = \frac{1}{|\mathcal{S}_{\text{ramp}}|} \sum_{(i,t)\in\mathcal{S}_{\text{ramp}}} |y_{i,t} - \hat{y}_{i,t}|.$$

Because the ramp subset is derived entirely from observed power, all models are evaluated on the exact same timestamps regardless of their predicted values.

The nine quantiles also allow forecast uncertainty to be assessed within the framework of strictly proper scoring rules [10]. The continuous ranked probability score (CRPS) approximation combines their pinball losses into one score, equal to twice the mean loss across quantile levels, where lower values indicate superior performance. Interval coverage at nominal level α is the fraction of observed targets that fall between the corresponding lower and upper quantiles. At each quantile level, the gap is between its stated probability and the fraction of targets that actually fall below it. Quantile expected calibration error averages these gaps across levels.

1.3 Research questions

Under the plant-disjoint split and scoring rules above, the central question is whether combining a time-series foundation model with a visual foundation model improves six-hour forecasts at unseen plants. The visual arms are fitted on other UK plants but receive no fitting on test plants. The numerical comparison tests whether adapting the backbone on training plants improves on that baseline. The visual comparisons ask whether recent, plant-aligned imagery adds information beyond power history and permitted covariates, and whether the fusion interface affects how much of that information reaches the forecast. In addition, the benchmark evaluates retrieval mechanisms and parameter-efficient adapters as non-visual baselines to assess how much transfer can be achieved without image processing.

An inference-only evaluation then applies the UK-trained models to eight US plants across three matched training seeds. It tests whether the within-UK ranking and visual contribution persist when both region and satellite source change.

1.4 Contributions

The thesis makes three contributions:

1. **Corpus.** It aligns plant power, numerical context, and satellite imagery across UK and US tracks while retaining sample-level provenance.

2. **Benchmark.** It defines a plant-disjoint evaluation under explicit input and scoring rules. The benchmark compares reference forecasts, supervised models, time-series foundation models, retrieval and adaptation methods, and multimodal systems.

3. **Fusion analysis.** It compares pooled and spatially addressable visual interfaces. Same-weights and content-substitution controls show reliance on recent, site-specific imagery.

1.5 Scope and wider relevance

Photovoltaic power is the testbed for a broader transfer problem: forecasting for an entity absent from training using a short local history and a spatial observation of its environment. A newly commissioned wind farm presents a similar combination of turbine records and approaching weather. An ungauged basin can pair a short streamflow record with satellite observations of upstream conditions. New fields, monitored districts, substations, and traffic links may also have thin local histories and spatial context. These examples share the information structure; the thesis does not measure transfer to those domains.

The scope of this thesis centers on solar power generation forecasting. Upstream surface irradiance modeling and downstream grid applications, including market bidding, economic dispatch, and inverter control, remain outside the empirical evaluation.

1.6 Thesis structure

Chapter 2 reviews statistical, deep sequence, and foundation-model approaches to cross-plant transfer. The UK and US multimodal datasets are described in Chapter 3, followed by the MMTSFM fusion ladder in Chapter 4. Chapter 5 sets out the plant-disjoint evaluation protocol and diagnostic interventions, then reports the benchmark rankings and ablation findings. The discussion in Chapter 6 examines visual reliance. Chapter 7 presents the conclusions and future directions.

2 Background and related work

Photovoltaic power forecasting relies on a broad range of time-series techniques, developed across operational statistics and modern machine learning. In practice, these methods face a shared challenge: separating the smooth diurnal cycle determined by solar geometry from sudden fluctuations driven by localized cloud cover. This chapter reviews the most established and widely used approaches for the task, organizing them by how they represent temporal dependencies and adapt to unfamiliar installations.

The survey spans four distinct modeling paradigms. It begins with operational statistical baselines, including persistence and clear-sky index forecasts, which define standard performance floors across utilities. It then covers machine learning models built on tabular feature representations, where gradient-boosted decision trees remain the prevailing operational benchmark. Next, it examines modern deep learning sequence architectures designed specifically for time series, comparing linear trend-residual decompositions, patched transformers, gated variable-selection networks, and cross-variable attention mechanisms. Finally, it reviews recent time-series foundation models adapted from large-scale pretraining, visual encoders, and multimodal systems that fuse numerical histories with sky observations. Each section outlines the structural assumptions of the corresponding method and its ability to transfer to unseen plants under disjoint evaluation.

2.1 Numerical forecasting models

The benchmark begins with methods that make different assumptions about temporal structure. Persistence, Smart Persistence, climatology, and seasonal naive forecasts establish how much of the target can be explained by the current state and the daily cycle. They matter because photovoltaic power has a strong astronomical envelope. A method that improves only on this envelope may appear accurate while missing the difficult cloud-driven changes. Among these references, Smart Persistence carries the observed clearness relationship forward while the deterministic solar envelope evolves [13]. It therefore separates the predictable envelope from the unpredictable sky and serves as the denominator of the skill score used throughout this thesis.

The benchmark also includes tabular and sequence predictors. Some learn from the training fleet; others bring an externally pretrained prior. Tabular models map the forecasting window into a flattened feature representation, testing whether non-sequential tree ensembles provide sufficient inductive bias. Supervised sequence models process temporal order directly, varying in patch tokenization, channel independence, and cross-variable attention mechanisms. Section 5.1.3 details the configuration of each baseline.

2.1.1 Smart Persistence: clearness-index baseline

Standard persistence assumes that future power equals the latest observed measurement, which fails as the sun rises or sets across the forecast horizon. Smart Persistence accounts for this predictable geometric progression by holding the ratio between observed power and clear-sky expectation constant. At forecast origin t, it calculates the clearness index $k_t = y_t / y_{\mathrm{cs},t}$, where $y_{\mathrm{cs},t}$ represents expected generation under clear-sky conditions from a deterministic solar model [13]. For future lead step $t+h$, the forecast scales the expected clear-sky generation by this recent ratio:

$$\hat{y}_{t+h} = k_t \, y_{\mathrm{cs},t+h} = y_t \, \frac{y_{\mathrm{cs},t+h}}{y_{\mathrm{cs},t}}.$$

This formulation separates atmospheric attenuation from astronomical motion. Because clear-sky irradiance is known in advance from solar geometry, Smart Persistence isolates how well a model anticipates actual cloud variation. In solar forecasting literature, it serves as the standard baseline for calculating skill scores [36]. Over short intervals up to an hour, local cloud conditions often persist and

make this baseline competitive. Across multi-hour horizons, cloud fields move and evolve, so constant-clearness projections degrade. In this thesis, Smart Persistence provides the reference denominator for all skill score evaluations.

2.1.2 LightGBM: gradient boosting over a feature table

LightGBM [16] grows an ensemble of decision trees over lagged normalized power, calendar variables, and permitted meteorological covariates. Each split is an axis-aligned threshold, allowing the model to capture nonlinear cutoffs and interactions. In this benchmark, LightGBM provides a feature-table reference for testing how much predictive information the engineered inputs expose.

The transfer setting strains these assumptions. Temporal relationships are fixed before learning begins. LightGBM can discover that a particular lag matters, but it cannot construct intermediate states as an attention-based or recurrent model does. A learned threshold transfers only if its feature has a similar meaning at a held-out plant. LightGBM tests whether the engineered table is sufficient for accurate transfer.

2.1.3 TabPFN: tabular foundation model

TabPFN [12] uses the same feature table without fitting a predictor on the target data. It applies a transformer pretrained across synthetic tabular tasks and predicts from a bounded in-context set of example rows. That setup is relevant when a new plant has little history and fitting a separate model is poorly posed.

TabPFN's bounded context means that a long, densely sampled corpus must be summarized into the rows and features the model accepts. The row-selection rule is therefore part of the method. Section 5.1.3 specifies how it is applied across plants. The result tests how far a pretrained tabular prior can forecast from engineered numerical features without learning directly from the original sequence.

2.1.4 DLinear and PatchTST: linear decomposition against patched attention

DLinear [43] decomposes each series into trend and seasonal components, projecting each with a single linear layer. It provides an efficient linear baseline that establishes how much predictive skill stems from capacity normalization and linear trend projection prior to applying transformer attention.

PatchTST [27] segments univariate histories into temporal patches, reducing token count while capturing local sub-series dynamics. It processes channels independently, preventing spurious cross-variable correlations from overfitting to specific training sites. While channel independence aids cross-plant transfer, it precludes direct modeling of weather-power interactions. Patch tokenization also aggregates observations into multi-hour tokens, smoothing intra-patch fluctuations during steep ramp intervals.

2.1.5 The Temporal Fusion Transformer: gating and variable selection

In contrast to PatchTST's channel-independent design, the Temporal Fusion Transformer [18] combines recurrent processing, gated residual connections, explicit variable selection, and interpretable multi-horizon attention. It was designed for supervised settings in which known future inputs, observed past inputs, and static metadata coexist and vary in quality. This setup resembles a photovoltaic fleet, where some covariates are reliable at one site and noisy at another. The model's gates can in principle learn to suppress unhelpful covariates. It is included to test whether explicit input selection and a purpose-built multi-horizon decoder compensate for the absence of broad pretraining.

2.1.6 iTransformer: variables as tokens

iTransformer [21] inverts the usual transformer convention. Each physical variable becomes a token whose embedding spans the whole input window. Self-attention operates across variables rather than across time. In photovoltaic forecasting, generation, solar position, cloud cover, irradiance, temperature, and wind share substantial structure. The inversion lets the model relate these variables explicitly without inferring their relationship through a shared temporal representation. The token layout is shown in Figure 2.1.

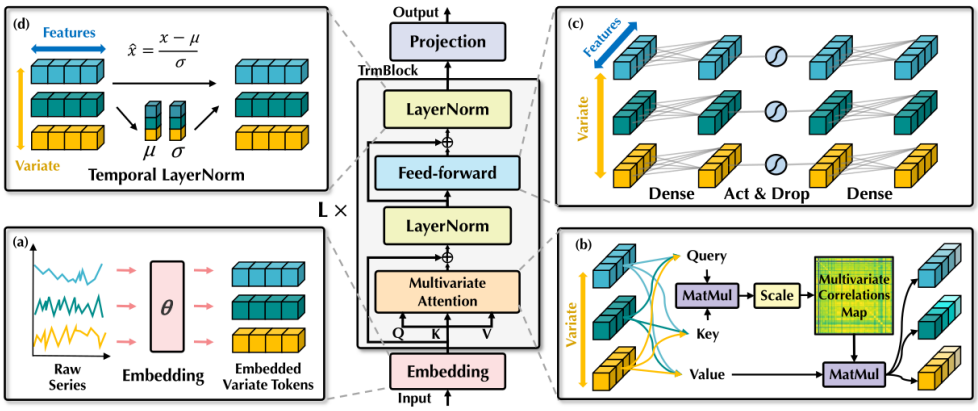


Figure 2.1: iTransformer architecture. Each variable is embedded as a token spanning the input window, and self-attention models relations across variables. Reproduced from the source architecture [21].

2.2 Time-series foundation models

Time-series foundation models are pretrained on many series before zero-shot use or task adaptation. Chronos-2 [1], TimesFM [11], TiRex [3], and Tiny Time Mixer (TTM) [6] cover several architectural families. A model exposed to a broad corpus may bring useful temporal representations to a series it has never seen.

Their treatment of the input differs. Patch-based models compress a long context into blocks of observations. This makes long histories tractable, but can blur rapid behavior within a patch. Covariate support and output representations also differ across families, from point forecasts to quantile distributions.

Pretraining does not remove the need for a controlled comparison. Photovoltaic power has daily zeros, seasonality, cloud-driven ramps, and plant-to-plant variation that may not be represented in a general pretraining corpus. A zero-shot result measures the match between prior knowledge and this task. A fine-tuned result measures both that prior and the information obtained from the training fleet. This distinction is carried into every result table in Section 5.2, because the two quantities support different claims.

2.2.1 Chronos-2: the shared numerical backbone

Chronos-2 [1] has a dual role in this thesis and therefore deserves a fuller description than the other checkpoints. It tokenizes a numerical history into fixed-length patches, processes them with a pretrained encoder, and emits quantiles for the forecast horizon in one pass. Figure 2.2 shows the original architecture. Patching keeps the sequence short enough for additional visual tokens, while the quantile head supports accuracy and calibration evaluation. The visual arms insert tokens into a backbone that was originally trained for numerical inputs.

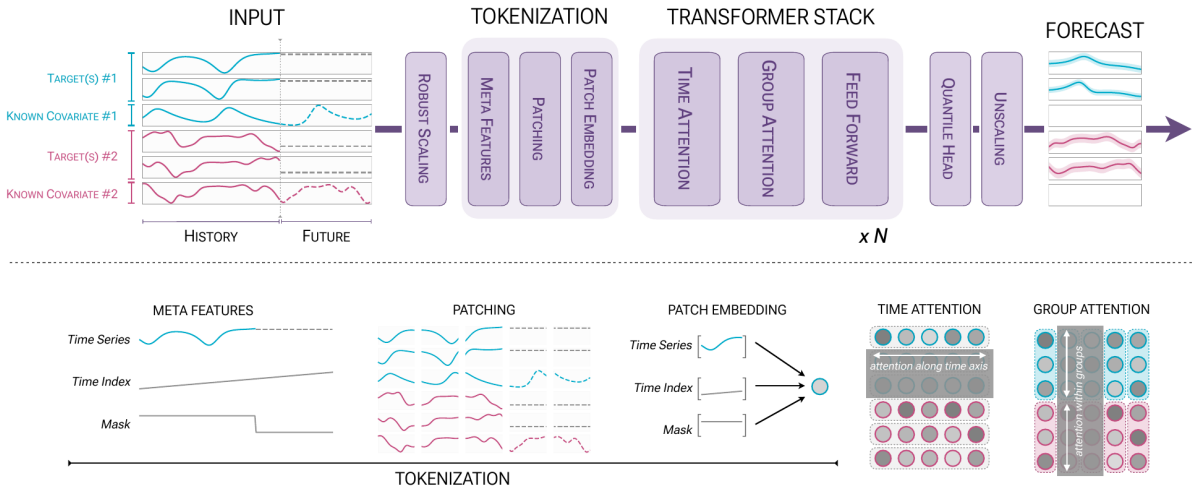


Figure 2.2: Original Chronos-2 architecture, showing numerical input processing, the transformer backbone, and quantile forecasting. Reproduced from Figure 1 of the source paper [1].

The same checkpoint appears in the benchmark twice. Its zero-shot run tests how the released model performs on held-out plants before exposure to the training fleet. Fine-tuning it on the training plants produces the numerical baseline. The difference between these runs measures the benefit of in-domain adaptation from that pretrained starting point; it does not measure the value of pretraining itself. The numerical baseline is also the vision-free anchor of the fusion ladder. Each visual arm starts from its seed-matched adapted checkpoint.

2.2.2 TimesFM, TiRex, and Tiny Time Mixer: breadth of temporal pretraining

The remaining foundation models provide task-level context across several architectures, showing how far off-the-shelf temporal pretraining reaches on this forecasting problem. TimesFM [11] uses a decoder-oriented design intended for direct zero-shot forecasting at arbitrary horizons. TiRex [3] emphasizes in-context learning across varied horizon lengths. Tiny Time Mixer [6] replaces attention with lightweight mixing blocks. It targets a much smaller parameter budget, keeping both pretraining and adaptation cheap.

Together they show how released time-series checkpoints behave on plant-disjoint photovoltaic forecasting before fitting to the fleet. Photovoltaic power combines nightly zeros, a strong astronomical envelope, and ramps driven by an external process that the series records only after it reaches the plant. A weak zero-shot result therefore says more about the relation between the pretraining corpus and this task than about one architectural choice.

2.3 Adaptation without a new model

Some methods adapt a pretrained time-series model without fully retraining it. Fine-tuning updates some or all backbone weights on the target fleet. Retrieval conditions a frozen forecaster on similar historical windows drawn from an external datastore. Alternatively, parameterized memory approaches such as MEMTS [42] internalize domain-specific patterns directly into adapter parameters without requiring external database lookups during inference. Covariate adapters offer another route by injecting external inputs into an established representation through a small trainable path. These adaptation strategies offer lightweight alternatives to full end-to-end multimodal fine-tuning by modifying small parameter subsets or conditioning on retrieved instances.

2.3.1 TS-RAG and Cross-RAG: retrieval as conditioning

TS-RAG [28] and Cross-RAG [17] retrieve related historical windows and make them available to a frozen or lightly adapted forecaster. Figure 2.3 shows the external-store interface in TS-RAG. Retrieval can supply examples of rare steep transitions at inference time without requiring the model weights to memorize them. The store can also grow without retraining the forecaster.

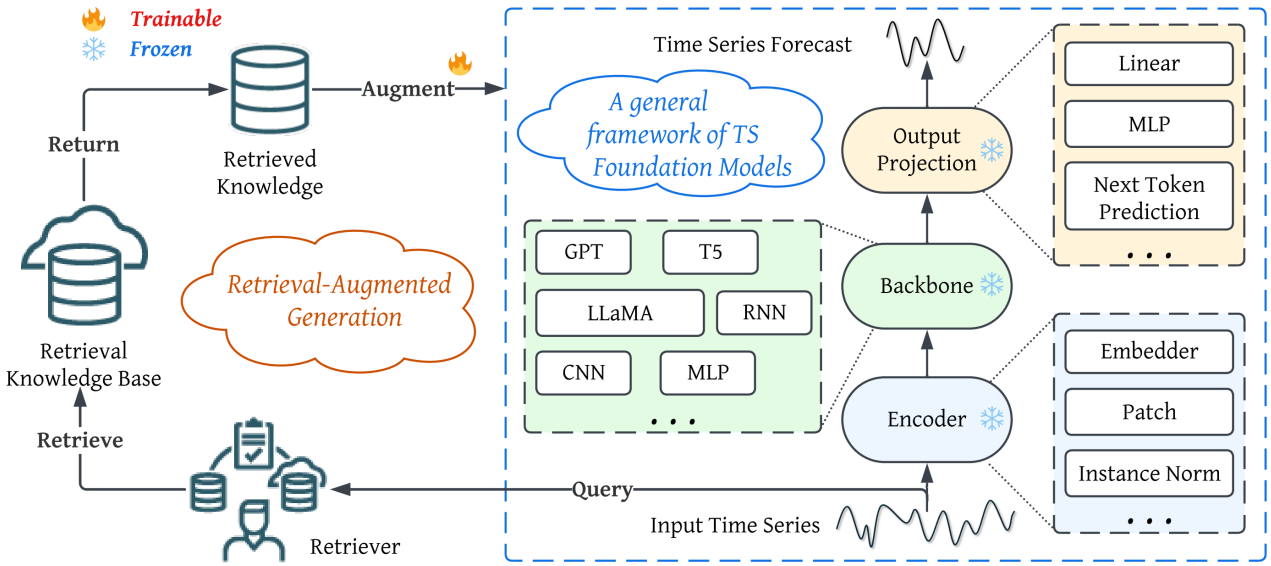


Figure 2.3: TS-RAG overview from the source paper: a retriever supplies knowledge from an external store to a time-series foundation model while its backbone remains frozen [28].

Retrieval poses a specific risk for cross-plant evaluation. The method's effective training information is determined by the store's contents. A plant-disjoint protocol remains valid only if the store, the query construction, and the reported aggregate all respect the same entity boundary as the forecasting model.

2.4 Multimodal forecasting interfaces

Multimodal forecasters use visual inputs in two ways. Time-series renderings let a vision encoder reinterpret measurements already present in the numerical history. Sky-camera and satellite images can add observations of the atmosphere that have not appeared in the power trace. The distinction matters when a forecast improves: a rendered image tests a new representation, while a real image may add new information whose use still needs to be measured.

2.4.1 Time-VLM and VisionTS++: rendered time-series images

Time-VLM [44] and VisionTS++ [35] render numerical history as an image and encode it with a visual backbone. Time-VLM also uses statistical text prompts; its paths appear in Figure 2.4. The rendered image may expose temporal patterns to a visual encoder, but it contains no observation beyond the numerical history used to construct it.

This distinction matters when interpreting a benchmark. A pseudo-image model can be a strong forecasting baseline because it tests a legitimate representation choice. It evaluates the value of re-encoding a history rather than the value of an independently observed modality.

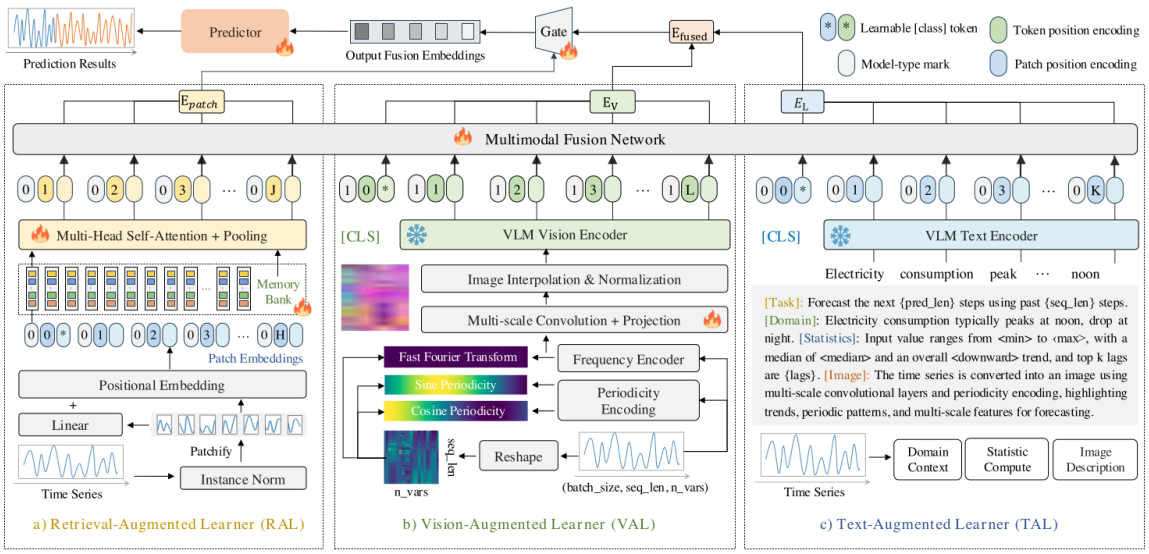


Figure 2.4: Time-VLM architecture. Historical time-series values are rendered as an image and combined with statistical text prompts before forecasting. Reproduced from the source architecture [44].

2.4.2 Aurora: pretraining across derived modalities

Aurora [39] extends this line of work to multimodal foundation-model pretraining. It combines modality-specific encoders with a cross-modality encoder that uses token distillation and modality-guided self-attention, as shown in Figure 2.5. Its training uses numerical series, images derived from those series, and text descriptions. The resulting model tests whether these representations improve general forecasting behavior and transfer.

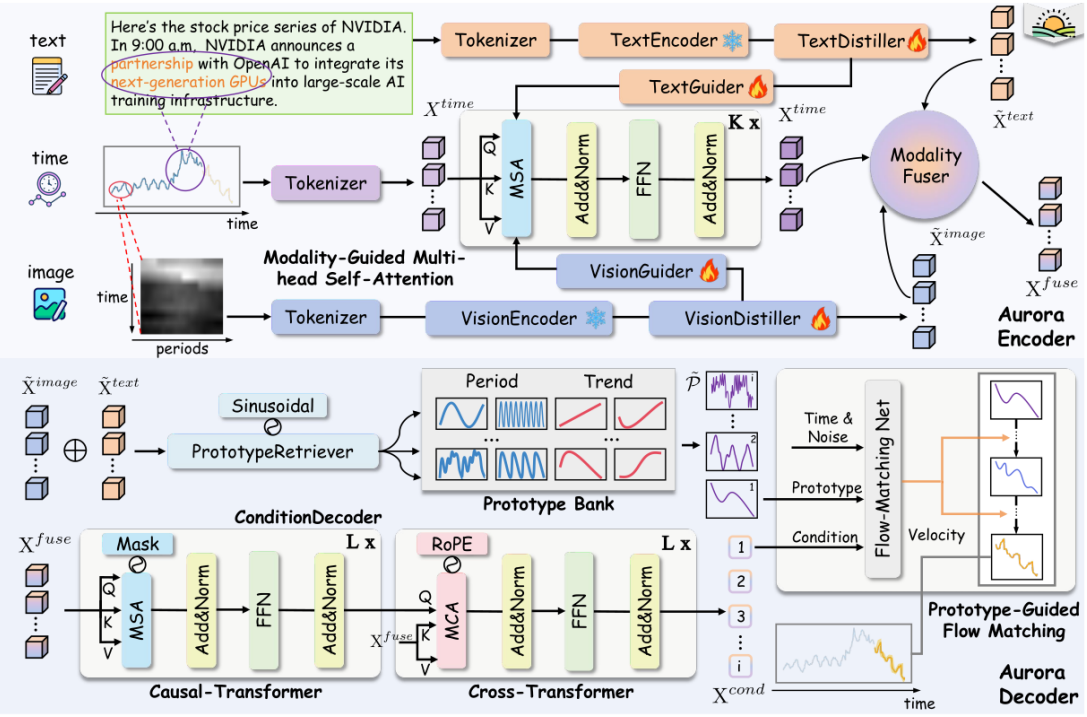


Figure 2.5: Aurora architecture. Modality-specific encoders feed a cross-modality encoder that uses token distillation and modality-guided self-attention. Reproduced from the source architecture [39].

2.4.3 SUNSET: a photovoltaic convolutional reference

SUNSET was an early end-to-end model for forecasting photovoltaic power from ground-based sky-image sequences and recent power output [37]. The forecasting architecture stacks successive frames along the image channel dimension and extracts visual features with two convolutional blocks. It inserts the power history after image processing and uses fully connected layers to predict 15-minute-ahead output [38]. Its released implementation has made it a recurring reference in image-based forecasting comparisons, including the SKIPP'D benchmark [25], providing an established convolutional baseline for comparison with modern foundation-model interfaces. Subsequent studies expanded ground-based sky observation using generative video modeling in SkyGPT [26] and multimodal vision-language architectures in PV-VLM [19], while multi-scale fusion networks such as M3S-Net [29] combine local telemetry with multi-scale image features.

2.4.4 Solar-VLM: gated late fusion

Solar-VLM [9] combines a visual pathway for satellite crops with meteorological text and a time-series pathway. As shown in Figure 2.6, transformer blocks and weighted temporal aggregation summarize the visual sequence. A KNN/GAT pathway models relations among sites, and cross-site attention integrates that spatial context. The resulting visual and meteorological representation is fused with the time-series state through a learned gate. The architecture therefore includes temporal and cross-site processing in addition to station-level fusion. For the present comparison, the relevant distinction is that the forecaster receives an aggregated representation rather than a sequence of individually addressable satellite patches.

Figure 2.6 shows the pooled station representation and gate in Solar-VLM.

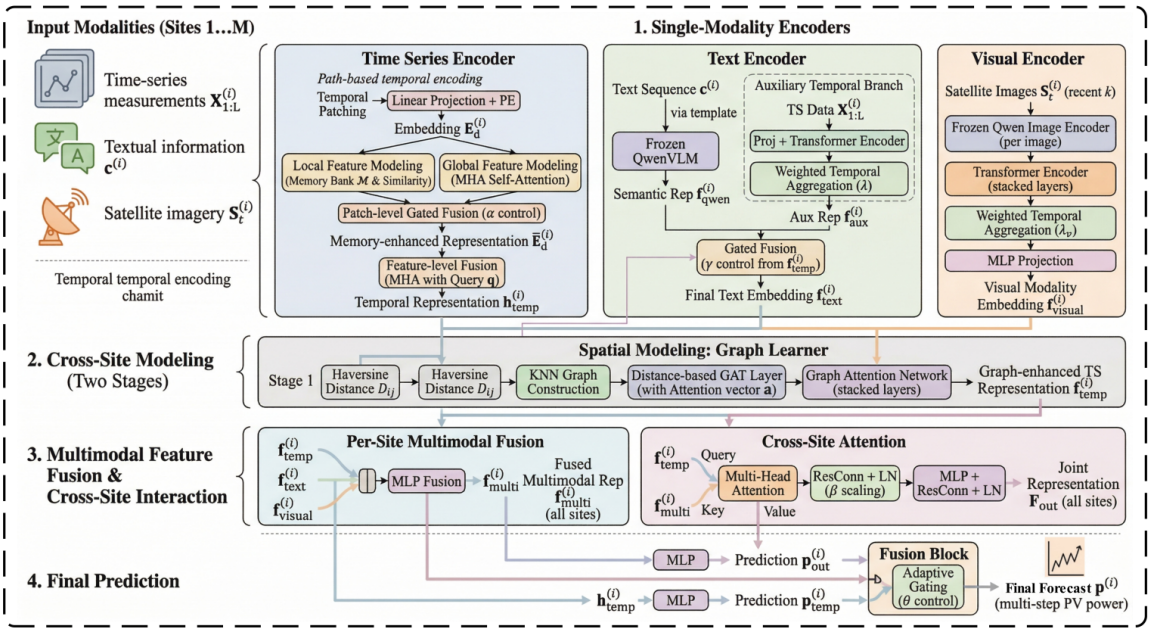


Figure 2.6: Solar-VLM architecture. The diagram shows visual temporal aggregation, KNN/GAT cross-site processing, cross-site attention, and gated fusion with the time-series branch. Reproduced from the source architecture [9].

2.4.5 PV-LLM: satellite and text prompts inside a language backbone

PV-LLM [15] carries the prompting idea into a pretrained language model, replacing the time-series backbone. It uses a frozen GPT-2 and three input channels. A temporal-semantic encoder maps power and numerical weather history into embeddings aligned with the word space. A text prompt contains plant coordinates, forecast start time, dataset and task descriptions, and input statistics. A satellite encoder uses a context compressor to reduce the patch grid of each frame to a single vector. Figure 2.7 shows these separate paths. Satellite features enter every self-attention layer through adaptive prompt-tuning blocks, whose gates start at zero. In its reported benchmark, the system is trained and evaluated on MMSP(S), a ten-plant Shandong subset of the same MMSP corpus released

with FusionSF, with Himawari imagery and ECMWF forecasts at hourly resolution. It reports lower error than unimodal transformers, MLP mixers, Time-LLM, and FusionSF.

The context compressor leaves one vector per frame, so image locations are no longer available as separate tokens inside GPT-2. The vector may still encode spatial information. The paper's prompt ablation finds larger gains from forecast start time and plant coordinates than from its other text components. Those gains concern metadata conditioning and do not establish which satellite content the model uses. Although coordinates can be available for a held-out plant, PV-LLM's reported evaluation does not test plant-disjoint transfer; its results are for easy and hard test windows. The cross-track result in this thesis evaluates different MMTSFM arms on eight held-out US plants. It does not change the scope of PV-LLM's published evidence.

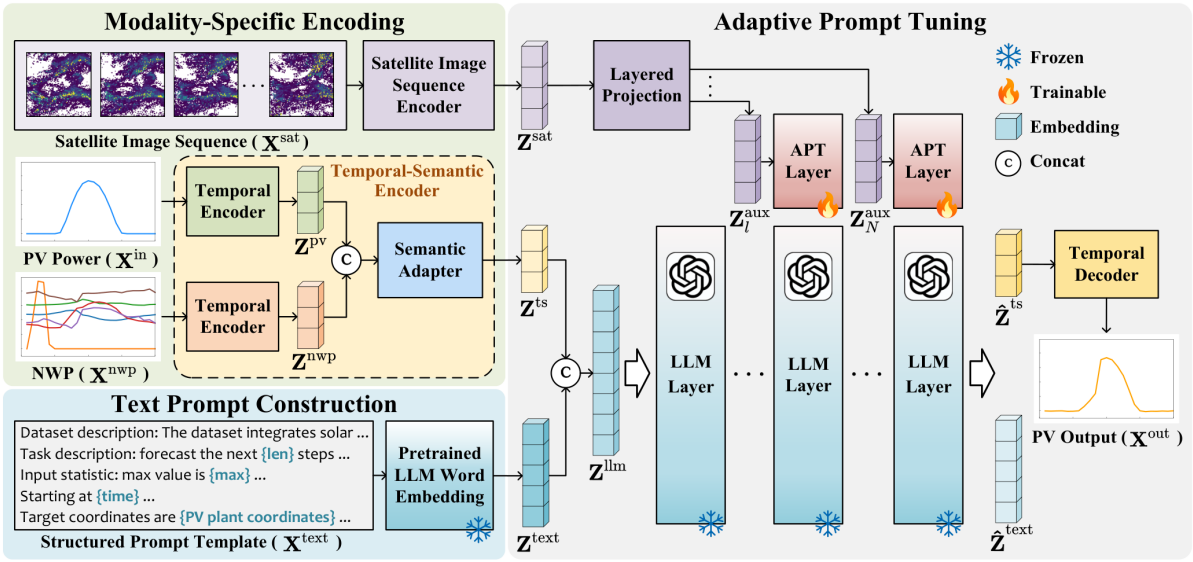


Figure 2.7: PV-LLM architecture. Satellite frames, power and numerical weather history, and a structured text prompt are encoded separately. The text and time-series embeddings enter a frozen language model directly, while satellite embeddings are injected at each layer through adaptive prompt-tuning blocks. Reproduced from the source architecture [15].

2.5 Nemotron 3 Nano Omni: visual token selection

Nemotron 3 Nano Omni [30] is a multimodal model outside photovoltaic forecasting. Its visual pathway reduces features by spatial downsampling and then selects tokens from regions that change over time, as shown in Figure 2.8. Efficient Video Sampling [5] supplies the selection step by pruning temporally redundant video tokens. This thesis draws on the resulting interface principle: reduce the visual token count while keeping selected locations distinct for the downstream backbone.

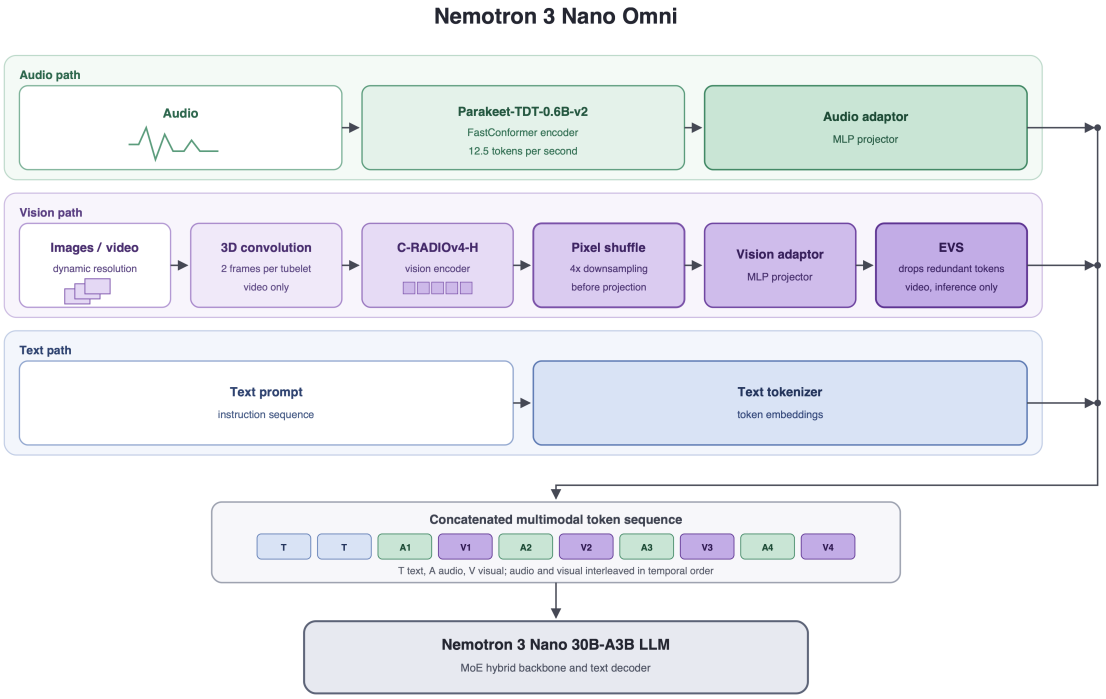


Figure 2.8: Nemotron 3 Nano Omni visual pathway. The visual encoder produces tokens that are spatially reduced and then sampled before projection into the shared backbone. Reproduced from the source architecture [30].

2.6 V-JEPA 2.1: dense video representations

V-JEPA 2.1 [23], built on the self-supervised joint-embedding predictive architecture introduced in V-JEPA 2 [2], is trained on large collections of images and video. Its dense predictive objective trains on both visible and masked tokens, with additional supervision at intermediate encoder layers. The resulting representations retain spatial structure and temporal information while supporting global scene understanding. Unlike a single embedding for each frame, its patch features leave individual image regions available to a downstream model.

Figure 2.9 shows the training architecture. Separate image and video tokenizers feed a shared context encoder; a predictor learns from both masked and visible tokens against targets from a teacher encoder.

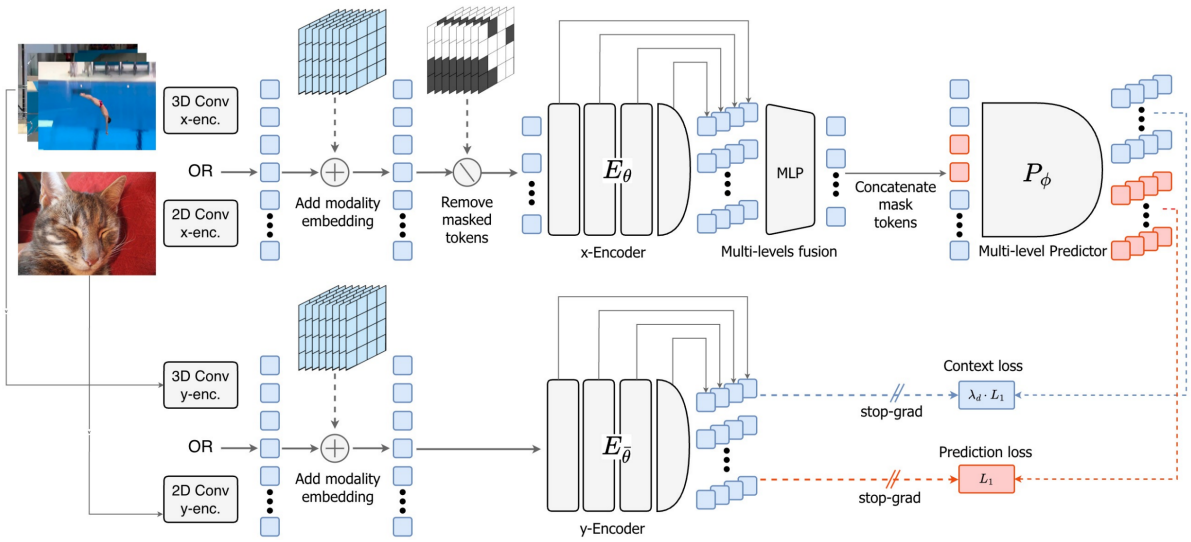


Figure 2.9: V-JEPA 2.1 training architecture. Image and video inputs use separate tokenizers before the context and target encoders. Multi-level predictions are supervised at masked and visible token positions. Reproduced from the source architecture [23].

This thesis pairs a frozen V-JEPA 2.1 encoder with Chronos-2. Nemotron motivates the selection interface, while V-JEPA 2.1 supplies the visual features.

3 Dataset construction

The research questions formulated in Chapter 1 define the requirements for the evaluation dataset. Testing whether multimodal foundation models improve cross-plant generalization requires power histories and numerical weather covariates synchronized with satellite observations available before each forecast origin. The test also requires strict preservation of plant identities, quality labels, and data provenance to prevent information leakage across splits. This chapter describes how records from the United Kingdom and the United States were assembled into a shared sample contract, which places disparate numerical and visual sources onto a common physical timeline. Chapter 5 details the plant partitions and scoring rules that evaluate these samples.

3.1 Challenges in building a transfer dataset

The evaluation in Chapter 5 holds out entire plants. Each record must therefore remain traceable to its source and site after the numerical streams are standardized. Otherwise, windows from the same plant could enter both training and test partitions, making a transfer result impossible to interpret.

The power record also has to distinguish faulty measurements from genuine low generation. An outage, a stuck inverter, and cloud cover can produce similar readings, but they describe different conditions. Quality flags and missing-value masks preserve that distinction; extended gaps remain unavailable instead of becoming interpolated targets that no instrument measured.

Finally, the image paired with a forecast must depict the right location at the right time. A frame captured after the forecast origin would reveal future conditions, while a crop from another site would describe a different scene. The corpus therefore retains site coordinates and frame timestamps, selects images available before the origin, and marks absent frames explicitly. These requirements guide the source selection and construction steps that follow.

3.2 Published photovoltaic datasets

This thesis uses measured plant power as the forecasting target. An irradiance measurement describes incoming sunlight, whereas a power measurement records the output of a particular installation. A forecasting model must learn from that installation's observed response, including differences between plants that remain after capacity normalization. The survey therefore focuses on datasets with plant-level production records and identifies which also provide imagery.

3.2.1 Production records without imagery

These collections provide measured generation and numerical context, but no paired image stream. The UK PV and PVDAQ releases supply the power records used in this thesis; the others show the range of available numerical sources.

Open Climate Fix's UK PV dataset [31] provides half-hourly metered generation for more than 30,000 systems across Great Britain from 2010 to 2025; a smaller subset also has five-minute readings. It includes approximate site locations, capacities, panel orientation and tilt, and flagged bad-data periods, but no imagery.

The NREL PVDAQ archive [24] publishes systems across the United States at sub-hourly resolution, from residential rooftops to utility installations, each with system metadata.

HKUST Rooftop PV [20] covers 60 rooftop systems in Hong Kong at five-minute inverter resolution, with on-site weather and equipment metadata. It is the densest of these collections in both time and instrumentation.

PVOD [41] covers ten plants in China at fifteen-minute resolution. It pairs measured power with local weather observations and numerical weather predictions, and it records orientation and location for each plant.

The Ausgrid solar home collection [4] records rooftop generation and household consumption for 300 homes in Sydney at half-hourly resolution over three years, but it lacks weather and plant metadata.

3.2.2 Production records paired with images

These datasets link measured plant output to visual observations of the sky or surrounding cloud field. Their image sources and site coverage determine which forecasting and cross-plant transfer tests they can support.

SKIPP'D [25] pairs one California system with ground-based sky images at one-minute resolution. The imagery is the closest available view of the sky above an array, which suits nowcasting; a single site cannot express a transfer task.

MMSP, released with FusionSF [22], is the closest published comparison to this work. It combines capacity-normalized power from 88 plants in China with Himawari satellite imagery and numerical weather data at hourly resolution. It evaluates forecasting on plants held out from training.

3.2.3 Positioning among existing datasets

Table 3.1 places these collections beside the corpus built here.

Dataset	Region	Sites	Cadence	Span	Inputs
UK PV [31]	GB	30,000+	5/30 min	2010–2025	P, M
HKUST PV [20]	Hong Kong	60	5 min	2 years	P, W, M
PVOD [41]	China	10	15 min	1 year	P, W, M
Ausgrid [4]	Sydney	300	30 min	3 years	P, L
PVDAQ [24]	US	1865	Variable	Variable	P, M
SKIPP'D [25]	California	1	1 min	2 years	P, C, G
MMSP [22]	China	88	1 h	1.5 years	P, W, S
This corpus	UK, US	110	15/30 min	2019–2020	P, W, M, S

Table 3.1: Published photovoltaic datasets. GB denotes Great Britain. Inputs: P, power; W, weather; M, metadata; L, household load; C, capacity; G, ground sky images; S, satellite images. UK PV has a five-minute subset within its thirty-minute collection. This corpus spans 2019–2020 in the UK and January–September 2019 in the US, with sample-level provenance. MMSP is the closest comparison through its multi-plant power, weather, satellite imagery, and plant-held-out evaluation.

3.3 The corpus built here

Among the collections surveyed here, none provides the combination of cross-plant evaluation, native-cadence power and imagery, and sample-level provenance used in this thesis. The corpus joins two source tracks under one data contract to supply that setting.

The dataset has two parts. A flat Parquet table holds 1,337,654 numerical records across 35 columns. A packed HDF5 archive holds 4,103,892 image frames in 110 site groups. Keeping them apart leaves the numerical data easy to process while preserving timestamp-exact access to large visual payloads. The first track, in the United Kingdom, contains 100 residential photovoltaic systems from 1 January 2019 through 31 December 2020 [31]. Power is sampled every 30 minutes. The systems have capacities from 1.5 to 4.0 kW and are distributed across the United Kingdom. Satellite frames are available from EUMETSAT sources [7].

The second track combines ten United States NREL PVDAQ systems [24] with GOES imagery. It covers January through September 2019 at fifteen-minute cadence, with capacities from 1.8 to 408 kW. That spread brings a different scale of deployment, a different sensor product, and a different geographic context. The results chapter reports an inference-only transfer evaluation on eight quality-filtered held-out US plants after resampling power to the same 30-minute cadence used in the UK benchmark. The primary multi-plant benchmark remains the UK evaluation. Table 3.2 summarizes both.

Compared with MMSP, the corpus covers two geographic tracks and preserves each source's native cadence: the interval at which its observations were collected. UK power remains half-hourly, US

Track	Sites	Rows	Power	Images	Capacity
UK	100	1,232,862	30 min	15 min	1.5 to 4.0 kW
US	10	104,792	15 min	15 min	1.8 to 408 kW

Table 3.2: The two data tracks. Power and Images report sampling intervals.

power remains fifteen-minute, and satellite frames retain their own timestamps. A common sample definition specifies which plant, history interval, and future interval belong to a forecast; it does not force all sources onto one sampling frequency. This distinction lets a fourteen-day history or six-hour horizon have the same physical meaning in both tracks. Interpolated values are not treated as new measurements.

Each selected image is linked to its capture time and plant location, allowing its observation time to be verified against the numerical history and forecast origin. The US track spans a wider capacity range than the UK track, alongside different satellite sensor characteristics (GOES-16 versus SEVIRI). Eight held-out US plants provide a cross-region and cross-satellite evaluation, testing whether models trained on UK data generalize when deployed under unfamiliar solar regimes and atmospheric dynamics.

The corpus defines each forecast example as one aligned record. At a forecast origin, it combines recent generation, atmospheric and astronomical covariates, and a visual record of the nearby cloud field. Source identity, UTC timestamps, site coordinates, quality flags, and the local frame index link these inputs to the same plant and forecast window. These fields make their alignment auditable.

Figures 3.1 and 3.2 show representative weeks on their native grids. The common corpus lets windows be defined in physical time and resolved at native cadence while preserving differences between the underlying processes.

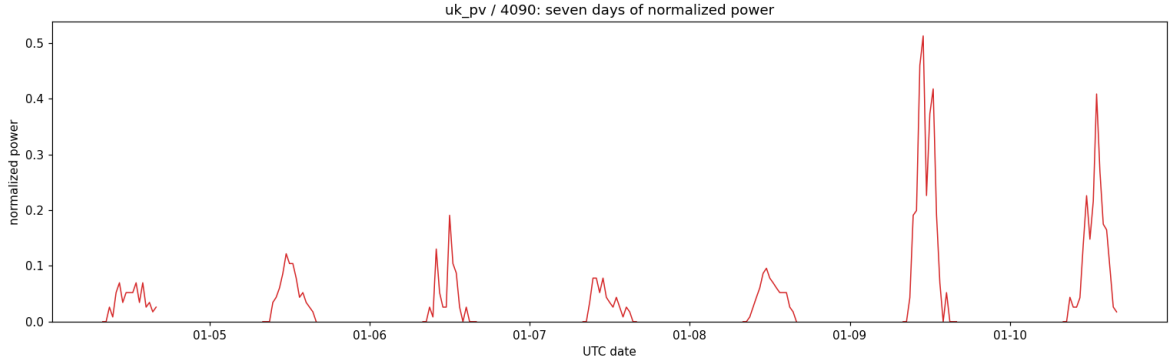


Figure 3.1: A representative week from the UK track. The daily envelope is clear, while departures from it vary sharply from one day to the next.

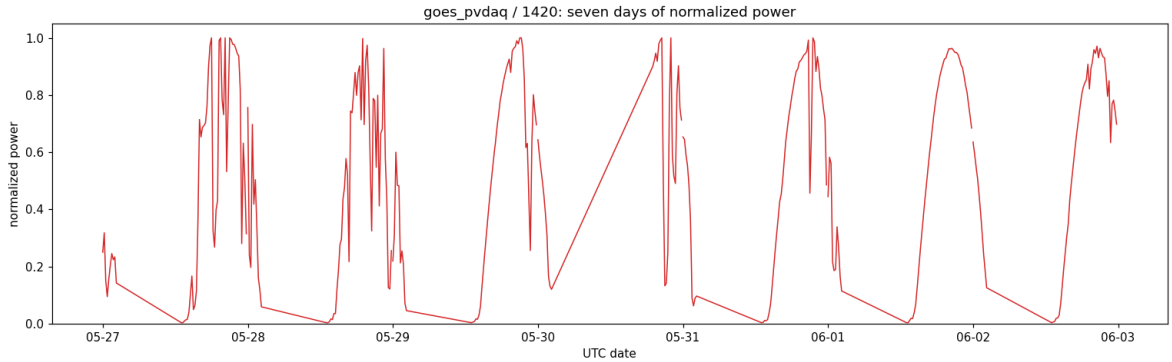


Figure 3.2: A representative week from the United States track. The different cadence, capacity range, and data span make this a complementary transfer setting.

3.4 Dataset provenance and harmonization

A corpus assembled from several archives is only credible if every field can be traced to its origin and if the transformations applied along the way are stated. This section names the public sources, then describes the harmonization steps that determine whether records from different countries and instruments can be compared. These steps convert heterogeneous power readings to a common capacity-normalized scale and place every power value, covariate, and satellite frame on one physical timeline.

3.4.1 Public sources

The corpus draws on four public source families. The UK track combines domestic generation measurements [31] with EUMETSAT satellite imagery [7]. The US track pairs NREL PVDAQ records [24] with GOES image crops. The Open-Meteo archive provides atmospheric covariates [32]. Table 3.3 identifies the role of each source.

Track	Source	Records
UK power	UK PV	Residential power (30 min), metadata
UK imagery	EUMETSAT	Site-centered satellite crops
US power	NREL PVDAQ	Power (15 min), metadata
US imagery	GOES	Site-centered RGB crops
Weather	Open-Meteo	Temperature, radiation, clouds, wind, precipitation

Table 3.3: Public sources combined in the experiment dataset.

The sources do not arrive on a common clock, grid, or storage format. Each source is checked, cleaned, and standardized before its records are stored in the common dataset format. Dataset construction preserves what was actually observed: power remains at native cadence, images remain in a separate packed archive, and the numerical record stores the metadata needed to select an aligned visual clip.

3.4.2 Power and capacity harmonization

The UK source reports energy over its sampling interval. For the forecasting record, this is converted to power in watts using the interval duration. The United States data already provide a compatible power measure. Both tracks retain measured power and a capacity-normalized target, computed by dividing measured power by the audited installed capacity.

Installed capacity is audited per site and stored with the numerical record. Capacity normalization scales power outputs to the unit interval while preserving local efficiency variations, clipping behavior, and plant-specific shading patterns.

Extended missing intervals remain recorded as unobserved timesteps through quality masks. Avoiding synthetic interpolations prevents imputation artifacts from biasing model training or evaluation.

3.4.3 Time, location, and image alignment

Every numerical row is timestamped in UTC and identifies the plant through its coordinates and source identifiers. The row also contains a direct reference to that plant's imagery. The referenced image archive stores each site's frame sequence with its original timestamps. For a forecast window, selection begins at the numerical forecast origin and retrieves visual slots that precede it.

In the UK track, images arrive at fifteen-minute intervals while power is recorded half-hourly. The loader retrieves the closest valid satellite frame preceding the forecast origin within a 15-minute window, storing the exact image timestamp in the sample record to verify temporal alignment.

3.5 Numerical observations

Each numerical record identifies the dataset, site, UTC timestamp and geographical location. It stores measured power in watts and capacity-normalized power, obtained by dividing the measurement by the audited installed capacity. Outage and stuck-signal observations keep their quality labels, while their normalized targets remain missing rather than being fabricated.

The atmospheric covariates are temperature, shortwave radiation, direct and diffuse radiation, direct normal irradiance, cloud cover, wind speed, and precipitation. Solar and calendar fields include

zenith, azimuth, a clear-sky estimate, seasonal sine and cosine terms, and solar time. Table 3.4 lists the full record; Chapter 5 specifies which of these fields each method may use at a forecast origin.

Component	Fields
Identity	Dataset, site, UTC time, location, station/camera IDs
Target	Measured and capacity-normalized power
Weather	Temperature, radiation, cloud cover, wind, precipitation
Solar/calendar	Geometry, clear-sky estimate, season, solar time
Quality	Outage, stuck-value, night, capacity, site flags
Image reference	Timestamp-exact archive index

Table 3.4: Logical structure of a numerical record.

3.6 Dataset contract

The dataset contract defines the contents of one forecasting sample. It preserves the native power and image grids, resolves windows in physical time, and carries the timestamps and validity masks needed to reconstruct each sample. Chapter 5 specifies how those samples are partitioned and scored.

Each logical sample contains normalized power history, a future target, numerical covariates, visual frames, timestamps, plant identity, and masks. The contract defines these fields independently of the model that consumes them; the input choices for each model family are stated in Table 5.2.

The contract also separates values from their validity. A missing power reading, a missing visual slot, and a future target that should not be scored are represented by different masks. Table 3.5 summarizes the fields and their temporal extent. In the model window, a measured zero at night is a valid observation; an absent nighttime row or a faulty reading is not. The loader must preserve that distinction when it constructs the physical-time grid, since replacing missing values with zero would confound absence with observed production.

Element	Extent	Role
Power history	14 days	Local plant context
Target	6 hours	Future power
Covariates	History + horizon	Archived weather, solar and calendar fields
Visual clip	Recent window	Cloud structure
Masks/times	Whole sample	Validity, alignment, provenance

Table 3.5: Logical elements of a forecasting sample.

Since the dataset has two power cadences, a fixed number of observations does not mean the same physical history in each track. Defining the task in physical time avoids an accidental advantage for a model evaluated at a coarser cadence. Fourteen days span 672 half-hour positions on the UK grid and 1,344 quarter-hour positions on the US grid. A six-hour horizon spans 12 or 24 positions, respectively. These are model-grid positions; the stored daytime records and missing intervals remain subject to their validity masks.

3.7 Exploratory characterization

This section characterizes variation in the forecasting task: the daily and seasonal structure of power, rapid changes that make forecasts difficult, and differences in behavior and coverage across plants. It then examines how far recent power history and numerical covariates explain the target, and whether the image stream provides a distinct, time-varying observation worth testing. These observations motivate the evaluation choices and multimodal design introduced later; they do not establish model performance.

3.7.1 Daily envelope and residual uncertainty

The two figures separate the regular structure of generation from the variation that remains around it. In Figure 3.3, each cell is the mean normalized power for one calendar month and solar hour; lighter colors indicate higher mean output. Reading across a row shows the daily production profile, while comparing rows shows how that profile changes through the year. Both tracks show the expected rise after sunrise, a middle-of-day peak, and a decline toward sunset. Calendar and solar geometry can therefore provide a useful baseline for expected production, but they cannot determine the realized output at a particular forecast origin.

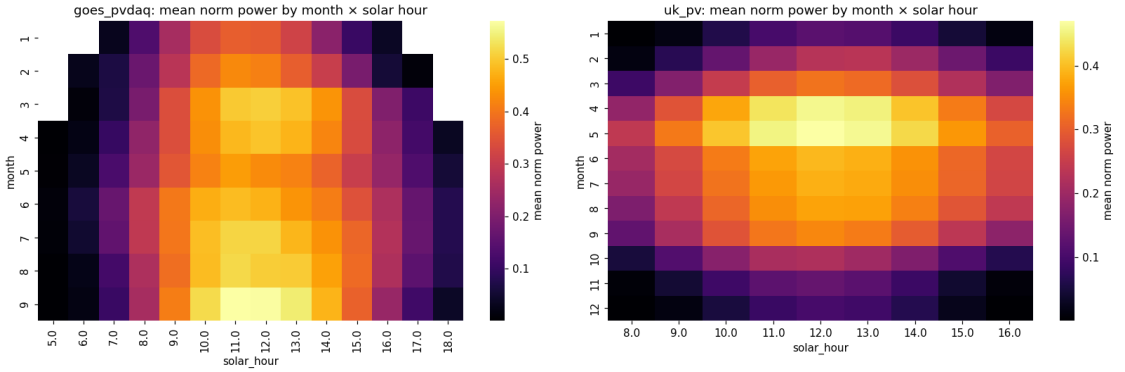


Figure 3.3: How to read the heatmaps: rows are calendar months, columns are solar hours, and each cell gives mean normalized generation, with lighter colors denoting higher output. The panels show the regular daily profile and its seasonal change in each track.

Figure 3.4 shows the remaining uncertainty from another view. Each panel groups daytime observations by one numerical feature and normalized power, with color indicating the logarithmic number of observations in each hexagon. The denser regions reveal broad relationships between power and radiation, cloud cover, or temperature. The substantial vertical spread at many feature values shows that no one of these variables fixes the output. A forecaster must capture the regular solar envelope and respond to the local conditions that move power above or below it. An average daily error alone can obscure failures in this second task.

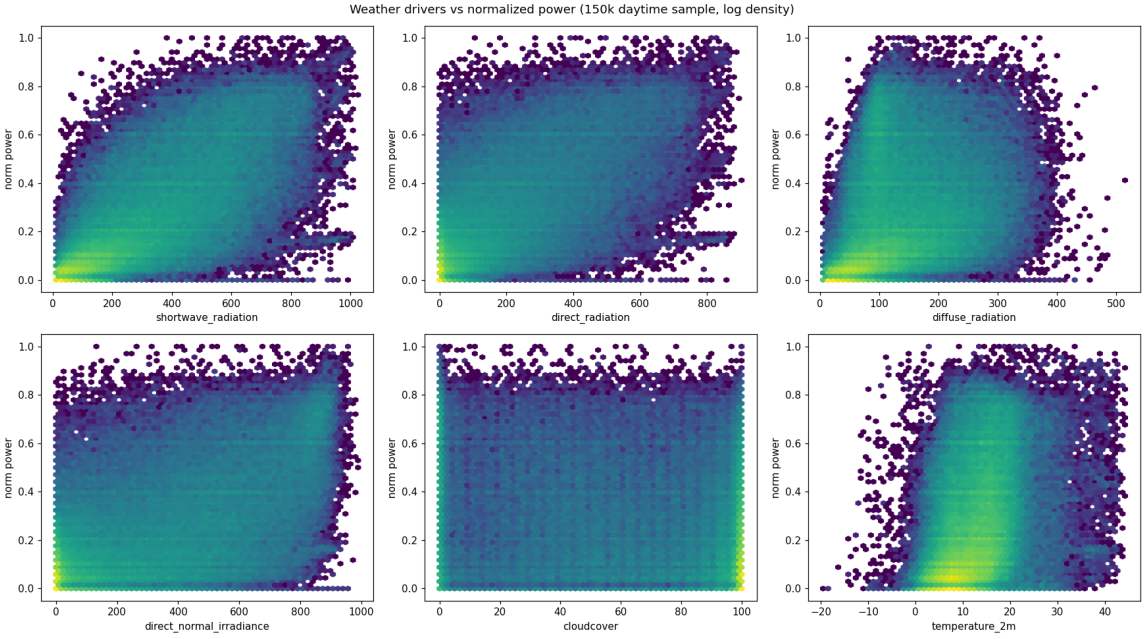


Figure 3.4: How to read the hexbin plots: each panel groups daytime observations by a numerical feature and normalized power. Color denotes the logarithmic number of observations in a bin. Broad vertical spread at a given feature value shows residual uncertainty beyond that individual feature.

3.7.2 Ramps and the role of visual context

Ramps are the signed changes in normalized power between successive valid daytime observations. Figure 3.5 has one panel for each track. The horizontal axis gives the change per native power step: negative values are drops in production, positive values are rises, and zero means that production changed little. The vertical axis counts observations on a logarithmic scale, showing the large mass of small changes and the relatively rare steep changes. Values outside the displayed interval are placed at -0.5 or 0.5; the outermost bars therefore include all larger changes.

Both distributions peak sharply around zero, showing that most adjacent observations are similar. Their tails show the less frequent ramps that are harder to forecast and can dominate short-horizon errors. While this exploratory visualization displays signed transitions to show directional symmetry, Chapter 5 defines how these observations become a scored ramp subset based on absolute transition magnitude.

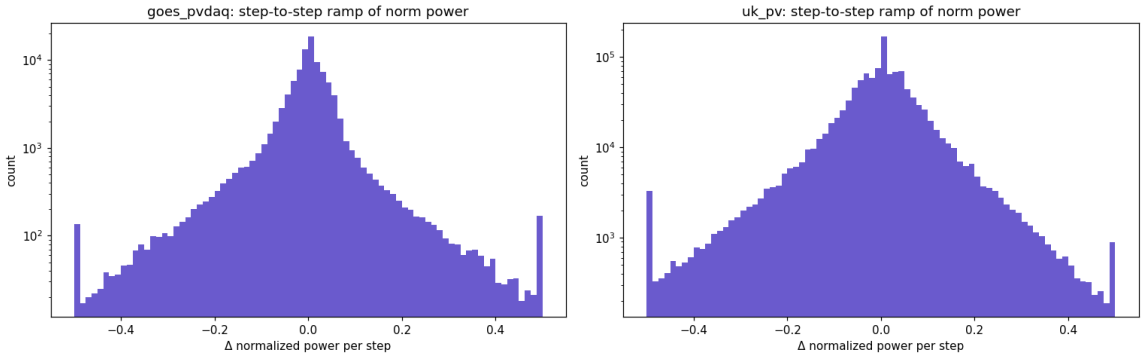


Figure 3.5: How to read the ramp distributions: the horizontal axis is the signed change in normalized power between successive daytime observations. The vertical axis is the logarithmic count of those changes. Values beyond -0.5 and 0.5 are placed at the endpoints. The central peak contains stable intervals; the tails contain rapid changes.

A satellite frame does not make every ramp predictable. A cloud may form outside the crop, move in a direction the model cannot infer, or affect a panel differently from its surroundings. The figure nevertheless defines a narrow test: if recent visual structure adds information beyond numerical history, its benefit should be most apparent during these rapid transitions.

3.7.3 Spatial and operational heterogeneity

Capacity normalization places power on a common scale while preserving differences between plants. Figure 3.6 shows these differences in two ways. In the capacity histograms, the horizontal axis is installed capacity and the vertical axis is the number of sites. UK systems occupy a narrow residential range, while the US track spans a much wider range. In the maps, each point is a plant; point size increases with the square root of capacity, and color gives mean normalized power. The sites are distributed across distinct geographic settings, and some UK plants lie close enough to share broad weather systems.

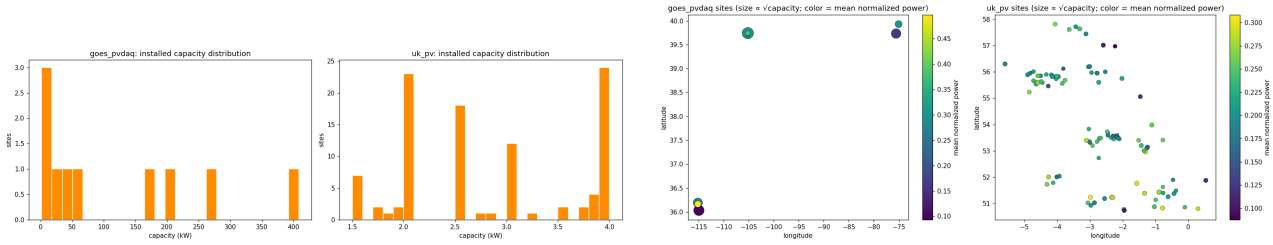


Figure 3.6: Capacity distributions occupy the left block and plant maps the right block. Within each block, US sites appear on the left and UK sites on the right. Map-marker size is proportional to the square root of capacity; color denotes mean normalized power.

Figure 3.7 compares the daytime normalized-power distributions of all 10 US plants and a fixed

random sample of 15 UK plants. Plants are ordered by their median output. Within each box, the center line is the median and the box covers the middle half of observations. The different levels and spreads show that capacity normalization standardizes units without removing differences in local conditions and plant behavior.

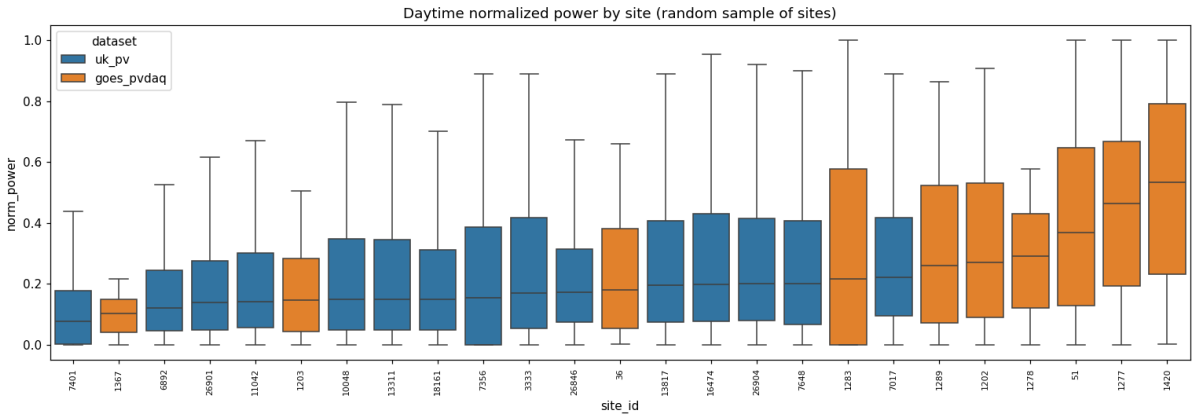


Figure 3.7: Daytime normalized-power distributions by plant, showing all US sites and a fixed random sample of UK sites. Plants are ordered by median normalized power.

Figure 3.8 summarizes the numerical inputs across all daytime observations. Rows and columns identify variables, and each cell gives the correlation for that pair; warmer colors indicate positive relationships and cooler colors negative ones. The left matrix uses Pearson correlation to measure linear association. The right uses rank-based Spearman correlation. Radiation variables are strongly positively associated with normalized power, whereas cloud cover is negatively associated. These pooled relationships provide useful context. Site-specific variation motivates plant-disjoint evaluation, while the next analysis examines temporal persistence within individual series.

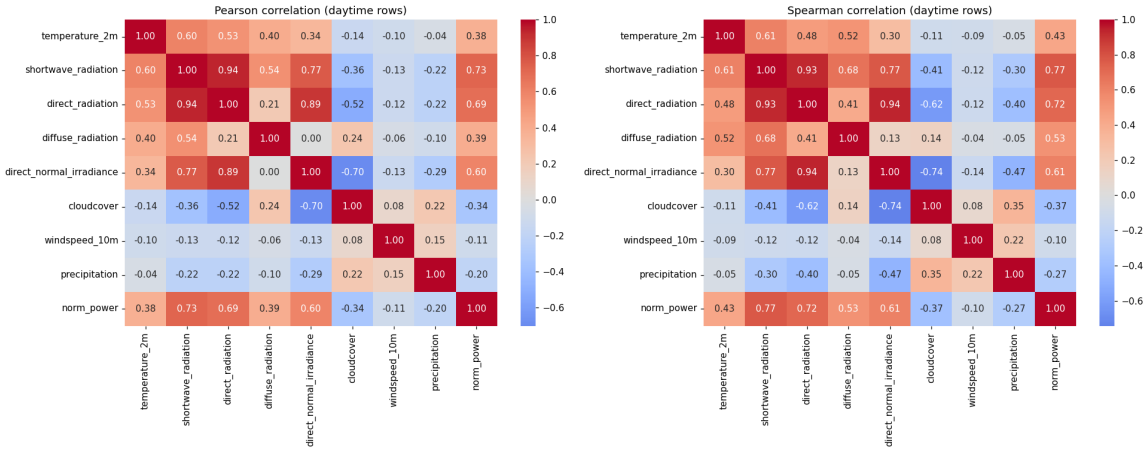


Figure 3.8: Pearson (left) and Spearman (right) correlations among numerical covariates and normalized power over daytime observations.

3.7.4 Persistence and the choice of context length

Figure 3.9 has two views of autocorrelation for PVDAQ plant 1420 and UKPV plant 6966. The left panel covers the first six hours, with lag on the horizontal axis and autocorrelation on the vertical axis. Both lines begin close to one, so persistence is a demanding reference forecast at short horizons. The US correlation reaches zero at about three hours and then becomes negative as the comparison crosses to the opposite side of the daily cycle. The UK correlation declines more gradually. It reaches about one half after three hours and remains near one third at six hours.

The right panel extends the same calculation over the full fourteen-day history. Both tracks show repeating daily peaks. At a one-day lag, autocorrelation is 0.822 for PVDAQ and 0.609 for UKPV.

At fourteen days, it is 0.783 and 0.530, respectively. The recurring pattern supports the use of several daily profiles in the context window. The intraday decline shows why recent power alone is insufficient for the later forecast horizon.

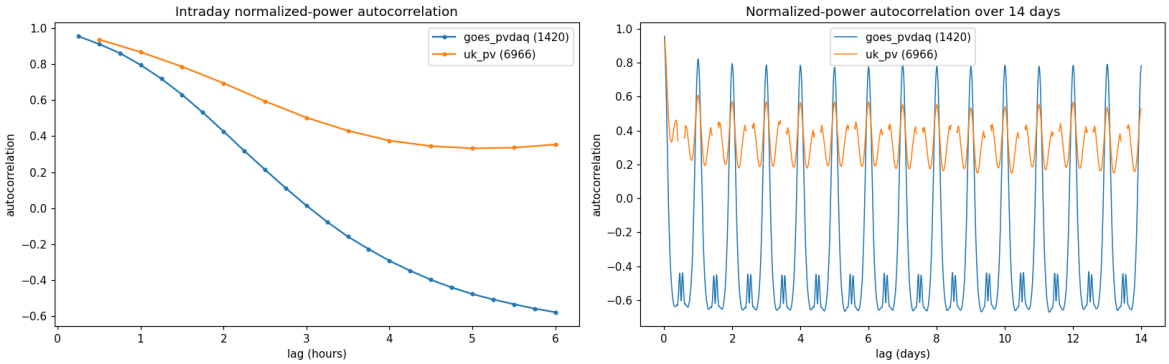


Figure 3.9: Normalized-power autocorrelation at PVDAQ plant 1420 and UKPV plant 6966, shown over the first six hours (left) and the full fourteen-day context window (right).

The model therefore receives fourteen days of history. This window gives it repeated observations of the same plant at the same clock time under different weather conditions. It still provides context when the most recent intervals are missing. For the UK track, fourteen days contain 672 half-hour observations. Chronos-2 reduces them to 42 temporal tokens before fusion. A longer window would add more tokens and missing intervals; a shorter one would make the forecast depend more heavily on a single recent daily profile. The ablations measure whether this context improves forecasting.

3.7.5 Unequal contribution across plants

Plants contribute unequal amounts of data because their records do not cover the same number of calendar days. Site coverage is measured from a plant's first to its last timestamp: it is the fraction of days in that span for which the corpus retains at least one observation. This measure is appropriate for the daytime-only record, where nights are absent by design. Figure 3.10 groups plants by that percentage, with each bar giving the number of plants in a coverage range. UK coverage is tightly concentrated between 99.3% and 100%, whereas PVDAQ coverage ranges from 71.8% to 100%.

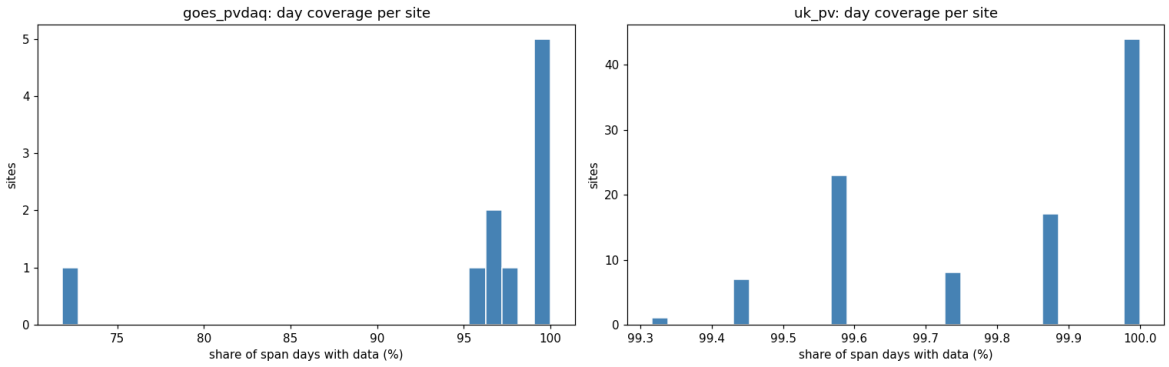


Figure 3.10: Share of active-span days containing observations for each plant in the US and UK tracks.

Figure 3.11 summarizes two aspects of daytime behavior by plant. The left histogram gives mean normalized power, and the right gives its Pearson correlation with shortwave radiation. The distributions show that plants differ in typical output and in their numerical relationship with radiation. This variation motivates the plant-level aggregation rule in Chapter 5.

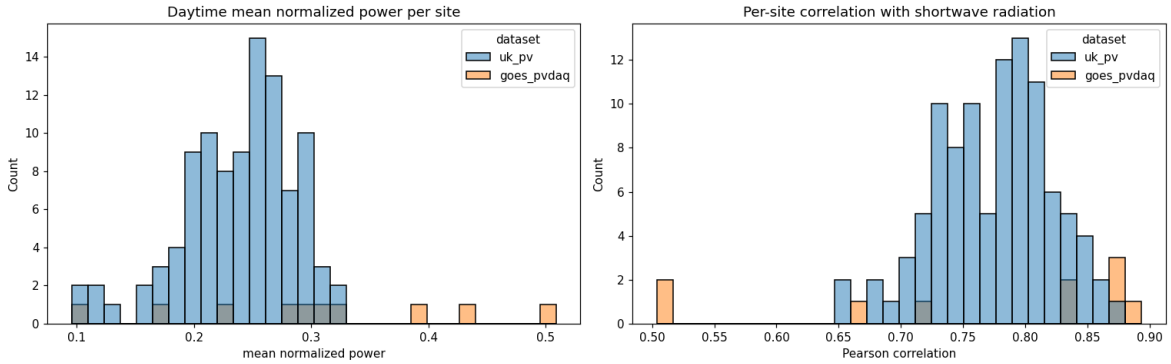


Figure 3.11: Per-plant distributions of daytime mean normalized power (left) and Pearson correlation between normalized power and shortwave radiation (right).

3.7.6 Seasonal and target imbalance

The two tracks cover different seasonal ranges. UKPV includes all of 2019 and 2020, whereas PVDAQ ends in September 2019. The target distribution therefore combines the daily production cycle with the months represented in each source.

Figure 3.12 shows this distribution in three forms. The first histogram includes all retained rows and has a large concentration near zero. The second restricts the sample to producing rows, revealing the distribution once near-zero values are excluded. The third shows absolute power on a logarithmic scale. This view makes the wide difference in physical scale across plants visible. High normalized output occurs less often than low output.

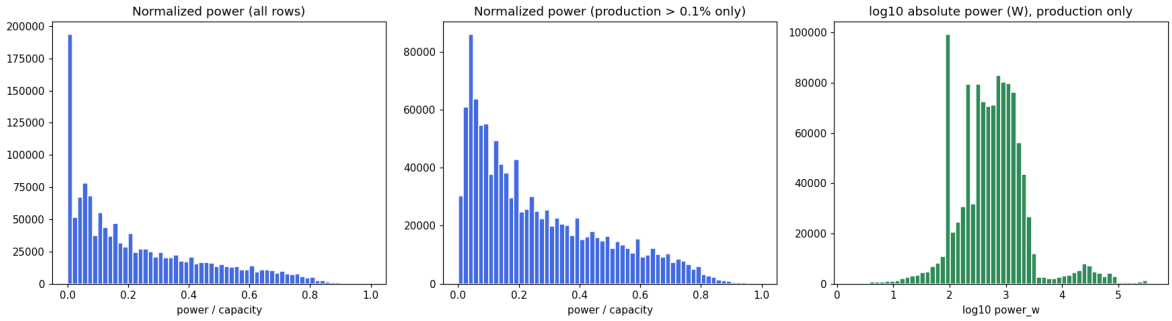


Figure 3.12: Normalized-power distributions over all retained rows (left) and producing rows (center), with the distribution of absolute power on a logarithmic scale (right).

Figure 3.13 plots mean normalized power for each calendar month. The UK line traces two full annual cycles, while the PVDAQ line covers January through September 2019. Production rises through spring, is higher during summer months, and falls again toward winter. The two lines should be read in light of their different geographic settings and time coverage. The next analysis narrows from these seasonal patterns to whether cloud cover explains variation within irradiance ranges.

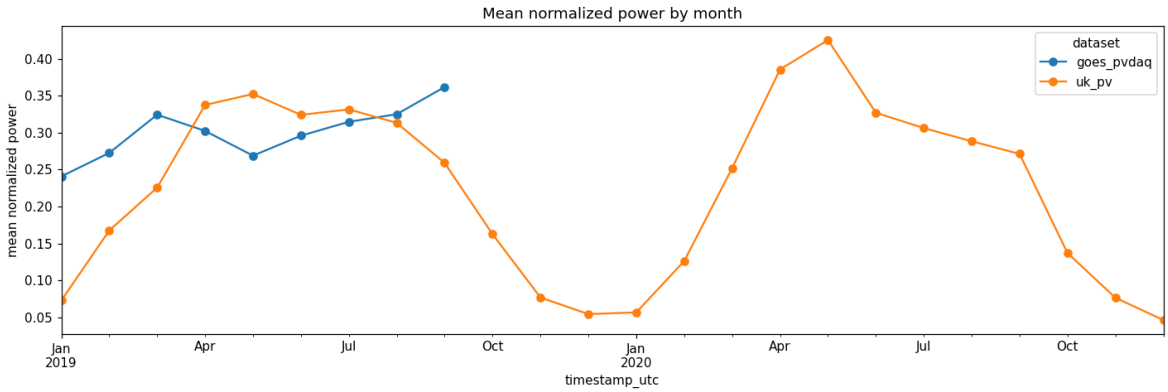


Figure 3.13: Mean normalized power by calendar month for UKPV and PVDAQ.

3.7.7 What numerical cloud cover leaves unexplained

The numerical record already includes cloud-related covariates. The relevant question is whether uncertainty remains after those covariates are known. Figure 3.14 conditions on shortwave radiation by grouping observations into five irradiance ranges. Within each range, it separates reanalysis cloud cover into clear, mixed, and overcast categories. Each box summarizes normalized power in one group: the center line is the median and the box spans the middle half of observations.

Within the displayed irradiance ranges, clearer categories generally have higher median output than overcast categories. This descriptive association motivates testing cloud-related inputs alongside irradiance. The broad bins and pooled plants leave any independent predictive contribution unmeasured. The boxes also overlap broadly, showing that these scalar covariates leave realized output variable.

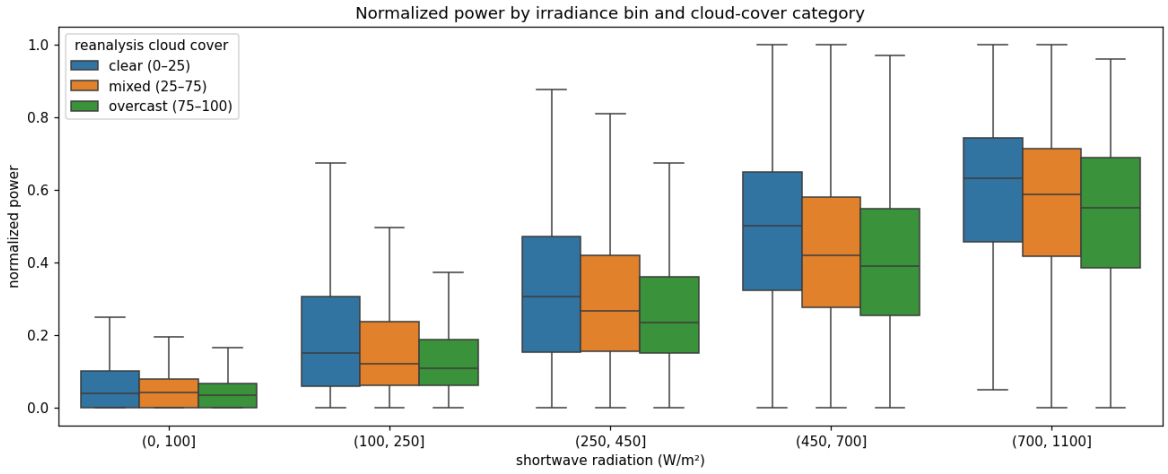


Figure 3.14: Daytime normalized power within shortwave-radiation ranges, separated by reanalysis cloud-cover category.

The boxes summarize observed groups after conditioning on irradiance and cloud cover. Their residual spread may reflect plant behavior, measurement error, or weather outside the image crop. The method and evaluation chapters test whether an independent visual input explains part of that variation.

4 Method

The dataset contract in Chapter 3 defines each forecast origin as an aligned combination of plant history, numerical context, and recent satellite imagery. This chapter turns that contract into a sequence of model variants, called arms. Each arm starts from the same numerical lineage and visual source, then changes one part of the visual pathway: its compression, token selection, placement in the sequence, or temporal coverage.

A power trace tells the model what has already happened at a plant, but it does not show a cloud boundary approaching from outside the local measurement. A recent satellite clip may carry that information. It is a dense spatiotemporal field, while the numerical forecaster works with a short sequence of compact tokens. Every arm therefore has to reduce the visual field enough to make computation feasible. It must also keep the location and evolution of the scene parts that could matter for the next hours of production.

4.1 Overview and shapes

The dataset contract supplies every arm with the same forecasting example: fourteen days of normalized power history and numerical covariates. Single-anchor visual arms use one eight-frame satellite clip recorded before the forecast origin. Each multi-anchor arm uses five such clips at distinct daily anchors. Horizon weather covariates are archived measurements used as proxies for forecasts, as defined in the problem setting in Chapter 1. For the primary United Kingdom track, that fourteen-day history contains 672 half-hour observations and the forecast covers the next 12 observations, or six hours. Each forecast returns nine quantiles at every lead time, allowing the model to express a wider interval when the next cloud evolution is uncertain.

The arms form a ladder of increasingly direct visual access. The numerical baseline receives no images. The pooled-image baseline reduces the visual field to a learned summary before fusion. Spatial-token fusion retains selected locations and inserts them into the recent numerical sequence. Multi-anchor applies the same idea at five daily anchors. Full payload is a separate extension that retains the full spatial-token payload at each anchor. The standard ladder arms share the partition, forecast origins, loss, and scoring rules. The full payload extension uses only origins with all five cached entries, as described in its subsection. Differences in dropout and mask handling are retained in the training recipe because they affect how individual comparisons should be read.

Table 4.1 summarizes the four ladder arms and the full payload extension before the sections below describe their shared components and individual pathways.

Arm	Visual input	Placement	Role
Numerical	None	None	Numerical control
Pooled-image	Learned summary	Auxiliary row	Summary bottleneck
Spatial-token	98 spatial tokens	Recent context	Spatial access
Multi-anchor	20 tokens/anchor	Five daily patches	Repeated daily evidence
Full payload	98 tokens/anchor	Five daily patches	Detailed daily views

Table 4.1: Fusion ladder. The standard arms change how the same satellite source reaches the forecaster. The full payload extension is evaluated on origins with all five required cached entries.

The representations in Table 4.2 determine the sequence length seen by attention. Chronos-2 groups the 672 half-hour context values into $T_{\text{ctx}} = \lceil 672/16 \rceil = 42$ numerical patches. A single future-query token requests all 12 lead times, and the quantile head returns $[B, 12, 9]$ for batch size B.

Multi-anchor has a similar target-row length to spatial-token fusion, although it distributes 100 visual tokens across five anchors instead of placing 98 at one recent anchor. The full payload extension

Arm	Context + query	Visual tokens	Row length
Numerical	$42 + 1$	None	43
Pooled-image	$42 + 1$	1 (auxiliary row)	43
Spatial-token	$42 + 1$	98 (one anchor)	141
Multi-anchor	$42 + 1$	20×5	143
Full payload	$42 + 1$	98×5	533

Table 4.2: Target-row token budgets of the numerical and visual arms. The pooled-image baseline has a 43-position target row and a separate one-position visual row; it is therefore not a 44-position temporal sequence.

expands that budget to 490 visual tokens across the same five anchors.

The pooled-image baseline compresses the 784 cells in the cached visual field to one learned summary before the numerical forecaster receives visual information. That summary is computed independently of the numerical state, and its auxiliary row cannot preserve an addressable image location. Spatial-token fusion retains 98 selected cells with their spatial identity. The later results therefore test whether that direct access to selected visual locations matters more than a global learned summary.

4.2 Design principle: decoupled resolution

The numerical and visual sources are intentionally kept at different resolutions. The numerical branch has a long fourteen-day history because it must infer daily structure and the recent behaviour of an unfamiliar plant. The visual branch has eight high-cadence frames near the forecast origin because it is intended to describe recent cloud evolution. Resampling the power history to image cadence would obscure the long context that supports cross-plant transfer. The architecture joins the two sources only after each has been represented at the resolution appropriate to its task.

4.3 Shared numerical backbone: Chronos-2

Every arm uses the same Chronos-2 numerical backbone. Visual-arm initialization follows the seed-matched numerical baseline defined below; the recipe and mask differences remain part of each comparison. This section describes the shared path from the fourteen-day numerical history to the quantile forecast.

Chronos-2 receives normalized power history and the numerical covariates permitted at the forecast horizon. It groups the 672 historical values into non-overlapping patches of 16, producing 42 context tokens that each span eight hours. A further position represents the future patch to predict. Patching preserves the full fourteen-day history without requiring attention over every half-hour observation.

The loaded checkpoint has width 768, twelve encoder blocks, twelve attention heads, and a 3072-dimensional feed-forward layer. These checkpoint dimensions are retained during loading. In the visual arms, the pretrained numerical backbone is frozen except for the shared quantile head, the new fusion components, and the final three encoder blocks. Those trainable components give the visual pathway a route into the forecast while retaining most of the representation learned by the numerical baseline. Within each block, temporal attention relates positions in the history, and group attention relates the target and covariate rows. The feed-forward layer updates their representations. The quantile head reads the final target representation and predicts all 12 horizon steps together.

The method proceeds from the numerical baseline to visual pathways with progressively less compression, all built on this shared backbone.

4.4 Numerical baseline

The numerical baseline is the vision-free control and seed-specific initialization for the visual arms.

The numerical baseline is trained first. Each visual arm starts from the numerical checkpoint with the matching random seed. Its bridge is therefore learned on top of a forecaster already adapted to the training plants. Figure 4.1 shows this numerical path without a visual stack.

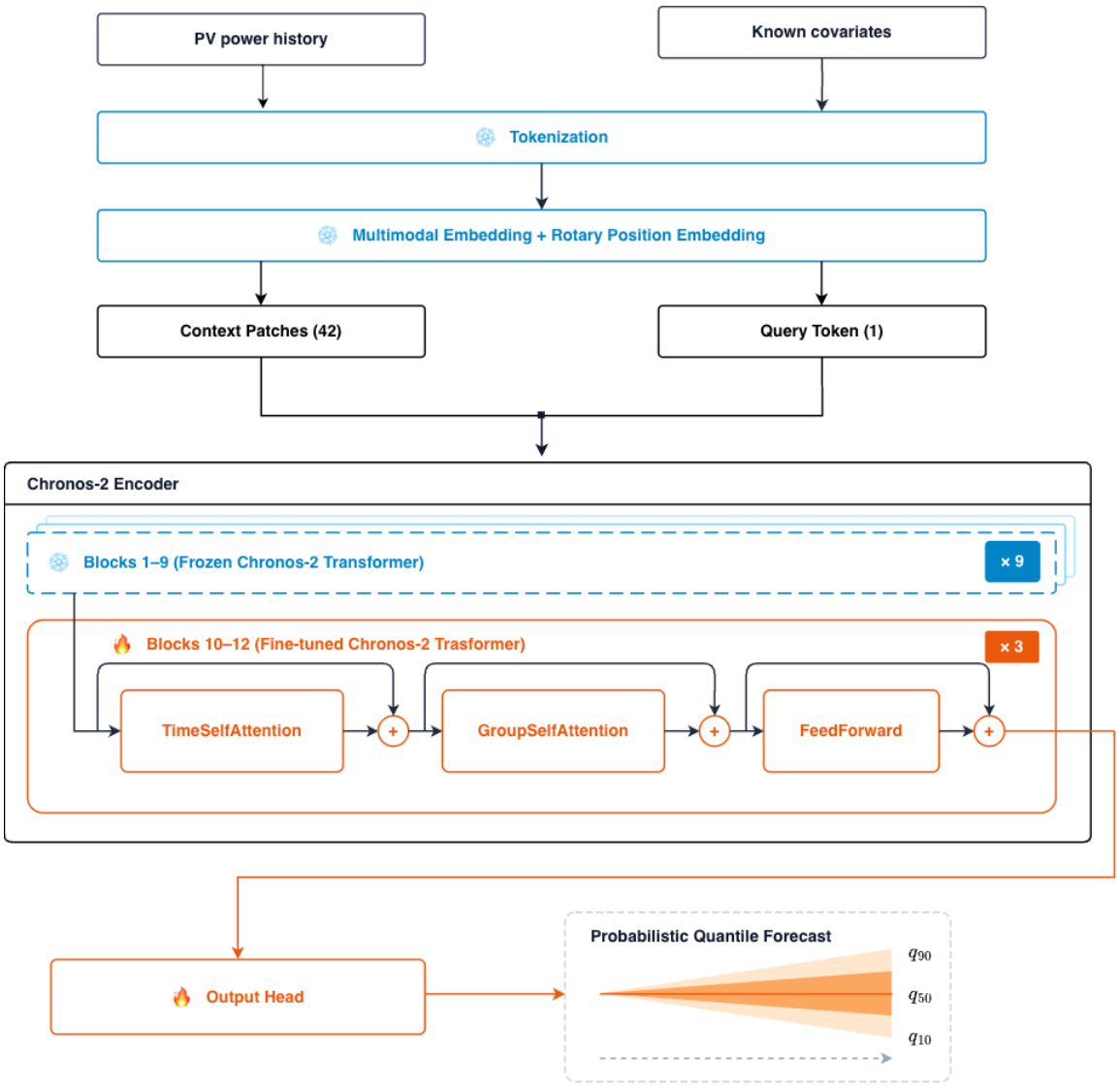


Figure 4.1: Power history and numerical covariates are converted into temporal tokens and processed by Chronos-2. This arm has no image input and provides the common starting point for the visual arms.

This first stage adapts the inherited forecasting components on the training plants, starting from general time-series pretraining rather than random weights. The resulting seed-specific checkpoint is the learned numerical base passed to every later arm.

4.5 Shared visual backbone: frozen V-JEPA 2.1

The visual encoder is shared by every visual arm and is never trained, so it is described once here. Its latent geometry, mismatch with the numerical sequence, and cached execution determine the visual interfaces that follow.

The visual pathway begins with V-JEPA 2.1 ViT-L/16. It processes the eight-frame satellite clip as a short video rather than as eight unrelated images. The encoder produces four temporal latent slices. Each slice contains a 14 by 14 grid of spatial patches, giving 196 patch features of width 1024. The complete visual representation therefore has the shape $[B, 4, 196, 1024]$.

A video encoder is chosen on the expectation that sky evolution carries forecasting information.

The visual latents are extracted before model training and read from a cache. Repeated seeds and multi-anchor layouts are feasible because the video encoder does not run in every training step. The visual-validity mask travels with the cached features. A missing frame is therefore represented as missing evidence, not as a synthetic clear-sky image. The architecture can continue from numerical evidence when the image clip is incomplete.

4.6 Pooled-image baseline, late fusion through a learned bottleneck

The pooled-image baseline tests whether one learned visual summary helps Chronos-2 while leaving its numerical sequence unchanged. A causal Perceiver-style resampler [14] attends over the V-JEPA patch field and produces one learned summary token. The cross-modal adapter projects it to Chronos-2's 768-dimensional width and places it on an auxiliary visual row. Group attention processes this row with the numerical rows.

Figure 4.2 follows the visual field through this bottleneck.

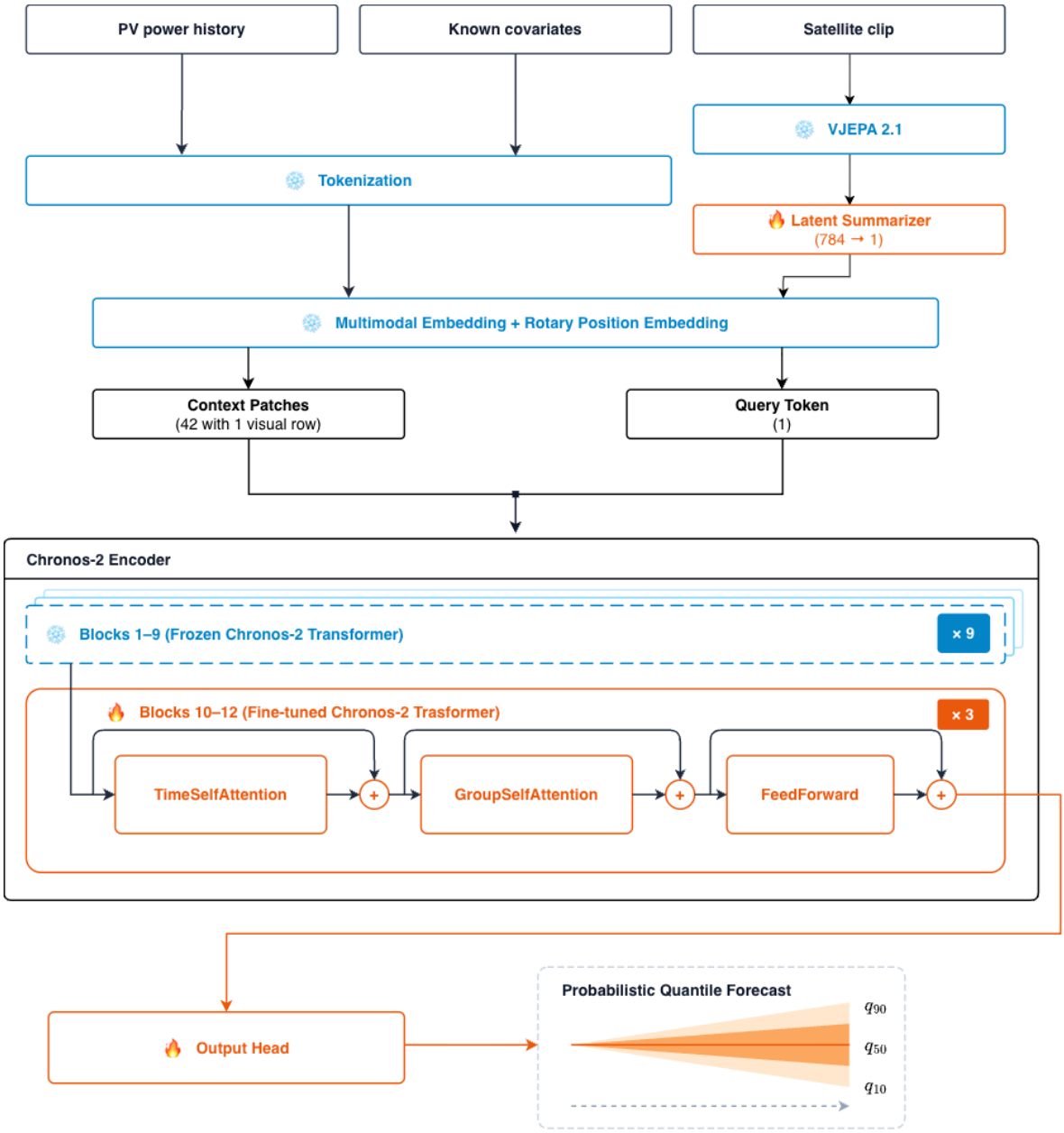


Figure 4.2: Pooled-image baseline. The learned resampler converts the complete visual field into one summary token, which enters Chronos-2 as a separate visual row.

The resampler maps 784 patch cells to one token, so spatial locations are not exposed as separate entries to later layers. The baseline tests whether this global summary carries enough information before the next arm preserves selected locations as addressable tokens.

4.7 Spatial-token fusion, single-anchor interleaving

Spatial-token fusion replaces the pooled-image baseline's global summary with selected visual tokens. Those tokens are inserted into the final numerical context patch, where the shared encoder can process

them with the recent power history. This arm introduces the resampler-free path at one anchor.

The visual pruning and interleaving draw inspiration from the visual-token pathway in Nemotron 3 Nano Omni [30], discussed in Section 2.5. Its spatial reduction and token selection motivate keeping a compact set of spatially distinct visual features. Here, that idea is adapted to forecasting: selected visual tokens are interleaved at their observation times with the numerical sequence, so the shared temporal encoder can process both sources together.

The visual field first undergoes a 2×2 spatial-to-channel rearrangement. Each block of four neighboring patches in the original 14 by 14 grid becomes one cell, leaving a 7 by 7 field of 49 cells. Concatenating the four 1024-dimensional patch features gives each cell 4096 features. A projector then maps each cell to Chronos-2's 768-dimensional width. Selection retains the 49 cells from the first temporal slice, ranks later cells by cosine change from the preceding slice, and retains 49 additional candidates. The resulting 98 tokens enter the numerical sequence at their observation positions. The tensor path is $[B, 4, 196, 1024]$ for the cached field, $[B, 4, 49, 4096]$ after rearrangement, and $[B, 98, 768]$ after projection and selection, where B is the number of forecast examples processed together. The other axes denote temporal slices, spatial cells, and feature dimensions before selection; afterward, the middle axis counts selected tokens. The final tensor is interleaved with 42 numerical context tokens and one future-query token.

The projector matches the visual width to the numerical width without mixing information across image locations. A learned coordinate embedding identifies each of the 49 cells, while the sequence position records its observation time. The model can therefore distinguish where a change occurred from when it was observed.

Figure 4.3 shows the reduction, selection, and interleaving path.

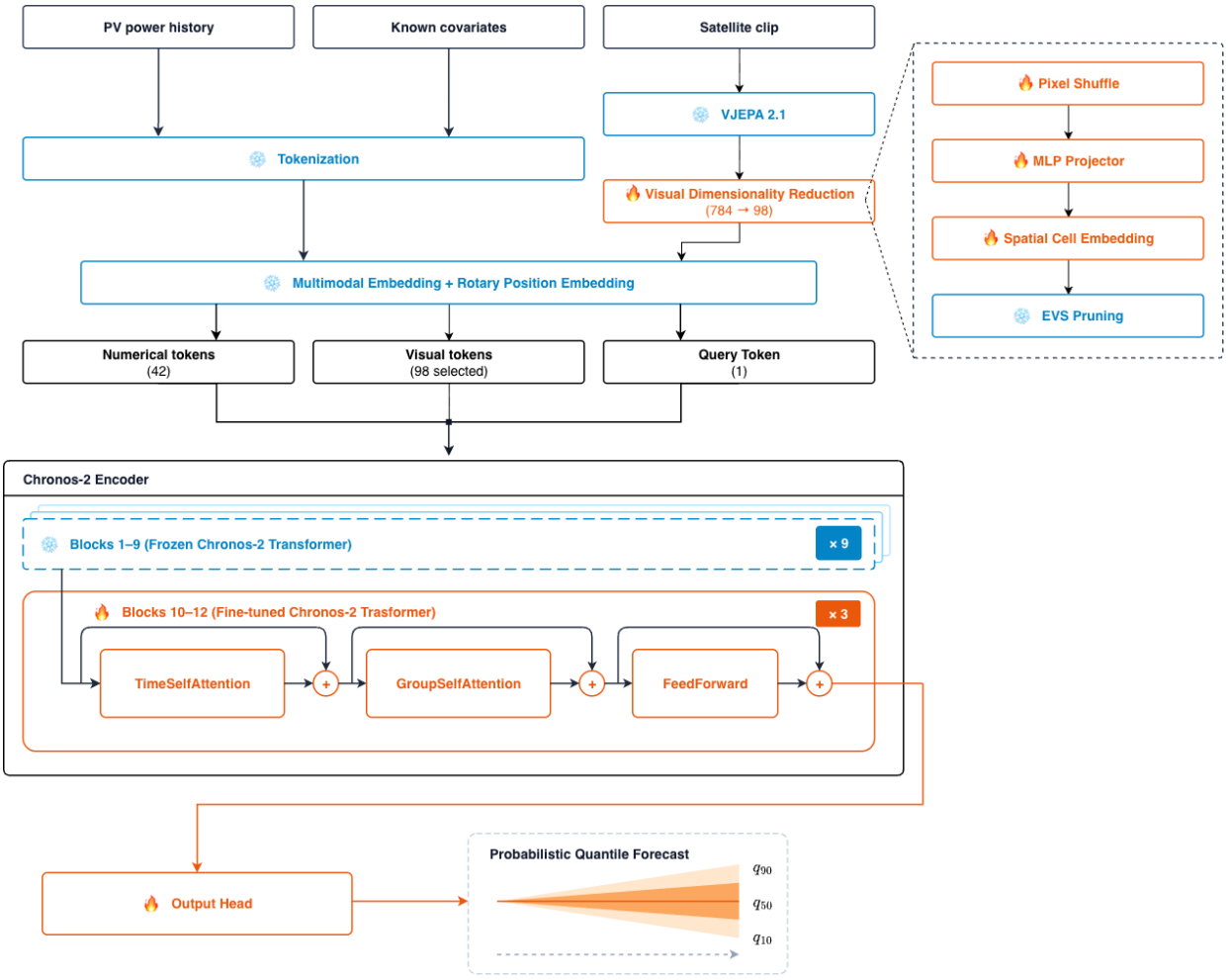


Figure 4.3: Spatial-token fusion, resampler-free interleaved fusion. Forty-two numerical context tokens, 98 selected visual tokens, and one query form a 141-position target row. Visual cells are reduced spatially, projected, selected by temporal change, and interleaved at their observation positions.

The cached field contains 196 cells: 49 spatial cells in each of four latent slices. All 49 cells from the first slice are retained as an anchor field. The remaining 147 cells are ranked by cosine dissimilarity from the corresponding cell in the preceding slice, and the highest-scoring 49 complete the 98-token payload.

The selected tokens take positions associated with their observation times inside the final numerical context patch, producing the 141-position sequence in Table 4.2. The shared temporal mixer processes visual and numerical tokens together, so a retained visual location can affect the numerical representation in every encoder block. The arm has no pooled-image resampler or adapter, so the selected locations remain directly available to the encoder.

4.8 Multi-anchor

Multi-anchor extends spatial-token fusion from one recent anchor to five anchors, one on each of five consecutive days. The anchors share the same clock time, so their solar geometry is broadly comparable. The model can then see several examples of the relation between local sky appearance and power at that time of day.

Each daily anchor is represented by a cached clip whose four temporal latent slices are available to the encoder. For multi-anchor, the selector compares two successive slices from that cached representation and retains both endpoints for each selected spatial trajectory. These two slices describe one anchor; they do not create two additional daily anchors.

Figure 4.4 shows where the five visual blocks enter the numerical history.

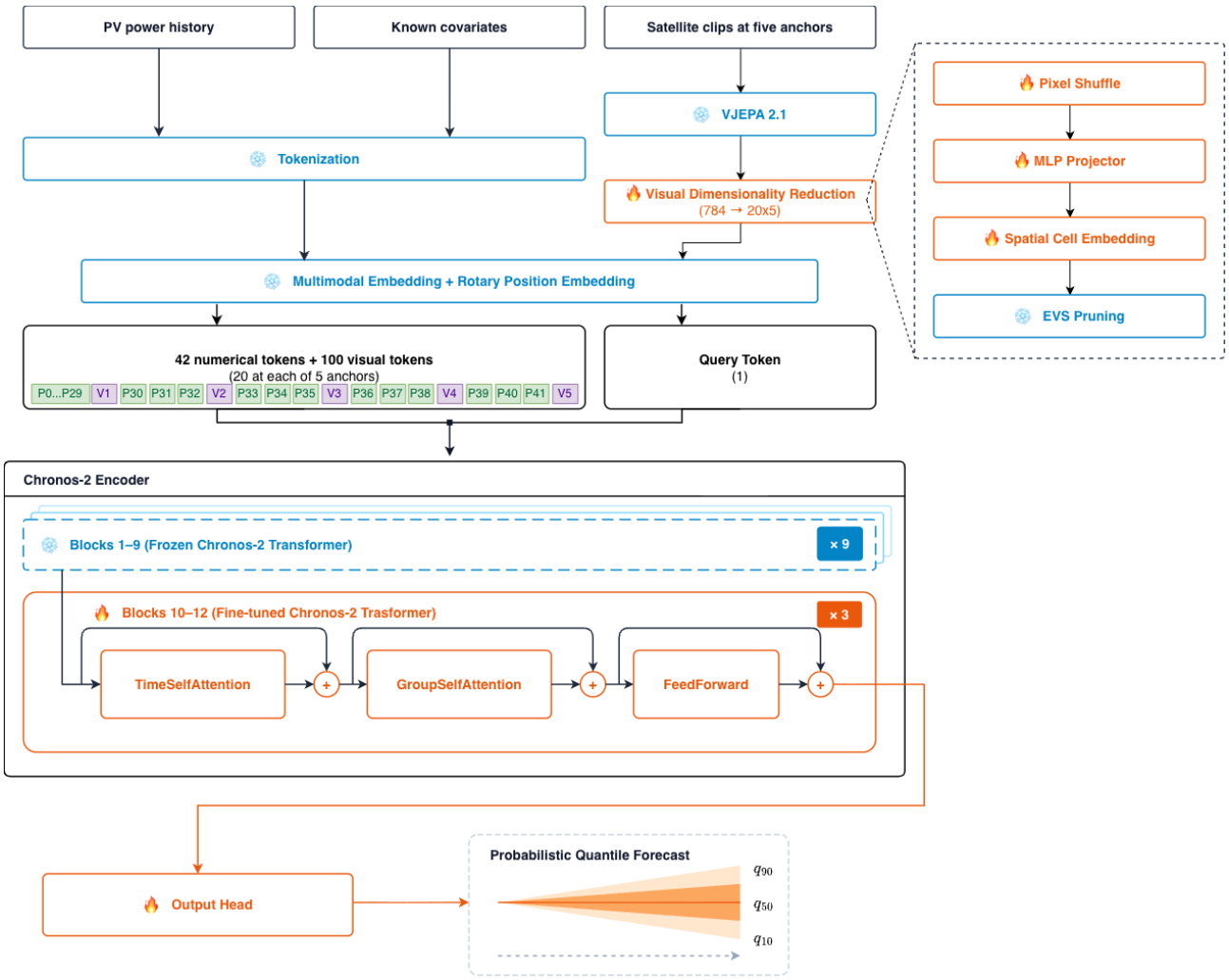


Figure 4.4: Multi-anchor. Five blocks of 20 visual tokens are interleaved with 42 numerical context tokens and one query, forming a 143-position target row. Each block represents one daily anchor.

Within each anchor, visual token selection ranks the 49 spatial cells by cosine change between

latent slices and retains the 20 highest-scoring candidates, yielding 100 tokens overall. A paired-novelty variant that retains both endpoints of the ten most dynamic trajectories is evaluated as a selector ablation in Section 5.3.

Each visual block shares the integer position of the numerical patch at its daily anchor. The resulting sequence has 42 numerical context tokens, 100 visual tokens, and one future-query token, for a total of 143 positions as shown in Table 4.2.

4.9 What each arm reuses and what it learns

The arms share numerical and visual foundations and differ through the visual bridge. Distinguishing inherited components from newly trained ones makes the comparison boundary explicit and clarifies which parts can account for differences in outcome.

Every arm loads the same Chronos-2 components (patch tokenizer, twelve encoder blocks, positional machinery, group attention, and multi-quantile head). Visual-arm initialization follows the numerical-baseline procedure above. Each arm also reads the same pretrained V-JEPA latents from cache. The video encoder never runs inside the training loop, so image augmentation, crop changes, and visual-backbone gradients are outside the training process. The bridge built on top distinguishes the visual arms, as summarized in Table 4.3.

Arm	Components
Numerical	None; inherited components adapted
Pooled-image	Causal summarizer, cross-modal adapter, shared head
Spatial-token	Projector, modality embeddings, shared head
Multi-anchor	Projector, integer positions, shared head

Table 4.3: Newly initialized and trainable components by arm. The visual encoder remains frozen; the final three Chronos-2 blocks and shared quantile head train in visual arms.

The ladder is a controlled comparison of visual interfaces, not a claim that every training detail is identical. Table 4.4 separates the common conditions from the choices that still vary and must therefore remain part of the interpretation.

Aspect	Shared	Varies
Evaluation	Plant split, horizon, scoring	-
Initialization	Seed-matched Chronos-2 warm start	Bridge and trainable parameters
Vision	V-JEPA cache and validity mask	Pooling, selection, anchor count
Sequence	42 context patches + one query	Auxiliary row or interleaved tokens
Training	Frozen V-JEPA; final three Chronos blocks trainable	Visual dropout, numerical masks

Table 4.4: Comparison boundaries for the fusion ladder.

4.10 Computational scaling of the interfaces

The visual information retained by each arm also determines the work required of the shared transformer. This section compares sequence sizes and then describes the per-sample forward path.

Sequence length sets the main attention cost. Self-attention compares pairs of positions, so its activation cost grows approximately with the square of the sequence length. Spatial-token fusion has 141 positions, compared with 43 for the numerical baseline target row, and therefore has about 10.8 times as many position pairs. Multi-anchor has a similar cost at 143 positions.

Sending all 784 visual candidates beside the numerical sequence would make the row longer still, especially with more than one anchor. The visual field must therefore be summarized or selected before it enters attention. The pooled-image baseline learns that compression, whereas spatial-token fusion and multi-anchor retain a fixed number of selected cells.

Arm	Length	Relative pairs	Memory
Numerical	43	1.0 ×	Reference
Spatial-token	141	10.8 ×	Spatial tokens; smaller batches
Multi-anchor	143	11.1 ×	Distributed anchors

Table 4.5: Approximate pair-count scaling relative to the 43-position numerical baseline target row. The values describe sequence geometry, not measured end-to-end runtime.

4.11 From one dataset window to one forecast

A forecast example in the primary UK setting illustrates data construction, arm-specific token assembly, model fitting, and inference. The same sequence applies to validation and test windows within their plant partitions.

Table 4.6 distinguishes inputs tied to the forecast origin from fields available later in the window. Power history and satellite frames come from observations at or before the origin. Calendar and solar geometry are known over the horizon; archived atmospheric measurements are supplied there as proxies for weather forecasts. Future power is reserved for training and evaluation.

Input	Availability	Use
Power history	Observed before origin	Numerical context
Satellite	Strictly before origin	Timestamp-matched frames
Calendar/solar	Deterministic	History + horizon
Weather	Archived horizon observations	Forecast proxy
Future power	Unavailable	Masked target; evaluation reference

Table 4.6: Information available to a forecast window.

4.11.1 Constructing the numerical and visual inputs

For the primary UK track, the loader resolves fourteen days of half-hourly power history and the following six-hour target window. It works in timestamp space, so missing intervals remain at their physical times instead of shifting later samples forward. Installed capacity converts measured power to the normalized target. Power history, covariate rows, and future targets retain separate validity masks. A measured zero remains an observation, while a missing patch remains unavailable context. The target mask also excludes invalid future values from training loss and evaluation.

The numerical input contains a power row and covariate rows. Calendar and solar-geometry values are known through the forecast horizon. Archived meteorological measurements are also supplied across the horizon as proxies for operational weather forecasts. Chronos-2 patches the 672 historical power samples into 42 tokens. Covariate rows share a group identifier with the power row. Group attention then relates the variables after temporal encoding.

For a visual arm, the plant identifier and frame timestamps link the forecast origin to the satellite observations for that location. Frames are selected before the origin using the timestamp-matching rule described in the dataset chapter. Their precomputed V-JEPA 2.1 representations are then retrieved. The frozen visual encoder processes each clip before forecaster fitting, so its output can be reused across training epochs. Each eight-frame clip supplies four temporal latent slices, each containing a 14 by 14 grid of 1024-dimensional patch features. If a frame or cached representation is unavailable, the corresponding visual validity mask marks it missing; no artificial image is substituted.

4.11.2 Building the model sequence

Every arm starts with 42 numerical context tokens and one future-query token. The numerical baseline sends these rows through Chronos-2 without images. It provides the adapted weights used to initialize the visual arms, with a separate checkpoint for each random seed.

The pooled-image baseline sends the cached patch field to a causal Perceiver-style resampler. The resampler condenses its 784 patch cells to one learned summary, and a cross-modal adapter maps that vector to Chronos-2's 768-dimensional width. The summary enters on an auxiliary visual row in

the same group as the numerical rows. Group attention can combine the summary with power and covariates while the 43-position target row stays unchanged.

Spatial-token fusion keeps selected locations as separate entries. A 2×2 spatial-to-channel rearrangement combines each block of four neighboring patches into one of 49 cells. A projector maps each 4096-dimensional cell to the 768-dimensional Chronos-2 width. The selector retains all 49 cells from the first latent slice, then ranks cells from later slices by cosine change relative to the preceding slice and selects 49 additional candidates. Their coordinate embeddings identify the spatial cells, while sequence positions identify observation times. The resulting 98 visual tokens enter the target row at those times, giving the row 141 positions including numerical context and query.

Multi-anchor allocates 20 visual tokens independently at each daily anchor, selecting cells with the highest temporal feature change to reach 100 tokens overall. Each block is placed at the numerical patch for its anchor. Full payload uses the single-anchor selector at every daily anchor. It contributes 98 tokens per anchor and 490 overall. Its target row has 533 positions.

After assembly, temporal attention processes positions along each row, while group attention relates the target, covariate, and, for pooled-image fusion, auxiliary visual rows. In the interleaved arms, spatial tokens share the target row with numerical context and are processed through the same temporal blocks. The future-query token contains no future target value. It carries the request for all twelve forecast steps through the encoder, so later predictions do not consume earlier generated predictions.

4.11.3 Fitting the arms

Training proceeds in two stages. The numerical baseline adapts Chronos-2, including its quantile head, on training-plant windows and supplies a seed-specific checkpoint. Each visual arm then initializes from the checkpoint with the matching seed. The cached V-JEPA encoder remains frozen; the arm-specific bridge, shared quantile head, and final three Chronos-2 blocks train on the masked quantile objective. The target mask removes invalid future values, and the objective averages pinball loss over valid target steps and the nine quantile levels, as defined in Section 1.2.2. During fitting, visual and numerical dropout expose the visual arms to missing inputs, ensuring that the model maintains forecasting capabilities when visual observations are unavailable.

4.11.4 Inference and scoring

At inference, the selected checkpoint receives a window assembled by the same timestamp and mask rules. For an unseen plant, the model uses its fourteen-day history, covariates, and any available pre-origin visual frames without updating its weights on that plant. Missing visual slots remain masked, so the numerical pathway can still produce a forecast when an image is unavailable.

The encoder processes the assembled sequence once. Its future-query representation feeds the quantile head, which returns a tensor of shape $[B, 12, 9]$: B forecast examples, twelve half-hour leads, and nine quantile levels from 0.1 through 0.9. The median supplies the point forecast; the full quantile set supports interval, CRPS, and calibration measures. The model does not impose monotonic ordering or apply post-hoc rearrangement, so predicted quantiles are not guaranteed to increase with their stated probability. Training loss and evaluation metrics use the validity masks, with plant weighting and ramp selection specified in Chapter 5.

The vision-on/off control reruns this same inference pass with visual embeddings set to zero while retaining the fitted weights, numerical embeddings, token positions, origins, and masks. Its score difference tests whether a trained forecast depends on its visual pathway. Stale-image and wrong-plant substitutions test whether that dependence is tied to recent, site-specific content; the intervention details and limits appear in Section 5.3.

4.12 Training recipe

The fitting procedure for the baseline and visual arms is described above. This section specifies the learning-rate schedule and modality-dropout settings relevant to comparisons among them.

The visual projector uses a 500-step learning-rate warm-up. Warm-up changes the learning rate at the start of training; it does not change which parameters are trainable. Lightning [8] and Hydra [40] manage the configuration, and training uses distributed bf16 execution on Leonardo HPC.

Modality dropout exposes visual arms to missing inputs during training. The pooled-image baseline uses visual dropout of 0.3 and numerical dropout of 0.1. The interleaved arms use visual dropout of 0.5 and the same numerical dropout.

4.13 Implementation and compute

Hydra composes dataset, model, stage, trainer, logger, and ablation settings; command-line overrides identify each arm and seed. Lightning manages fitting, validation, checkpointing, and testing. The data module constructs plant-disjoint windows, and the model module applies masked quantile loss and exports protocol metrics. Gradient accumulation reduces the per-step GPU batch while retaining an effective batch of sixteen. The training entry point distinguishes a weights-only warm start from a full-state resume. Weights & Biases records training metrics through Lightning; it runs offline on Leonardo compute nodes, with resolved configurations saved at run start for later synchronization.

Experiments used one or more 64 GB NVIDIA A100 GPU per training job with mixed bfloat16 precision. Frozen V-JEPA 2.1 latents were extracted to a separate cache before model fitting, so the video encoder did not process source frames at every optimization step. The default microbatch of four with four accumulation steps exceeded available device memory for full payload. That arm used a microbatch of two with eight accumulation steps to keep the effective batch at sixteen. The shorter arms used the standard setting.

Table 4.7 reports the mean elapsed runtime for each stage across completed runs on a single NVIDIA A100 GPU. These numbers record observed execution time, whereas the scheduler wall-time limits were set higher to allow for checkpoints and queue timeouts.

Job	Mean elapsed time
V-JEPA latent extraction	7 h
Numerical	18 h
Pooled-image	20 h
Spatial-token	22 h
Multi-anchor	23 h
Full payload	45 h
Evaluation	30 min

Table 4.7: Mean elapsed execution time on a single NVIDIA A100 GPU across completed runs.

4.14 Full payload

The multi-anchor arm distributes a fixed visual budget across five days. The full payload arm removes this budget constraint and tests the effect of retaining the full visual detail available to the single-anchor pathway at each daily anchor.

Full payload is a separate high-capacity arm that tests the upper end of the multi-anchor design. It retains five daily anchors, spaced 24 hours apart, and gives each anchor the full spatial-token fusion payload: eight input frames, four temporal latents, and 98 selected visual tokens. The arm therefore supplies five detailed examples of the local sky-power relationship instead of distributing a 100-token budget across five anchors. Its sequence contains 42 numerical context tokens, 490 visual tokens, and one future-query token, for a total of 533 positions.

The arm tests whether retaining more visual detail at each anchor helps the model exploit repeated daily evidence. Visual payload and sequence length increase together, leaving the effect of retaining more detail coupled with added sequence capacity. Attention has about fourteen times as many position pairs as in spatial-token fusion, and training takes longer. The larger interface therefore adds capacity alongside visual evidence.

The arm reuses existing full-depth cached representations, avoiding a new forty-frame video encoding for every forecast origin. For an origin at time t, the loader reads cached embeddings across the five daily origins from $t - 96$ to t hours, excluding forecast windows where any of the five daily anchors is missing.

Memory, rather than cache storage, is the limiting factor. The default microbatch of four examples with four accumulation steps exhausts a 64 GB A100. The launch therefore uses a microbatch of two with eight accumulation steps, preserving the effective batch size of 16 while reducing the memory required by each forward pass. This adjustment is necessary to run the arm, and it is another reason to interpret it as a separate high-capacity experiment rather than as a direct replacement for the shorter fusion interfaces.

5 Evaluation protocol and experiments

5.1 Evaluation protocol

The dataset contract in Section 3.6 defines the forecast sample. This section specifies how those samples are compared. It defines which plants are held out, what each model may observe at the forecast origin, how errors are aggregated, and how visual reliance is tested. Within each dataset track, direct comparisons use the same forecast origins, target masks, and horizon.

5.1.1 Plant-disjoint partitions

Plant identity is the unit of partitioning within each dataset track. The fixed United Kingdom assignment contains 69 training plants, 15 validation plants, and 14 test plants. Table 5.1 gives the quality-valid power observations in each group before forecast-window and daylight filtering. Any evaluation on another track requires its own plant-disjoint partition under the same separation rule. The benchmark results section reports the UK test partition.

Split	Plants	Valid steps	Use
Training	69	846,633	Fitting and adaptation
Validation	15	182,809	Early stopping and tuning
Test	14	171,543	Final evaluation

Table 5.1: United Kingdom plant partition after quality-controlled exclusions. Each installation belongs to exactly one partition.

The evaluator advances by one forecast horizon between origins and applies the target-validity and daylight masks to the windows defined in Section 3.6. For the United Kingdom partition, this leaves 19,964 test windows and 165,295 scored target steps. The 171,543 test steps in Table 5.1 count valid power observations before those filters.

Forecast origins denote the timestamps separating historical observations from future horizon steps. Eligible evaluation points are governed by two filters: a target-validity mask excluding sensor faults and missing intervals, and a daylight mask restricting evaluation to solar elevation angles above zero. Evaluating all models across identical masked steps guarantees that performance differences reflect true forecasting quality under uniform test periods.

Training plants supply fitted parameters, including adapter weights, and any retrieval datastore. Validation plants determine checkpoints and hyperparameters, including retrieval weights when tuned. Test plants are used only for final reporting. This boundary also applies to learned normalization and feature selection. Each implementation declares its fitted statistics. The loader checks that plant assignments do not overlap.

5.1.2 Scenarios and input parity

The primary benchmark is the United Kingdom plant-disjoint comparison. The cross-region experiment applies models trained on UK plants to all eight eligible US plants. Checkpoints from seeds 42, 43, and 44 remain fixed during US inference, and US power is resampled to 30-minute cadence. Thus, the two reported settings measure transfer to unseen plants within the UK and transfer from the UK to the US. Across these comparisons, each model receives the modalities its design supports. Persistence forecasts use no satellite frames. Numerical forecasters use power history and, where supported, covariates. Multimodal systems can also use recent imagery. Table 5.2 summarizes the inputs by model family. Each result identifies the fields used by the individual method.

Family	Power	Covariates	Images	Role
References	Yes	Varies	No	Daily/short-horizon reference
Supervised models	Yes	Varies	No	Numerical forecasting
Tabular FMs	Yes	Yes	No	Tabular pretraining
Time-series FMs	Yes	Varies	No	Time-series pretraining
Retrieval/adapters	Yes	Varies	No	Retrieval or adaptation
Multimodal models	Yes	Varies	Yes	Visual context

Table 5.2: Modalities available at the shared forecast origin. FMs denotes foundation models; Varies means method-specific covariates, where supported.

5.1.3 Baseline implementations and comparison boundaries

The models reviewed in Chapter 2 are compared under the track-specific plant split and input rules above. This subsection specifies each model's inputs, whether it learns from the training plants, and whether its reported score uses the same evaluation set as other methods.

The non-learned references use only permitted observations. Persistence repeats the latest valid power value across the horizon. Seasonal naive reuses the value from the same clock time on the previous day. Hourly climatology estimates a plant's typical output at each hour from its available history. Smart Persistence carries the latest observed clearness relationship forward as the deterministic solar envelope changes [13]. It supplies the denominator of the skill score.

Tabular models (LightGBM [16] and TabPFN [12]) operate on fixed feature tables of lagged power, calendar variables, and weather observations without satellite frames or future covariates. Supervised sequence baselines (PatchTST [27], TFT [18], DLinear [43], and an MLP baseline) train on the same plant partition using historical covariates. iTransformer [21] models cross-variable interactions directly, providing a non-visual multi-variable reference.

The time-series foundation-model entries start from released checkpoints. Chronos-2 [1] is evaluated both zero-shot and after adaptation on training plants as the numerical control of the fusion ladder. TimesFM [11] and TiRex [3] are used zero-shot. Tiny Time Mixer [6] appears in zero-shot and fine-tuned forms. CoRA [34] adds a covariate adapter to a mostly frozen backbone. TS-RAG [28] and Cross-RAG [17] condition the forecaster on windows retrieved from training plants. These entries receive no imagery. Their architectures, pretraining corpora, and adaptation strategies differ. Their rankings therefore compare benchmark performance across methods and do not constitute an architectural ablation.

Multimodal baselines encompass both satellite-conditioned models (MMTSFM, Aurora, Solar-VLM, SUNSET) and visual time-series rendering methods (Time-VLM, UniCast). All models are scored across the held-out test installations on valid daylight steps.

5.1.4 Scoring and reporting

Within each track, valid daylight targets are scored separately for every held-out plant before the plant scores are averaged. This gives installations equal weight even when their usable records differ in length. The two co-primary measures answer different questions. Skill score measures the reduction in NRMSE relative to Smart Persistence across the scored targets. Ramp NMAE measures error during rapid changes in observed power. A gain in overall skill therefore does not compensate for worse ramp accuracy. The formulas are given in Section 1.2.2.

Models that produce quantiles also report the CRPS approximation, 80% interval coverage, and quantile expected calibration error. These metrics use the same valid-daylight mask and plant weighting.

Training variation is reported alongside these scores. Learned methods use fixed random seeds for data ordering and parameter initialization.

For multimodal models, a score advantage alone does not show that the trained forecast uses images. Each trained checkpoint is therefore evaluated with vision active and again with its visual embeddings set to zero. The numerical embeddings and token positions remain in place. Both passes use the same weights, forecast origins, target masks, ramp thresholds, and aggregation. Their differences in normalized error and ramp NMAE measure sensitivity to visual input under that checkpoint.

This comparison keeps the training route, capacity, and regularization fixed. A separately trained numerical model would change those factors as well. A negligible vision-on/off difference can therefore coexist with a nominal advantage over a numerical model. Higher error with vision disabled indicates a positive contribution from the visual pathway under this checkpoint and intervention.

The vision-on/off pass cannot identify which part of the clip matters. The cached-latent-order, stale-image, and wrong-plant-image substitution interventions are specified in Section 5.3. They use the trained weights and, where possible, the same forecast origins. They are reported apart from standard model comparisons because they change the input to an existing model. A wrong-plant-image substitution needs a clip from a different test plant at the same time. Shuffled test batches allow the evaluator to find such a donor when series-major batches would contain only one plant.

Each reported result identifies the dataset track and plant partition, physical window lengths, input modalities, exact covariate fields, and forecast-origin rules. It also identifies the random seeds, numbers of windows and scored target steps, checkpoint-selection metric, tuning budget, and whether the comparison was prespecified or exploratory.

5.2 Benchmark results

This section reports benchmark performance for the forecasting methods considered in the thesis. The evaluation follows the protocol defined in Section 5.1 and includes both aggregate scores and examples of forecast trajectories.

5.2.1 Numerical results

This section compares skill score and ramp NMAE across reference forecasts, numerical models, time-series foundation models, and multimodal forecasters.

Reference forecasts

Table 5.3 reports three rule-based reference forecasts under the common UK protocol. They show how much skill follows from carrying forward recent power or exploiting its recurring daily pattern.

Method	Skill (%)	Ramp NMAE
Persistence	1.41	0.2550
Seasonal naive	10.68	0.2056
Hourly climatology	23.37	0.1665

Table 5.3: Rule-based reference forecasts on the primary cross-plant test set. Higher skill and lower ramp NMAE are better.

The scores rise from persistence through seasonal naive to hourly climatology, showing the value of recurring daily structure. Hourly climatology also lowers ramp error. Its fixed per-hour mean cannot adapt to local conditions that change from one forecast day to the next.

Tabular models

The models in Table 5.4 use engineered lagged numerical features. They test how far learned tabular representations can improve on the rule-based references without modeling the input as a sequence.

Method	Skill (%)	Ramp NMAE
LightGBM	38.54	0.1672
TabPFN	40.63	0.1631

Table 5.4: Tabular models on the primary cross-plant test set. Higher skill and lower ramp NMAE are better.

Supervised sequence models

Table 5.5 lists numerical sequence models trained on the benchmark training plants.

Method	Skill (%)	Ramp NMAE
iTransformer	52.57	0.1445
PatchTST	45.81	0.1543
TFT	42.64	0.1605
MLP baseline	42.19	0.1624
DLinear	32.31	0.1746

Table 5.5: Supervised numerical sequence baselines. TFT denotes Temporal Fusion Transformer.

Time-series foundation models zero-shot

Table 5.6 lists time-series foundation models evaluated without task-specific fitting.

Method	Skill (%)	Ramp NMAE
Chronos-2 zero-shot	47.37	0.1544
TiRex zero-shot	28.73	0.1826
TimesFM zero-shot	27.08	0.1902
TTM zero-shot	-8.07	0.2922

Table 5.6: Protocol-aligned zero-shot time-series foundation models.

Chronos-2 has the strongest zero-shot result in this table, with skill 47.37 and ramp NMAE 0.1544. TTM's negative zero-shot skill shows that pretraining alone does not guarantee transfer to this plant-disjoint task.

Adapted time-series foundation models

Table 5.7 compares models that adapt a pretrained time-series foundation model to the training plants, either by fine-tuning or with a covariate adapter.

Method	Skill (%)	Ramp NMAE
Chronos-2 fine-tuned	52.30	0.1506
CoRA	37.98	0.1624
TTM fine-tuned	36.01	0.1716

Table 5.7: Protocol-aligned time-series foundation-model adaptation results.

The fine-tuned Chronos-2 control has skill 52.30 and ramp NMAE 0.1506. Its score is 4.93 points above the zero-shot Chronos-2 result, and its ramp NMAE is lower by 0.0038. By comparison, iTransformer has similar skill but lower ramp NMAE.

Retrieval methods

Table 5.8 reports results for TS-RAG [28] and Cross-RAG [17]. Both methods retrieve related historical power sequences from an external store to condition the forecast on past similar production regimes.

Both retrieval methods achieve skill scores near 47.7%, placing them slightly above zero-shot Chronos-2 (47.37%) and above supervised sequence baselines such as PatchTST (45.81%) and TFT (42.64%). TS-RAG reaches slightly higher overall skill at 47.79%, while Cross-RAG obtains a lower ramp NMAE of 0.1545, matching the ramp error of zero-shot Chronos-2 (0.1544). Both methods score below fine-tuned Chronos-2 (52.30%) and iTransformer (52.57%), showing that retrieving similar historical trajectories provides useful reference context across plants without matching the accuracy of direct parameter adaptation.

Method	Skill (%)	Ramp NMAE
TS-RAG	47.79	0.1568
Cross-RAG	47.68	0.1545

Table 5.8: Retrieval-augmented forecasting models on the cross-plant test set.

Multimodal baselines

Table 5.9 reports benchmark results for MMTSFM and fine-tuned Aurora.

Method	Skill (%)	Ramp NMAE
Spatial-token	55.10	0.1440
Multi-anchor	54.09	0.1463
Pooled-image	52.58	0.1487
Aurora fine-tuned [39]	40.52	0.1555

Table 5.9: Multimodal benchmark results on the primary cross-plant test set. Higher skill and lower ramp NMAE are better.

The matched vision-off controls in Section 5.3 test whether the fitted models depend on their visual pathways.

Spatial-token fusion has the highest skill score among the MMTSFM arms. Within the ladder, its ramp NMAE is 0.1440, compared with 0.1487 for pooled-image fusion and 0.1506 for the numerical control. Fine-tuned Aurora scores 40.52% skill with 0.1555 ramp NMAE, providing a comparison from an atmospheric foundation model.

Full payload feasibility result

Full payload has one completed seed and a smaller eligible target set, with 163,559 scored target steps compared with 165,295 for the single-anchor arm. Because it requires cached imagery at all five daily anchors, origins missing a required entry are excluded. This smaller scored set can change the average error independently of the model. Its recorded score is therefore descriptive pending additional seeds and scoring both interfaces on their shared eligible windows.

Method	Skill (%)	Ramp NMAE
Full payload	54.49	0.1458

Table 5.10: Full payload feasibility result on its smaller eligible target set.

Alternative multimodal baselines

Table 5.11 reports results for five additional multimodal and vision-augmented forecasting models, spanning vision-language architectures, atmospheric foundation models, and convolutional networks.

Time-VLM achieves 54.04% skill, comparable to Multi-anchor (54.09%) and Spatial-token fusion (55.10%). It records the lowest ramp NMAE across all evaluated models at 0.1374, showing that rendering time series as visual plots allows the vision-language backbone to detect steep transitions and ramp events effectively. Solar-VLM attains 43.96% skill and 0.1514 ramp NMAE, outperforming supervised models such as TFT (42.64%) while trailing zero-shot Chronos-2 (47.37%).

Atmospheric and convolutional models record lower overall skill. Zero-shot Aurora obtains 23.24% skill and 0.2051 ramp NMAE. When fine-tuned on plant data, as shown in Table 5.9, its skill rises to 40.52% and ramp error drops to 0.1555, showing the importance of task adaptation for general weather representations. SUNSET achieves 21.62% skill and 0.1757 ramp NMAE; its convolutional image pipeline improves upon simple persistence and climatology references, yet falls short of modern transformer architectures on held-out plants. UniCast scores 12.11% skill and 0.2302 ramp NMAE, reflecting limited zero-shot transfer without installation-specific adaptation.

Method	Skill (%)	Ramp NMAE
Time-VLM [44]	54.04	0.1374
Solar-VLM [9]	43.96	0.1514
Aurora zero-shot [39]	23.24	0.2051
SUNSET [38]	21.62	0.1757
UniCast [33]	12.11	0.2302

Table 5.11: Alternative multimodal forecasting baselines on the cross-plant test set.

5.2.2 Zero-shot transfer from the UK to the US fleet

The preceding tables report the primary UK plant-disjoint benchmark. Table 5.12 reports the cross-region experiment: checkpoints trained on UK plants are evaluated on eight US plants with their parameters fixed. Plants 1283 and 51 are excluded by the dataset quality flags. All remaining US plants are unseen during model fitting. Power is resampled to 30-minute cadence, giving a twelve-step, six-hour horizon from fourteen days of local history. Each run scores the same 41,914 valid daylight targets. The four main arms use matched UK training seeds 42, 43, and 44, with plants weighted equally within each run and run scores then averaged across seeds.

Arm	Skill (%)	NMAE	Ramp NMAE
Numerical	8.90	0.1719	0.1864
Pooled-image	−3.03	0.1933	0.2008
Spatial-token	8.85	0.1745	0.1692
Multi-anchor	10.46	0.1657	0.1702
Full payload	3.46	0.1771	0.1786

Table 5.12: Inference-only UK-to-US transfer on eight plants and 41,914 valid daylight targets. Main-arm entries are three-seed means. Full payload has seed 42 only. Skill uses the US Smart Persistence reference.

Multi-anchor achieves the highest mean skill score at 10.46 percent and the lowest overall NMAE at 0.1657, improving on the numerical control at 8.90 percent skill and 0.1719 NMAE. It exceeds spatial-token fusion in skill across all three paired seeds. On ramp intervals, both structured visual architectures improve over the numerical baseline. Spatial-token fusion records the lowest ramp NMAE at 0.1692, followed closely by multi-anchor at 0.1702, compared with 0.1864 for the numerical control. The localized cloud information in spatial tokens and multi-temporal anchors helps the network anticipate steep power swings under unfamiliar US atmospheric conditions.

In contrast, pooled-image fusion yields negative skill at −3.03 percent and higher errors across the board, with an NMAE of 0.1933 and a ramp NMAE of 0.2008. Compressing the visual input into a single pooled token discards spatial gradients and makes the model sensitive to differences in sensor calibration between SEVIRI and GOES-16. By retaining spatial layout, spatial-token fusion matches the numerical control on overall skill (8.85 percent versus 8.90 percent) while achieving lower ramp error. Multi-anchor fusion gains further by distributing tokens across five observation times, providing temporal progression across the context window.

The US full payload configuration uses 70 selected tokens at each of five daily anchors, fewer than the 98 tokens per anchor in the UK experiment. Its single-seed run attains 3.46 percent skill and 0.1786 ramp NMAE, tracking ramps better than the numerical baseline (0.1864) and pooled-image fusion (0.2008). Across the main arms, calibration reflects the regional transfer. Mean nominal 80 percent coverage ranges from 0.409 for spatial-token fusion to 0.523 for pooled-image fusion, while multi-anchor reaches 0.459, lower than the UK coverage levels. These eight plants provide evidence across different climatic zones, though they represent a quality-filtered subset observed over a single evaluation period.

5.2.3 Visual inspection of the forecasts

The result tables show overall performance, but they do not show where errors occur. Figure 5.1 compares observed and predicted normalized power for fine-tuned Chronos-2, iTransformer, Light-

GBM, and fine-tuned Aurora. In each panel, observed power is on the horizontal axis and predicted power is on the vertical axis. Points on the dashed diagonal are accurate predictions. Points above it are overpredictions, and points below it are underpredictions. Colour shows where observations are most concentrated. The plots pool valid daylight observations from the 14 held-out plants. Benchmark scores weight plants equally, so the densest parts of a plot do not necessarily have the greatest influence on its score.

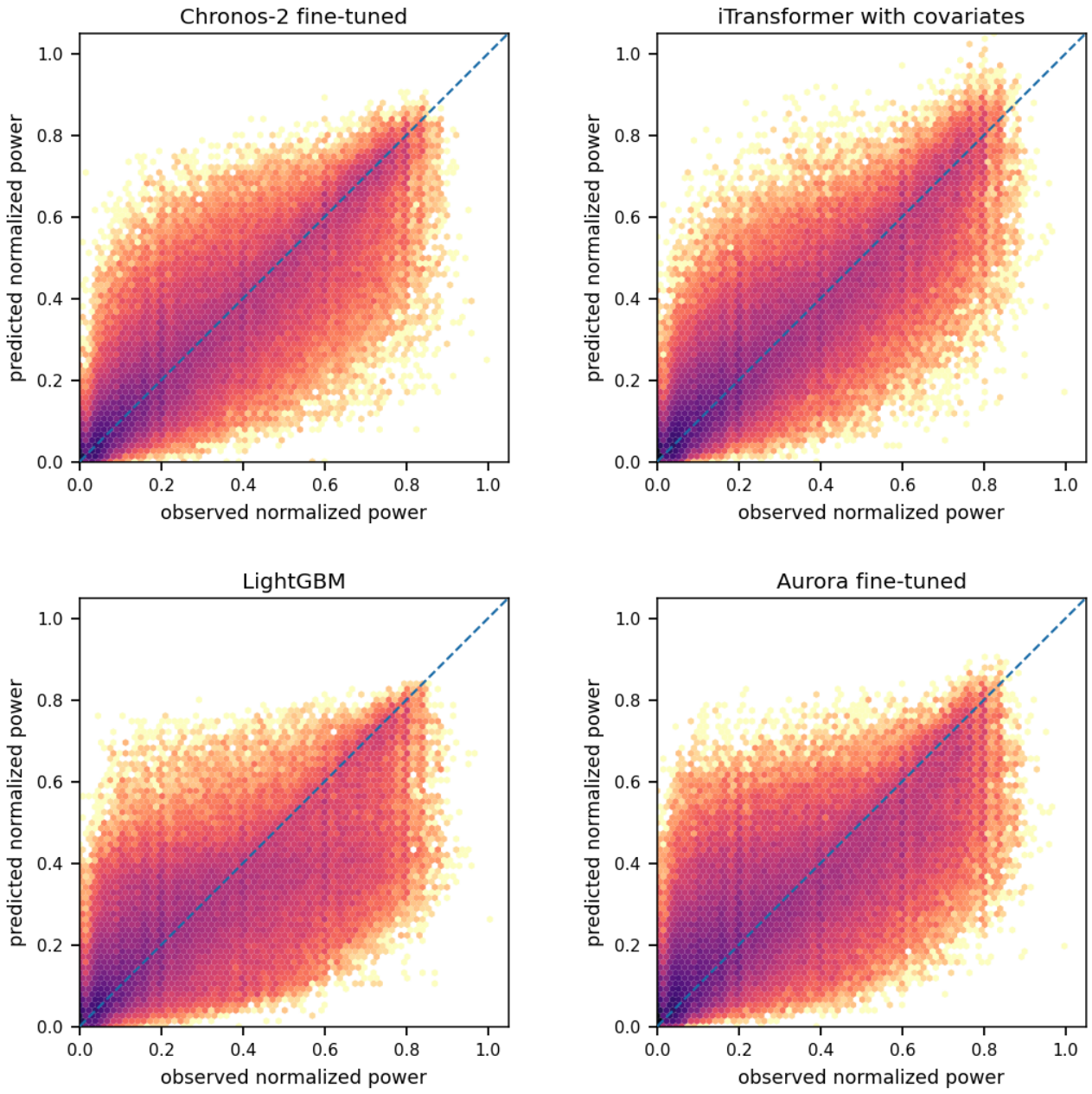


Figure 5.1: Predicted versus observed normalized power for four benchmark models. Colour shows observation density on a logarithmic scale. The dashed line marks perfect predictions.

In all four panels, the densest observations follow the diagonal at low and moderate output. Predictions spread more widely as observed power rises. High observed values are often predicted too low. This is clearest for LightGBM, whose predictions level off before the observed range ends. iTransformer reaches higher predicted values, although its scatter remains broad. Fine-tuned Chronos-2 and Aurora also show underprediction near the upper end. Each panel contains positive predictions when observed output is close to zero. These patterns show where the models make errors that their aggregate scores cannot locate.

Figure 5.2 compares the three visual fusion arms on the same daylight observations and axes. It

shows how the distribution of prediction errors changes across the fusion interfaces.

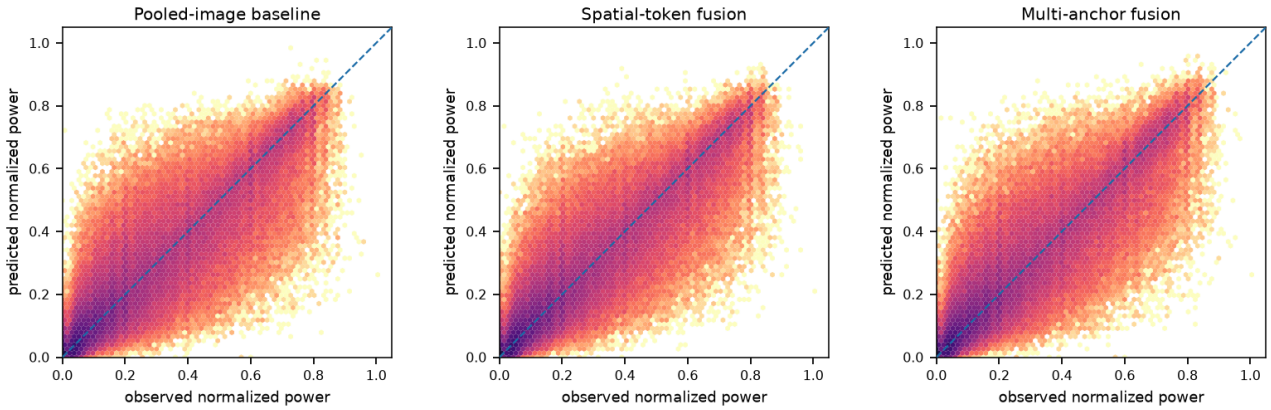


Figure 5.2: Predicted versus observed normalized power for the three visual fusion arms, shown with identical axes and daylight masking.

All three arms follow the overall rise in observed power. Their predictions spread apart at higher output. Pooled-image fusion has a broad cloud below the diagonal when observed power is high. Spatial-token fusion places more predictions near the diagonal in this range, although it still under-predicts some large values. The multi-anchor panel is more dispersed than the spatial-token panel and shows no comparable tightening around the diagonal. Each arm also makes some positive predictions when observed power is close to zero.

Figure 5.3 follows three ramp windows from an evaluation plant. Each row shows the same window in both columns. The left column compares fine-tuned Chronos-2, iTransformer, LightGBM, and fine-tuned Aurora. The right compares pooled-image, spatial-token, and multi-anchor. The pale grey line at negative hours is observed power from the previous day. The break in the horizontal axis skips the intervening night. The dotted line at hour zero marks the forecast origin. To its right, the dark grey line is the observed outcome, and the coloured dashed lines are six-hour forecasts. Their direction, timing, and height show which changes each model follows or misses.

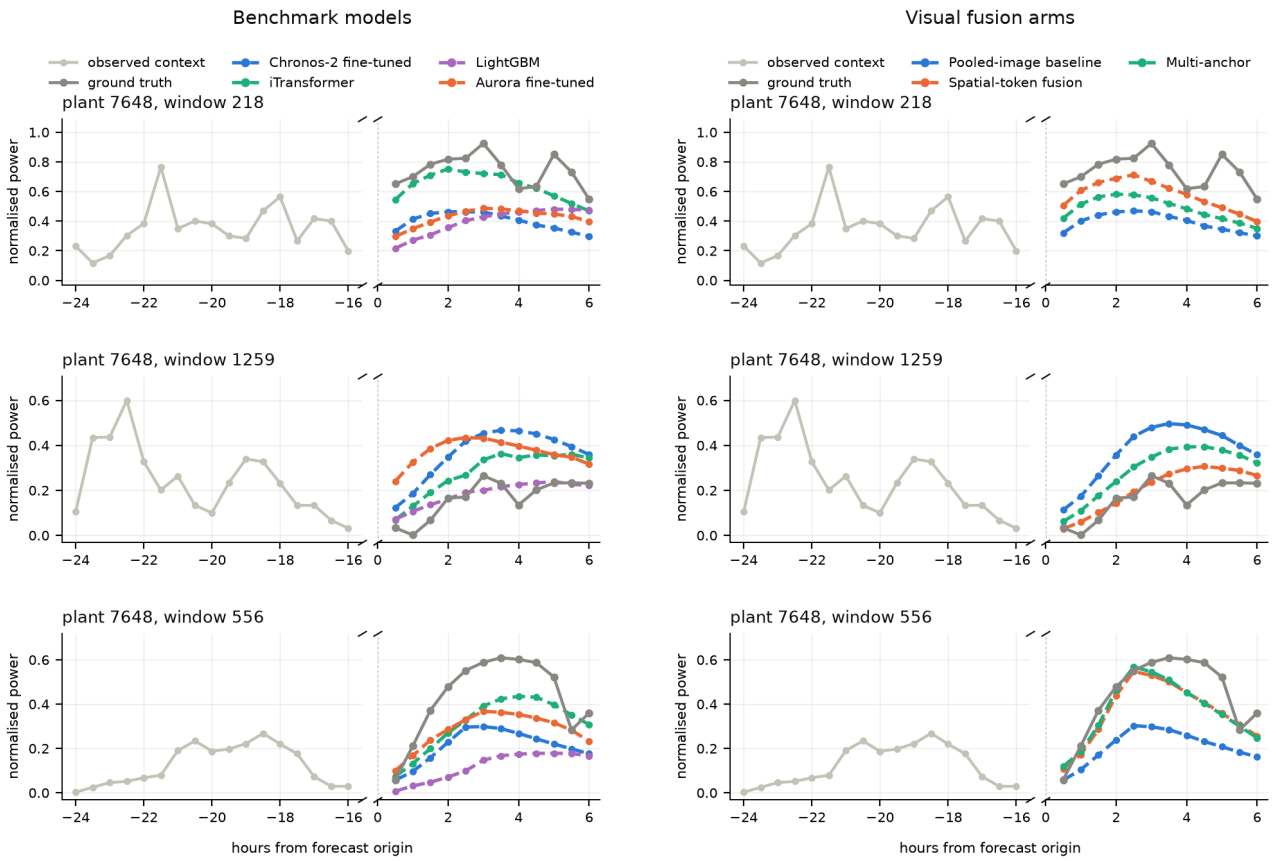


Figure 5.3: Forecast trajectories for three ramp windows from plant 7648. The same windows appear in both columns.

The three windows expose different errors. In window 218, the observed output rises above most forecasts. iTransformer comes closest to the peak among the benchmark models. Spatial-token fusion is closest among the visual arms. In window 1259, output remains low after a brief rise. LightGBM and spatial-token fusion follow that lower level more closely. Several other forecasts stay too high. In window 556, spatial-token and multi-anchor capture the sharp rise but still miss the peak. The pooled-image baseline and fine-tuned Chronos-2 produce flatter, lower forecasts for that window.

Figure 5.4 shows a pronounced upward ramp within one six-hour forecast. It uses the same broken time axis as Figure 5.3. The pale grey segment shows observations from the previous day. The dark grey line on the right shows the outcome after the forecast origin. The coloured dashed lines show the model forecasts.

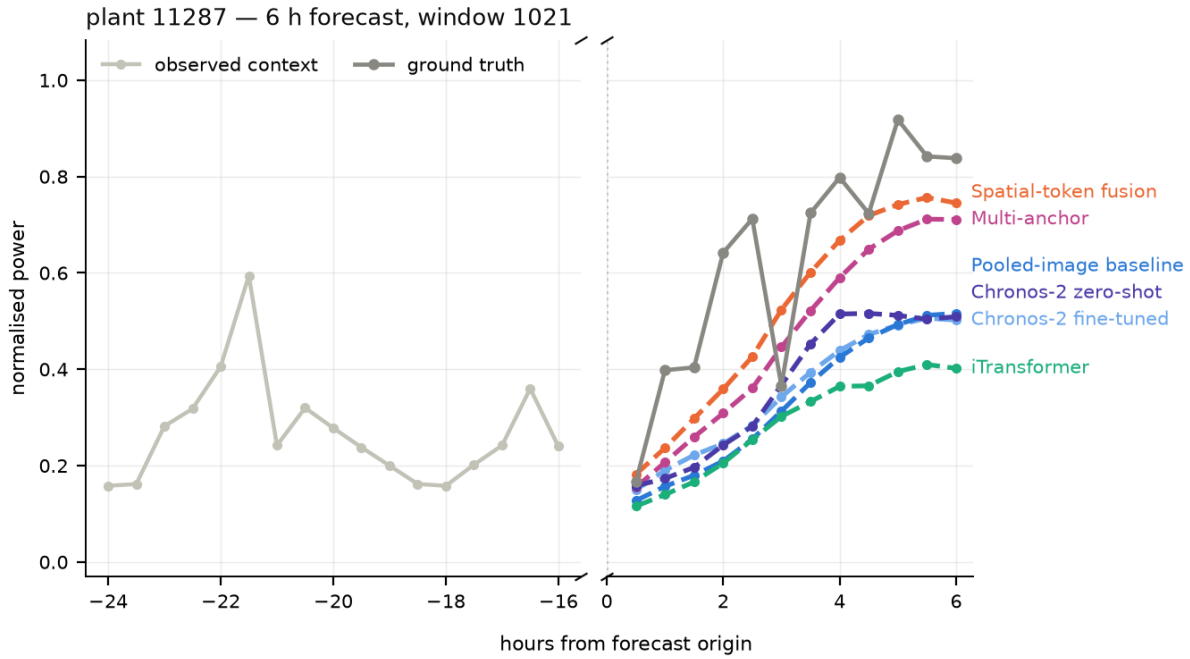


Figure 5.4: Six-hour forecast across an upward ramp at plant 11287, window 1021.

Observed power rises sharply after the forecast origin, with brief peaks around hours 2.5 and 5. Spatial-token fusion follows the overall climb but smooths those peaks; multi-anchor also rises, though at a lower level. The numerical control and pooled-image baseline flatten earlier. This trace shows how the forecasts differ during one ramp event.

The ramp comparisons above evaluate plants within the UK fleet where the models were trained. Figure 5.5 shifts focus to the cross-region transfer, displaying individual six-hour forecasts on unseen US plants. Each panel presents the daylight window with the greatest observed change between the first and last target step for the first three plants in the test partition.

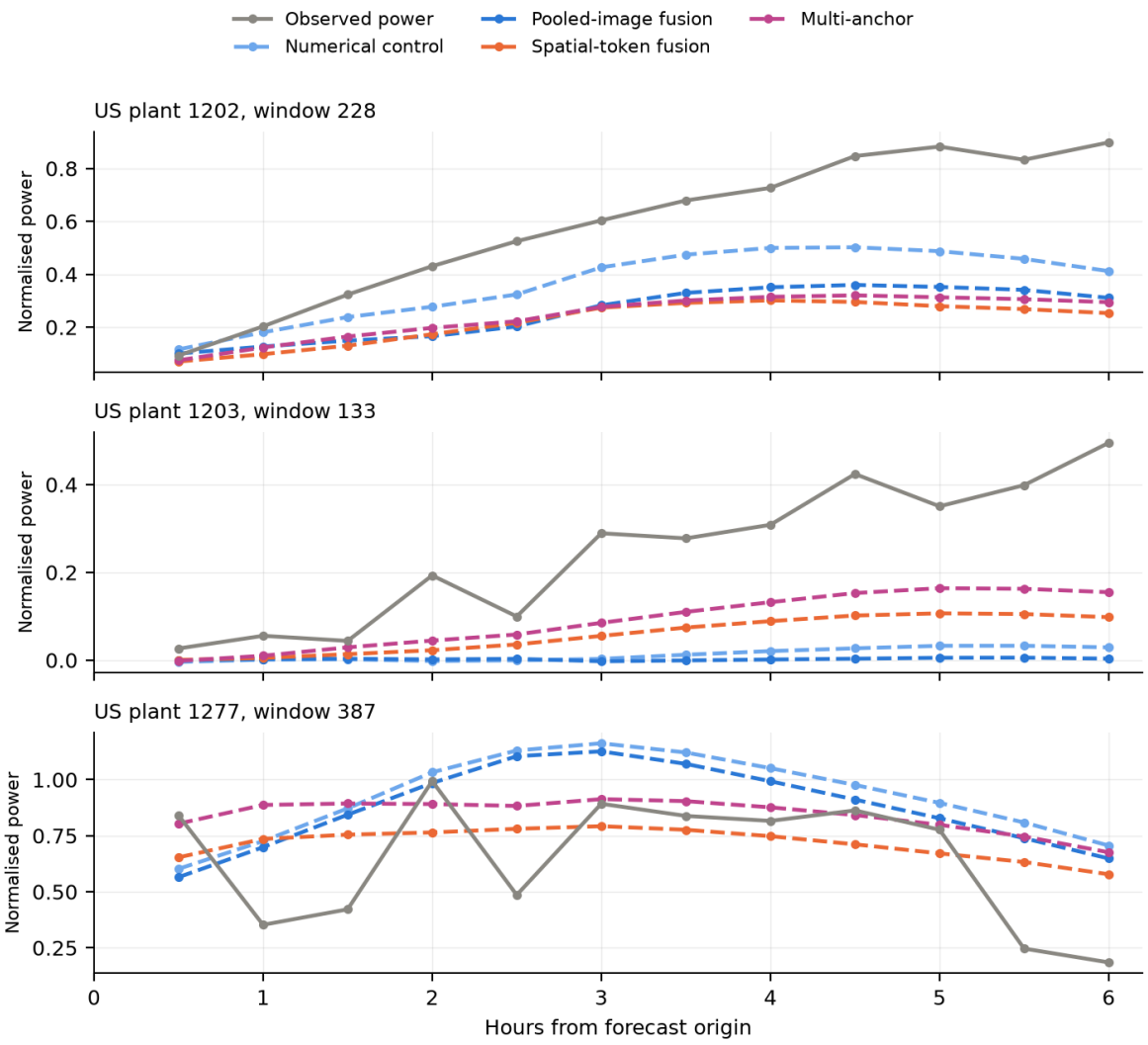


Figure 5.5: Six-hour US forecasts from the seed-42 prediction exports across plants 1202, 1203, and 1277. Each panel displays the fully valid daylight window with the largest absolute change between its first and last observed target. Dashed lines denote model forecasts and the solid line represents observed normalized power.

At plants 1202 and 1203, all four models underpredict the observed morning climb. The numerical control tracks more of the initial rise at plant 1202, while multi-anchor reaches closest to the peak at plant 1203. At plant 1277, the forecasts stay smooth through midday fluctuations and fail to capture the steep late-afternoon drop. These individual events illustrate how even models with competitive aggregate scores can struggle on localized, high-amplitude transitions when deployed across distinct geographic regions.

5.3 Results and ablations

The benchmark in Section 5.2 establishes the forecasting performance of the proposed models. The ablations examine what supports that performance: whether the fitted models use their visual inputs, which properties of those inputs matter, and how the fusion interface affects transfer to unseen plants. Unless stated otherwise, results use the UK cross-plant test, the self-attention temporal mixer, and seeds 42, 43, and 44. Interpretation follows the size and direction of the observed changes, their consistency across matched seeds, and the variables changed by each intervention. Small changes are informative when they narrow the explanation of a larger gain; losses reveal where an additional design choice fails to improve the tested configuration.

The main result is that spatial-token fusion improves point forecasts across most held-out plants and depends on timely, site-matched imagery. The controls refine this finding. Fractional token

placement has little observed effect, mean pooling improves skill while retaining spatially distinct cells, and freezing the numerical encoder largely preserves point performance at a calibration cost. The multi-anchor experiments extend the same visual interface to older observations and expose a trade-off between temporal coverage and the payload available at each anchor. Together, these comparisons provide an empirical account of the fusion design, with the scope of each conclusion tied to the intervention that produced it.

5.3.1 Interface and content controls

Tables 5.13, 5.14, and 5.15 list the controls and interventions. The following paragraphs report their results and clarify which interpretation each comparison supports. The main arms are defined in Chapter 4. Controls preserve the plant split, seed-matched numerical warm start, and scoring rules unless a difference is stated.

Control	Intervention
Anchor placement	Tokens at one shared position
Paired novelty	Alternative multi-anchor selector

Table 5.13: Fusion-interface controls. Outcomes are reported in the paragraphs that follow.

The anchor-placement check asks how much the forecast depends on the precise positions assigned to visual tokens. Moving the same 98 tokens to one shared integer position changes mean skill by -0.02 percentage points and ramp NMAE by $+0.0004$ across three paired seeds. Both differences change direction across seeds. The fitted interface therefore tolerates this positional simplification with little observed change in point performance. This narrows the explanation of the spatial-token gain: fractional placement is not supported as its main source. The check preserves the visual payload and tests one placement change, so its conclusion concerns positional precision within this interface.

Under paired-novelty selection, which retains paired endpoints of moving trajectories instead of individual dynamic cells, multi-anchor skill reaches 54.16 percent, a 0.07 percentage point increase over the default selector. Because changing the trajectory selection rule leaves the gap to spatial-token fusion (55.10 percent) largely unchanged, the multi-anchor shortfall is primarily associated with the reduced per-anchor token budget (20 tokens versus 98) and the dilution of visual capacity across multiple days, rather than the specific token ranking heuristic.

Control	Intervention
Stale images	Clip shifted six hours earlier
Wrong plant	Another plant's contemporaneous clip
Random tokens	Random selection; 98-token budget

Table 5.14: Content-grounding ablations.

Image timing produces a much larger response. Shifting the clip six hours into the past raises ramp NMAE by between 0.0156 and 0.0182 across three seeds and lowers skill by about 13.5 percentage points. The stale clip performs worse than disabling vision. Supplying an image can therefore hurt the fitted model when that image describes an earlier state. The consistent degradation shows that the model's use of vision depends on temporal relevance and supports matching the clip to the forecast origin.

Spatial correspondence matters as well. Substituting another plant's contemporaneous clip raises ramp NMAE by between 0.0163 and 0.0189 across three seeds and lowers skill from about 55 to 37 percent. Every seed degrades. Contemporaneous imagery from elsewhere cannot substitute for the site's own clip in the fitted model. Together, the timing and wrong-plant controls support content grounding: predictions respond to whether the supplied visual evidence matches the place and time being forecast. These substitutions identify the importance of that correspondence without identifying the individual visual features responsible.

At the fixed 98-token budget, random selection raises ramp NMAE by 0.0039 and lowers skill by 1.52 percentage points across two paired seeds. Vision-off scores stay unchanged. Holding the count

fixed makes the selected payload relevant to the result: equal-length visual inputs do not give equal performance. Novelty-selected tokens better serve the model trained with that selector. Because the intervention changes selection only at inference, the observed loss includes any mismatch with the training distribution; it supports retaining the trained selection rule without establishing its superiority under separate retraining.

Control	Intervention
Latent order	Permute temporal latents after encoding
Token budget	Vary token count without retraining
Spatial grid	Mean-pooled 4×4 vs 7×7
Mean pooling	Replace pixel-shuffle concatenation
Backbone freeze	Freeze adapted encoder blocks during visual training

Table 5.15: Mechanism, representation, and backbone controls.

Permuting V-JEPA's cached temporal latents after visual encoding changes ramp NMAE by no more than 0.00030 in any seed. The small response localizes the observed sensitivity: the downstream forecast depends little on this ordering of the already encoded latents. V-JEPA may have mixed motion information within each latent before the permutation, so this result concerns the fusion stage's access to latent order. Frame order before encoding remains unchanged. The check helps distinguish reliance on the encoded visual content from reliance on its explicit downstream temporal arrangement.

The token-budget sweep shows that increasing the available visual payload does not automatically improve a fitted model. Evaluating without pruning to the 98-token budget raises ramp NMAE by 0.0102 across three seeds. Keep settings of 0.3 and 0.7 raise it by 0.0022 and 0.0069. Error is lowest near the count used during training. These results support preserving the trained token budget at inference and show that the interface is sensitive to sequence length. Since the alternative settings are evaluated without retraining, 98 tokens is the best tested inference setting for these weights, with no claim of a universal budget optimum.

The spatial-resolution probe finds higher ramp R^2 with a mean-pooled 4×4 grid than with 7×7 at 30, 60, and 120 minutes. A coarser representation can therefore retain useful predictive information and outperform the finer grid under this probe. This result weakens the assumption that additional spatial cells necessarily improve the representation and motivates considering compression as a design variable. Both probe grids use mean pooling, whereas the deployed arm uses pixel-shuffle concatenation, so the comparison ranks the probe representations and leaves the corresponding end-to-end grid choice unresolved.

The end-to-end pooling check provides a more direct test of sub-patch detail. With the number and positions of selected cells fixed, replacing pixel-shuffle concatenation with mean pooling raises three-seed skill from 55.10 to 55.64 percent. The skill gain appears in every paired seed. Ramp NMAE falls from 0.1440 to 0.1435, and 80 percent coverage rises from 0.731 to 0.753. The larger change in skill and smaller change in ramp error show that the representation choice affects the reported metrics to different degrees. Spatially distinct cells remain available under mean pooling, so the result supports a compressed spatial payload and revises the motivation for preserving sub-patch detail. The original spatial-token gain does not require that detail in this comparison.

Freezing the adapted numerical encoder's blocks during visual-arm training yields three-seed skill of 55.29 percent and ramp NMAE of 0.1441, close to spatial-token fusion's 55.10 percent and 0.1440. Point forecasting therefore remains effective when visual training leaves those blocks fixed. The uncertainty measures reveal a separate trade-off: coverage falls from 0.731 to 0.710, while quantile ECE rises from 0.0359 to 0.0444. Allowing encoder adaptation has more apparent value for calibration than for the point scores in this check. The intervention changes the trainability of the blocks together and does not attribute the calibration difference to a particular block.

5.3.2 Fusion results and vision-off controls

Table 5.16 compares point forecasts across the fusion configurations and reports the change in error when vision is disabled in the same fitted weights. The numerical baseline provides the image-free

reference. Values for that baseline and the pooled-image, spatial-token, and multi-anchor arms summarize three seeds. Full payload has one completed seed on a smaller eligible target set.

Arm	Skill (%)	Ramp NMAE	ΔNMAE	ΔRamp NMAE
Numerical	52.30	0.1506	—	—
Pooled-image	52.58	0.1487	0.0008	0.0000
Spatial-token	55.10	0.1440	0.0047	0.0063
Multi-anchor	54.09	0.1463	0.0031	0.0045
Full payload	54.49	0.1458	0.0040	0.0041

Table 5.16: Fusion results on held-out UK plants. Skill is percent. Lower ramp NMAE is better. The Δ columns are error increases after vision is disabled in the same trained weights. Positive values indicate degradation.

Spatial-token fusion has the highest skill, 55.10 percent, and the lowest ramp NMAE, 0.1440, among the main fusion configurations in Table 5.16. Relative to pooled-image fusion, skill increases by 2.52 percentage points and ramp NMAE falls by 0.0047. Pooled-image fusion produces a smaller change from the numerical baseline: skill rises from 52.30 to 52.58 percent and ramp NMAE falls from 0.1506 to 0.1487. The comparison favors the spatial-token configuration for point forecasting on unseen UK plants. Since fusion path and training recipe differ between arms, the controls are needed to determine which parts of this advantage involve the visual input.

The vision-off columns test image use within each fitted model by holding its weights fixed. Disabling vision in pooled-image fusion raises all-daylight NMAE by 0.0008, while the mean ramp change rounds to 0.0000 and has no consistent direction across the three seeds. Its modest point-score improvement thus gives limited evidence of a visual contribution to ramp forecasting. In spatial-token fusion, the same intervention raises all-daylight NMAE by 0.0047 and ramp NMAE by 0.0063, with degradation consistent across seeds. This connects the better point forecasts to reliance on the visual pathway. The stale-image and wrong-plant controls further show that this reliance is sensitive to the supplied content. The larger absolute ramp change should be read alongside the different error baselines for ramp and all-daylight targets; it alone does not establish a concentration of visual benefit on ramps.

Multi-anchor fusion scores 54.09 percent skill and 0.1463 ramp NMAE. It improves on the numerical baseline, while remaining below spatial-token fusion on both point measures. Its mean vision-off increases of 0.0031 NMAE and 0.0045 ramp NMAE show that the model uses images within the distributed layout. Extending visual context across daily anchors therefore retains a visual contribution, but does not improve the UK point scores over the recent single-anchor configuration. This comparison jointly changes anchor count, daily spacing, latent depth, and per-anchor token budget. It measures the consequence of that combined design and cannot assign the gap to older observations alone.

The full payload feasibility run retains four latent slices and 98 tokens at each of five anchors. It reaches 54.49 percent skill and 0.1458 ramp NMAE, with vision-off increases of 0.0040 and 0.0041. This run demonstrates that a larger multi-anchor payload can be trained and used by the forecast. Its 533-position sequence is substantially longer than the 141-position single-anchor interface. One completed seed and a smaller eligible target set make its point-score ranking descriptive. The experiment establishes feasibility and visual reliance for this larger interface; the recorded results do not establish that restoring payload closes the gap to spatial-token fusion.

5.3.3 Variation across held-out plants

Table 5.17 separates the fourteen held-out UK plants to show the breadth and limits of the aggregate gain. The first four arm columns are three-seed means. Full payload multi-anchor values come from seed 42 on a smaller set of scored targets.

Spatial-token fusion exceeds the numerical baseline in the three-seed mean at thirteen of fourteen plants. Plant 7648 illustrates the difference between the visual interfaces: pooled-image fusion remains close to the numerical control, while spatial-token fusion improves skill substantially. The aggregate result therefore reflects improvements across much of the held-out fleet, including sites where pooled fusion contributes little. Plant 12642 is the exception and marks a limit to the consistency of transfer.

Plant	Numerical	Pooled	Spatial	Multi-anchor	Full payload
3333	47.12	47.64	51.67	50.89	51.79
6648	50.03	49.82	52.28	50.47	50.84
7648	42.88	43.22	49.06	47.72	48.50
7836	44.22	44.89	49.37	49.30	49.70
10793	62.87	63.43	65.93	65.09	65.35
11176	36.29	36.39	39.01	36.12	38.54
11287	48.89	49.34	49.80	49.44	49.36
12642	44.03	44.26	43.76	42.62	42.48
13388	63.56	64.00	64.87	64.34	64.52
13817	58.61	58.91	61.12	60.37	61.31
18989	61.81	62.65	66.96	66.17	66.69
26854	41.31	41.07	43.11	42.01	42.04
26933	58.03	57.95	60.88	59.79	59.14
27020	53.94	53.42	54.61	53.25	53.11

Table 5.17: NRMSE-based skill score in percent for each held-out UK plant. The first four arm columns are three-seed means. Full payload is seed 42 on fewer scored targets.

The smaller plant-level differences remain descriptive because the table reports means without paired spread. The breadth of the observed gains supports cross-plant generalization within this UK fleet, while the exception preserves the distinction between an aggregate improvement and a guarantee for an individual unseen plant.

Multi-anchor trails spatial-token fusion in mean skill at every held-out plant. The aggregate gap consequently reflects a fleet-wide pattern in these means, rather than a loss driven by one site. This consistency strengthens the empirical preference for the recent single-anchor configuration under the tested UK protocol. The cause remains shared among the changes to daily spacing, temporal depth, and token allocation. Full payload scores above spatial-token fusion at some plants and below it at others. With one seed and a different set of forecast origins, those entries document site variation in the feasibility run without establishing a plant-level recovery from restoring payload.

Each entry in Table 5.17 reports the skill score evaluated specifically on that installation's target subset. The unweighted arithmetic mean of these plant-level skill scores is 51.04% for the numerical baseline and 53.75% for spatial-token fusion. These averages differ from the headline benchmark scores in Table 5.16 (52.30% and 55.10%), which are computed from fleet-wide pooled NRMSE before calculating percentage skill against Smart Persistence. Evaluating skill at the fleet aggregate versus averaging per-plant skill ratios yields different numerical quantities because the skill score formula is a non-linear ratio. Macro-averaging scores across plants ensures equal contribution from each installation regardless of valid observation count. While spatial-token fusion improves accuracy across 13 of 14 installations, residual variations across sites reflect differences in local topography, cloud regime frequency, and sensor reliability.

5.3.4 Skill across the six-hour horizon

Table 5.18 reports NRMSE-based skill against Smart Persistence at each half-hour lead on valid daylight targets. For each arm, plant NRMSE is macro-averaged at each lead before comparison with the corresponding Smart Persistence value. The first four arms are then averaged across three seeds. These lead-specific values are computed from the saved in-process prediction exports. Their aggregate reconstruction matches the recorded overall skill scores. The full payload column has one seed and fewer eligible targets.

Every arm's skill rises toward three hours and then declines. Since skill compares each model with Smart Persistence at the same lead, the mid-horizon peak identifies where the fitted models gain the most ground relative to that reference. In particular, between 2.5 hours and 3.0 hours, skill scores exhibit an abrupt step increase of approximately six to seven percentage points across all arms. This jump reflects the degradation of the Smart Persistence denominator: beyond 2.5 hours, carrying forward the initial clearness index under evolving solar elevation incurs rapidly growing NRMSE,

Lead	Numerical	Pooled	Spatial	Multi-anchor	Full payload
0.5 h	43.25	43.74	43.79	42.11	45.19
1.0 h	46.33	47.08	49.07	47.48	48.96
1.5 h	49.14	50.04	53.07	51.51	52.41
2.0 h	50.20	51.38	54.43	53.27	53.88
2.5 h	52.04	53.00	56.12	55.23	55.59
3.0 h	59.18	59.33	62.16	61.49	61.85
3.5 h	58.29	58.25	60.99	60.45	60.50
4.0 h	57.06	56.91	59.60	59.01	58.74
4.5 h	55.34	55.22	57.62	56.94	56.78
5.0 h	53.47	53.14	55.80	54.90	54.84
5.5 h	50.21	49.90	52.53	51.45	51.25
6.0 h	47.98	47.85	50.16	49.19	49.06

Table 5.18: NRMSE-based skill score in percent by lead time on the fourteen held-out UK plants. The first four arm columns are three-seed means. Full payload is seed 42 on fewer scored targets.

which inflates the relative percentage skill of all learned forecasters at lead step six. Absolute error need not fall as skill rises. This shared shape provides the context for interpreting the differences between fusion arms across the horizon.

Spatial-token fusion is nearly tied with pooled-image fusion at the first half-hour step and separates from both pooled-image fusion and the numerical baseline from one hour onward. Its advantage persists at six hours. The better aggregate score thus extends across much of the forecasting horizon, including leads well beyond the most recent image. Pooled-image fusion stays close to the numerical control throughout. This pattern supports the forecasting value of the spatial-token configuration across lead times. The same-weights vision-off results establish visual reliance in aggregate; the lead-time table compares separately trained configurations and describes where their forecasts differ.

Multi-anchor fusion falls below spatial-token fusion at every lead, with the smallest gap around the middle of the horizon. Extending visual history therefore produces no lead at which the reduced-payload arm overtakes the single-anchor arm in the recorded means. The absence of paired uncertainty estimates limits the precision of this comparison, and the combined design changes limit its mechanistic interpretation. Full payload scores above spatial-token fusion at the first half-hour step and below it thereafter. This shape is an observation from one seed on a smaller eligible set, so it remains a descriptive feature of the feasibility run.

5.3.5 Seed-level stability

Table 5.19 reports three matched seeds for the numerical, pooled-image, spatial-token, and multi-anchor arms. Full payload has one completed seed on a smaller eligible target set.

The matched seeds provide a check on whether the configuration rankings depend on a particular numerical warm start. Visual arms with the same seed begin from the same adapted numerical checkpoint, although their subsequent training trajectories differ. Point metrics show limited variation within each arm across the three recorded runs.

Spatial-token fusion improves both skill and ramp NMAE over the numerical baseline in every paired seed. Multi-anchor has lower skill and higher ramp NMAE than spatial-token fusion in all three pairs. These rankings persist across the recorded initializations and agree with the per-plant and lead-time patterns. Three runs give a bounded view of training variability, but the agreement supports the observed configuration preference. Explaining the multi-anchor loss still requires separating anchor count, temporal depth, token allocation, and daily spacing.

The probabilistic measures assess different properties. CRPS summarizes quantile forecast error across the predicted distribution. Coverage counts how often observations fall between the predicted 10th and 90th percentiles; that interval should contain about 80 percent of observations. Quantile ECE measures departures between each quantile's stated probability and its observed frequency.

Calibration varies more clearly across the recorded configurations. Spatial-token coverage is below the nominal 80 percent in every seed. Its quantile ECE exceeds the numerical baseline's in every seed.

Arm	Seed	Skill (%)	Ramp NMAE	Coverage	ECE
Numerical	42	52.75	0.1494	0.770	0.0277
Numerical	43	51.97	0.1514	0.766	0.0317
Numerical	44	52.17	0.1509	0.779	0.0296
Pooled-image	42	52.53	0.1483	0.754	0.0324
Pooled-image	43	52.18	0.1498	0.757	0.0331
Pooled-image	44	53.03	0.1478	0.808	0.0243
Spatial-token	42	55.00	0.1433	0.731	0.0328
Spatial-token	43	54.97	0.1448	0.723	0.0416
Spatial-token	44	55.32	0.1441	0.738	0.0333
Multi-anchor	42	53.92	0.1458	0.730	0.0371
Multi-anchor	43	54.22	0.1456	0.711	0.0438
Multi-anchor	44	54.14	0.1474	0.738	0.0346
Full payload	42	54.49	0.1458	0.742	0.0355

Table 5.19: Seed-level fusion results on held-out UK plants. Skill is in percent. Lower ramp NMAE and quantile ECE are better. 80 percent coverage targets 0.80. The full payload run scores 163,559 targets versus 165,295 for the other arms.

Multi-anchor coverage is also below nominal in all three runs. Its ECE is higher than spatial-token fusion's in each matched seed. Spatial-token fusion has lower CRPS than the numerical baseline in each recorded run, with three-seed means of 0.0544 and 0.0574, respectively.

The lower CRPS shows that spatial-token fusion improves the reported distributional error as well as the point scores, while coverage and ECE identify a calibration cost. These measures capture different properties of the forecast and should inform different modeling decisions. The freezing control is relevant here: it leaves point scores close to the original arm while worsening calibration further. Interval widths and individual quantile predictions are unavailable for this analysis, so the undercoverage cannot be attributed specifically to narrow intervals, displaced intervals, or a combination of the two.

The full payload seed has higher skill and coverage than multi-anchor seed 42, with identical ramp NMAE when rounded to four decimals. It also scores fewer forecast targets. This is evidence from one larger multi-anchor run, so the apparent improvement cannot establish a repeatable payload effect. Pooled-image fusion illustrates a different form of run variation: coverage is below nominal in two seeds and slightly above it in the third, while point scores remain near the numerical baseline.

Taken together, the ablations support spatial-token fusion as the strongest tested main configuration for UK cross-plant point forecasting. Its improvement is accompanied by image reliance, sensitivity to timing and site correspondence, and gains distributed across held-out plants and forecast leads. The smaller-effect controls refine the design argument: precise fractional placement contributes little in the tested intervention, mean pooling preserves the useful spatial interface while improving skill, and encoder adaptation has a clearer benefit for calibration than for point prediction. The multi-anchor results document useful visual forecasts with a longer history, alongside a consistent UK point-score loss relative to the recent single-anchor arm. These findings establish which configurations worked and which simplifications remained effective, while keeping the causal interpretation at the level resolved by the completed experiments.

6 Discussion

Sections 5.2 and 5.3 show that spatial-token fusion improves forecasts for unseen UK plants and depends on recent, site-matched imagery. The UK-to-US evaluation extends the comparison to a new region and gives a different ordering of the visual interfaces.

6.1 Spatial information and visual reliance

Spatial-token fusion reaches 55.10 percent skill and 0.1440 ramp NMAE on the UK plant-disjoint test, compared with 52.30 percent and 0.1506 for the adapted numerical control. Both measures improve in every paired seed. Mean skill improves at thirteen of the fourteen held-out plants, and the advantage extends from one hour through the six-hour horizon. This supports a gain across much of the tested fleet and horizon, with no guarantee of improvement at every new plant.

Evaluating trained models with visual embeddings set to zero confirms that spatial-token fusion actively relies on its image inputs, incurring an NMAE increase of 0.0047 and a ramp NMAE increase of 0.0063 across all seeds. By contrast, pooled-image fusion shows negligible ramp change (0.0000), indicating minimal visual engagement during steep power transitions.

Stale clips and contemporaneous clips from another plant degrade spatial-token forecasts in every tested seed. The fitted model responds to whether imagery matches the time and place being forecast. These substitutions establish content grounding within the UK experiment without identifying the visual features responsible for the gain.

The pooling and placement controls refine the interface argument. Mean pooling retains spatially distinct cells and raises skill to 55.64 percent, with a smaller improvement in ramp NMAE. Preserving sub-patch detail through pixel-shuffle concatenation is therefore unnecessary for the gain in this comparison. Moving the same tokens to a shared integer position produces little change, so precise fractional placement also has limited measured value. The spatial interface tolerates these simplifications. Differences in training recipe between the main arms still limit attribution of their score gap to a single architectural choice.

6.2 Multi-anchor forecasting across regions

In the UK benchmark, spatial-token fusion outperforms multi-anchor fusion, reaching 55.10 percent skill compared with 54.09 percent. This gap reflects how visual capacity is allocated across time. Spatial-token fusion concentrates its entire token budget on the most recent observation at the forecast origin. Within the UK, where the fitted model operates on its native SEVIRI sensor and familiar maritime cloud regimes, high-resolution spatial features provide sharp, reliable depictions of approaching cloud boundaries. Multi-anchor fusion divides its token budget across five historical days, reducing the spatial detail retained at the forecast origin. For in-domain plants where sensor calibration and local spatial features are dependable, trading immediate spatial resolution for older historical anchors lowers overall point accuracy.

Under cross-region transfer to the US, the relative ranking of the two architectures reverses: multi-anchor fusion achieves the highest skill at 10.46 percent, while spatial-token fusion drops to 8.85 percent, matching the numerical control's 8.90 percent. This inversion stems from how the two interfaces interact with sensor domain shift. Transferring from SEVIRI to GOES-16 alters spatial resolutions, channel response functions, and viewing parallax. Spatial-token fusion relies heavily on fine spatial coordinates from a single snapshot; under unfamiliar sensor geometry, subtle coordinate distortions or spectral shifts degrade its point predictions. In contrast, multi-anchor fusion observes atmospheric states across multiple daily timestamps. Even if individual GOES-16 images contain localized sensor discrepancies, the sequence of daily anchors captures broader synoptic patterns, distinguishing persistent clear-sky periods from multi-day frontal systems. Multi-temporal anchors act as a structural

regularizer, providing recurrent diurnal context that stabilizes point forecasts against sensor noise.

Ramp error demonstrates that spatial tokens retain a distinct physical role under transfer. Spatial-token fusion achieves the lowest US ramp NMAE at 0.1692, followed closely by multi-anchor at 0.1702, and both substantially improve on the numerical control's 0.1864. Steep power ramps are driven by local cloud edges directly obscuring the solar array at the forecast origin. Even when cross-sensor domain shift reduces the point skill of single-anchor spatial features, immediate spatial gradients still provide useful geometric signals about incoming cloud boundaries that multi-day temporal averages cannot substitute.

Full payload provides a preliminary test of whether expanding the visual budget resolves this trade-off. By pairing five daily anchors with 70 to 98 tokens per anchor, it demonstrates the technical feasibility of retaining both multi-day history and fine spatial detail. Its UK run achieves 54.49 percent skill and 0.1458 ramp NMAE with positive vision-off degradation, while its US run reaches 3.46 percent skill and 0.1786 ramp NMAE. Because full payload has one completed seed and fewer eligible evaluation targets, additional seeds and matched target sets are required before drawing firm conclusions about its scalability.

6.3 Content grounding and temporal information

Content-grounding interventions demonstrate that the forecasting architecture actively extracts local physical signals from satellite imagery. Substituting contemporaneous imagery from an unrelated plant drops mean skill from 55.10 percent to 37.12 percent and raises ramp NMAE by up to 0.0189. Supplying stale imagery from six hours prior causes a comparable collapse, reducing skill by 13.5 percentage points and performing substantially worse than zeroing out visual embeddings entirely. These pronounced penalties reveal that the fusion mechanism relies on genuine spatio-temporal alignment between cloud fields and surface irradiance. When presented with geographically or temporally mismatched clips, cross-attention layers project non-existent cloud boundaries onto the target installation, actively misleading the forecast. Visual context provides an informative local observation that directly guides generation.

In contrast to the sensitivity observed under content substitutions, permuting cached temporal latents changes ramp NMAE by at most 0.00030 across all tested seeds. This indicates that downstream cross-attention layers in the fusion adapter do not depend on the sequential ordering of temporal slices. The model queries these latents primarily based on spatial and optical properties rather than directional frame order. Whether the upstream V-JEPA encoder internally encapsulates temporal motion within its tubelet representations, or whether a single frame would provide comparable information to a multi-frame clip, remains an open empirical question.

The token selection ablations reinforce this mechanistic division. At a fixed 98-token budget, replacing novelty-guided token selection with random sampling lowers skill by 1.52 percentage points and increases ramp error across all seeds. Novelty selection prioritizes dynamic cloud-clear sky boundaries and localized cloud formations over uniform clear skies or homogenous overcast layers. Furthermore, varying the token count without retraining increases ramp error, demonstrating that cross-attention weights calibrate to a specific visual token density during training. Maintaining structured novelty selection with a consistent token budget preserves the spatial boundaries that inform power ramp prediction.

6.4 Calibration

Spatial-token fusion improves point scores and lowers mean CRPS from 0.0574 to 0.0544 in the UK. Its nominal 80 percent interval nevertheless has coverage below nominal in every seed, and its quantile ECE exceeds the numerical control's in every paired run. Lower distributional error and better probability calibration are separate outcomes. Interval widths and individual quantile predictions are unavailable for this analysis, so undercoverage cannot be attributed specifically to narrow intervals, displaced intervals, or both.

Freezing the adapted numerical encoder preserves point scores close to the original spatial-token arm, while coverage falls and ECE rises. Encoder adaptation has a clearer measured benefit for

calibration than for point prediction in this check. Under US transfer, nominal interval coverage is lower than in the UK. Improving point forecasts therefore leaves calibration as a separate research problem.

6.5 Evidence from the fusion controls

Table 6.1 separates the observations from the explanations that remain open.

Question	Evidence	Unresolved
Content grounding	Stale and wrong-plant clips degrade UK forecasts	Visual features responsible
Interface	Mean pooling retains gains; placement has little effect	Individual causes of the main-arm gap
Visual use	Same-weights vision-off degradation in UK spatial-token fusion	Reliance under regional transfer
Temporal information	Cached-latent permutation has little effect	Motion information mixed before caching

Table 6.1: Evidence and remaining questions from the completed fusion controls.

Forecast accuracy reflects end-to-end model performance, whereas vision-off interventions isolate dependence on visual inputs and content substitutions test sensitivity to local scene context. Decoupling benchmark rankings from lesion controls verifies the presence of multimodal gains while clarifying the physical evidence supporting the forecasts.

6.6 Scope of cross-plant transfer

The visual arms are fitted on UK training plants and evaluated on plants excluded from fitting. Their parameters remain fixed at the new plant, where the forecast uses fourteen days of local history. This is transfer to unseen plant identities after in-domain training. The zero-shot Chronos-2 benchmark receives no task-specific fitting and measures a different use of pretraining.

Setting	Held out	Scope
Within UK	Fourteen plant identities	Transfer to unseen UK plants
UK to US	Eight US plants	Inference-only regional transfer

Table 6.2: Transfer settings evaluated in the thesis.

The US experiment evaluates fixed UK-trained checkpoints on eight quality-filtered plants, with the same valid daylight targets and three matched seeds for the main arms. It provides evidence beyond the UK fleet, while covering one evaluation period and a joint change in region and imagery source. Further regions and observation periods are needed to assess the breadth of this transfer.

6.7 Limitations and reproducibility

Several methodological boundaries characterize the current evaluation protocol. First, while the plant split is strictly disjoint across installations, all United Kingdom installations are observed across the 2019–2020 period. Because synoptic-scale weather fronts can span multiple adjacent installations simultaneously, spatial autocorrelation across co-temporal observations cannot be entirely ruled out. A multi-year spatio-temporal split (for instance, training on 2019 and evaluating on 2020) will provide an even more stringent test of geographical transfer without shared weather events. Second, horizon meteorological covariates in this benchmark are drawn from archived reanalysis observations. While this isolates the incremental contribution of satellite imagery from operational weather forecast noise, these inputs represent an idealized upper bound compared to operational numerical weather prediction (NWP) feeds. Third, the collapse observed under the wrong-plant image swap control (from 55.10% to 37.12%) reflects a combination of dynamic cloud field mismatch and static geographic disruption, as satellite crops inevitably capture local topography, water bodies, and coastal boundaries

that characterize specific installations. Finally, the strong performance of Time-VLM on ramp NMAE (0.1374) demonstrates that re-encoding numerical generation histories through visual transformers provides an effective smoothing mechanism that competes with satellite-augmented models during steep generation changes without requiring external imagery.

A comprehensive understanding of multimodal foundation models for solar forecasting requires expanding empirical validation across broader meteorological regimes. Evaluating models over multi-year continuous records will capture annual seasonal transitions, including low-solar-elevation winter regimes, snow cover, and summer convective activity. Testing should also encompass diverse climatic environments, such as arid regions characterized by high aerosol loads and mountainous regions where local topography drives complex microscale cloud dynamics. Systematically benchmarking performance during severe weather fronts and rapid squall lines will provide direct evidence of model stability under grid-critical ramp events.

Computational requirements present distinct trade-offs across training and operational deployment. To keep training efficient across multiple seeds, the architecture relies on precomputed, frozen visual latents from V-JEPA. This caching strategy eliminates repeated video encoding passes, enabling the processing of extensive histories within standard accelerator memory limits. However, freezing the visual backbone prevents end-to-end gradient updates from tuning early visual filters to solar irradiance dynamics. At inference time, downloading real-time satellite channels, cropping coordinates, and projecting multi-anchor tokens introduces data-loading latency and memory overhead. For transmission operators operating under sub-minute dispatch intervals, the quadratic complexity of cross-attention over combined temporal and spatial tokens requires careful budgeting of token length and anchor frequency.

Generalization across international fleets requires advancing transfer learning across sensor domains and heterogeneous plant designs. Deploying models trained on SEVIRI imagery to GOES-16 observations involves accommodating differences in spectral channels, spatial resolutions, refresh rates, and parallax distortion from satellite viewing angles. Beyond sensor variations, solar installations differ in tilt angles, inverter clipping thresholds, panel degradation rates, and tracking mechanisms. Zero-shot transfer currently relies on historical numerical power records for site normalization. Developing architectures that explicitly condition on installation metadata or learn sensor-invariant representations will strengthen cross-regional transfer across disparate grids.

Multimodal architectures also need to deepen their reliance on visual evidence during complex weather patterns. While spatial-token fusion responds to recent imagery during ramp transitions, the visual pathway should contribute more consistently under diffuse sky conditions and multi-layer cloud formations. Incorporating inductive biases tailored to atmospheric advection and cloud optical depth can prevent the time-series backbone from falling back on smooth diurnal statistics when cloud patterns are ambiguous. Furthermore, point-forecast improvements must be paired with calibrated uncertainty estimation. Cross-plant deployment introduces epistemic uncertainty from unfamiliar topographies and sensor noise. Future architectures should integrate distribution-free conformal calibration to ensure that prediction intervals provide certified reliability for storage management and grid stability.

Reproducible evaluation of multimodal foundation models depends on pairing benchmark score tracking with systematic lesion controls. Paired vision-off evaluations, coordinate permutations, and temporal shuffling confirm that forecast improvements reflect genuine physical atmospheric signals. Retaining these matched control protocols across future architectural comparisons ensures that empirical progress reflects authentic multimodal integration.

7 Conclusions

The thesis tests whether fusion of time-series and visual foundation models can improve forecasts for plants excluded from fitting. The UK benchmark and its controls establish a forecasting gain from the tested spatial-token interface. The UK-to-US experiment extends the evaluation to regional transfer and shows that the preferred visual configuration depends on the metric and setting.

7.1 Answer to the research question

Spatial-token fusion reaches 55.10 percent skill and 0.1440 ramp NMAE on the UK plant-disjoint test. The adapted numerical Chronos-2 control reaches 52.30 percent and 0.1506 under the same protocol. Both measures improve in every paired seed, and mean skill improves at thirteen of the fourteen held-out plants. The advantage extends from one hour through the six-hour horizon.

Disabling the visual pathway in the same trained spatial-token weights raises NMAE by 0.0047 and ramp NMAE by 0.0063. Stale and wrong-plant clips also degrade the forecasts consistently. These controls show that the fitted model uses recent, site-specific visual content. Pooled-image fusion shows a much smaller NMAE response and little measured ramp response when vision is disabled.

The ablations identify simplifications that preserve this result. Mean pooling retains spatially distinct cells and improves skill, while precise fractional placement has little measured effect. Freezing the adapted numerical encoder leaves point scores close to the original arm but worsens calibration. Permuting cached temporal latents produces negligible degradation (0.00030 NMAE), indicating that downstream cross-attention is insensitive to the ordering of pre-encoded visual latents.

7.2 What the transfer results establish

The UK result concerns transfer to unseen plant identities after training on other plants in the same corpus. The inference-only US experiment evaluates fixed UK-trained checkpoints on eight eligible plants, using three matched seeds for the main arms. Multi-anchor fusion reaches 10.46 percent mean skill and 0.1657 NMAE, improving on the numerical control's 8.90 percent and 0.1719. Spatial-token fusion records the lowest ramp NMAE at 0.1692, with multi-anchor close at 0.1702; both improve on the numerical control's 0.1864.

These results support structured multimodal fusion in the two tested transfer settings. They also distinguish overall point skill from performance on ramp intervals. The UK visual-use controls establish reliance within that experiment. The US score comparison extends the geographic evidence, while leaving the separate effects of region, sensor, and visual-history allocation unresolved.

Full payload is a feasibility result with one seed in each track. Its UK run reaches 54.49 percent skill and 0.1458 ramp NMAE and shows visual reliance in the same-weights control. The US run reaches 3.46 percent skill and 0.1786 ramp NMAE. Additional seeds are needed to establish a repeatable payload effect.

Point-forecast gains do not resolve probability calibration. Spatial-token fusion lowers UK CRPS, but its interval coverage and quantile ECE are worse than the numerical control's. Coverage is lower under US transfer. Calibration remains a limitation of the tested systems alongside the restricted geographic coverage and the unresolved motion mechanism.

7.3 Future work

The findings of this thesis suggest several technical extensions for multimodal solar forecasting. Research can expand along five directions: broadening geographic and seasonal coverage across international fleets, deploying agentic pipelines for large-scale dataset construction, optimizing the temporal allocation of visual context, improving probability calibration, and evaluating emerging multimodal

foundation architectures. The following subsections describe these priorities.

7.3.1 Broader geographic evaluation

The next evaluation should include more plants, longer observation periods, and additional regions. Holding out entire regions or sensor families would extend the present plant-disjoint test and help separate geographic transfer from a change in imagery source. The common plant-weighted evaluation and matched training seeds should be retained so that differences can be interpreted as the evidence grows.

7.3.2 Dataset construction and validation

Scaling the benchmark across international fleets requires structured processing of multi-terabyte satellite archives and ground sensor recordings. The next engineering phase will deploy an agentic data curation pipeline operating downstream of raw archive retrieval, dedicated to data cleaning, standardization, and corpus organization. Autonomous software agents will execute multi-stage cleaning routines, identifying sensor flatlines, filtering missing timestamps, and auditing quality flags across solar fleets. For standardization, the agents will harmonize differing UTC timestamp conventions, align satellite spatial coordinate projections with ground site locations, and compute uniform daylight elevation masks. The system will then organize cleaned multimodal streams into structured, shard-based formats and construct strict plant-disjoint evaluation splits with inspectable provenance logs. Automating these cleaning and organization workflows ensures reproducible sample reconstruction as the benchmark expands to global scale.

7.3.3 Visual history and payload allocation

The multi-anchor results motivate experiments that vary history length, anchor spacing, and per-anchor detail independently. The current comparison changes these choices together. Completing the full payload seeds would provide a firmer basis for evaluating a larger visual history. Each new configuration should include the same-weights vision-off control, followed by timing and site-substitution checks where a visual contribution is measured.

Evaluating lead-specific retrieval of visual evidence would clarify how different forecast horizons draw upon recent versus older cloud observations. Controlled masking across lead times can directly isolate which temporal anchors support near-term ramp prediction compared with longer-horizon trajectory forecasting.

7.3.4 Calibration and mechanism controls

Future work should assess calibration alongside point and ramp error. Interval widths and individual quantile predictions would help distinguish the possible causes of undercoverage. The encoder-freezing result also motivates a closer examination of how numerical adaptation affects probability calibration.

Temporal interventions before visual encoding would address a different question from the completed cached-latent permutation. They could test whether the original frame order contributes useful information that is already mixed into the cached representation. Selector and token-budget alternatives should also be retrained under matched conditions before claiming a general advantage for one payload construction.

7.3.5 Extending the external comparison

Further multimodal architectures should be evaluated under the same plant-disjoint setting. Comparing different ways of representing imagery and joining it to numerical history would test how broadly the observed fusion gains apply. Those comparisons should report forecasting quality together with visual-use controls, so that an improved score can be assessed alongside evidence that the additional modality contributes to the fitted forecast.

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